Bible of Economics of Technology and Management

Marco Marino

2025/2026

Summary

1 Introduction

1.1 Random concepts captured (24/02/2026)

Sustainability

The modern concept of sustainability was formalized in the 1980s in Northern Europe and politically promoted by Gro Harlem Brundtland, Prime Minister of Norway.

Definition 1.1 (Sustainable Development). *Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.*

Sustainability is based on three interconnected pillars:

- Environmental protection
- Economic growth
- Social improvement

Taxes

Definition 1.2 (Tax). *Compulsory payment imposed by the government that increases the final price paid by consumers beyond the original market price of a product or service.*

A tax creates a wedge between the price paid by consumers and the price received by producers.

Ministry of Made in Italy

Definition 1.3 (Ministry of Made in Italy). *Italian governmental institution responsible for promoting, protecting, and developing national production, industrial policy, and the international competitiveness of Italian firms.*

Greenwashing

Definition 1.4 (Greenwashing). *The practice of communicating or promoting environmentally sustainable actions or policies without actually implementing substantial or verifiable environmental improvements.*

It creates a misleading perception of environmental responsibility.

Competition

Definition 1.5 (Competition). *Strategic process through which firms seek to gain market advantage either by minimizing production costs or by differentiating their products from those of competitors.*

Cost minimization leads to price-based competition, while product differentiation leads to value-based competition.

Opportunity Cost of Capital

Definition 1.6 (Opportunity Cost of Capital). *The expected return that investors forgo by investing resources in a specific project instead of in the best alternative investment with comparable risk.*

It represents the minimum return required to justify the allocation of capital to a given investment.

Economies of Scale

Definition 1.7 (Economies of Scale). *They occur when the average cost of production decreases as the quantity of output increases.*

They arise because fixed costs are spread over a larger number of units and because larger scale production may improve efficiency and bargaining power.

Consider bottled water. The average price per liter is typically lower when purchasing a large pack (e.g., a case of water bottles) rather than a single small bottle. This happens because fixed costs such as packaging, transportation, and distribution are spread over a larger quantity of product. As a result, the cost per unit decreases when buying in bulk.

Equilibrium between Demand and Offer

Definition 1.8 (Market Equilibrium). *The condition in which the quantity demanded by consumers equals the quantity supplied by producers.*

Economics studies how prices adjust to balance demand and supply, leading to an equilibrium allocation of goods and resources.

Supply Risk

Definition 1.9 (Supply Risk). *The possibility that the production of a good is disrupted due to limited availability or restricted access to essential raw materials.*

For example, smartphones contain small but significant amounts of gold and other rare materials. If these resources are geographically concentrated (e.g., controlled or largely available in specific countries) and access becomes restricted, firms may be unable to produce the final product. Dependence on critical raw materials therefore represents a strategic production risk.

Customer Perceived Value

Definition 1.10 (Customer Perceived Value). *The subjective evaluation that a consumer forms about a product, which may differ from its objective or technical characteristics.*

Perception is influenced by branding, marketing, reputation, and personal expectations, and it often differs from the product's real intrinsic value.

Digital Product Passport

Definition 1.11 (Digital Product Passport (DPP)). *A structured digital record that contains standardized information about a product's origin, materials, production processes, environmental impact, and lifecycle.*

It enables transparency, traceability, and sustainability monitoring across the entire value chain, supporting circular economy practices.

Energy Impact of Digitalization

Observation 1.1. *Digitalization requires significant energy consumption, often supplied by non-renewable sources, particularly for data centers, network infrastructures, and digital devices.*

As a consequence, the expansion of digital technologies may increase carbon emissions unless energy is sourced from renewable alternatives.

Sustainable Development Goals (SDGs)

Definition 1.12 (Sustainable Development Goals (SDGs)). *A set of 17 global objectives adopted by the United Nations to promote sustainable development by 2030, integrating economic growth, social inclusion, and environmental protection.*

They provide a common framework for governments, firms, and institutions to align policies and strategies with sustainability principles.

Strategic and Speculative Investments

Definition 1.13 (**Strategic Investment**). *An allocation of capital aimed at generating long-term value, competitive advantage, or sustainable growth for a firm.*

Definition 1.14 (**Speculative Investment**). *An allocation of capital primarily aimed at obtaining short-term financial gains by exploiting expected price fluctuations.*

10-Point Scale

Definition 1.15 (**10-Point Scale**). *Survey evaluation method in which respondents assign a score from 1 to 10 to express their judgment or level of satisfaction.*

Compared to a 5-point scale, a 10-point scale provides greater granularity and allows for more differentiated evaluations. In a 1–10 scale, the intermediate value is 5.5, which represents the theoretical midpoint of the evaluation range.

Willingness to Pay (WTP)

Definition 1.16 (**Willingness to Pay (WTP)**). *The maximum amount of money that a consumer is willing to spend to obtain a good or service.*

It represents the monetary expression of the value that a consumer assigns to a product.

Open Dialogue

Definition 1.17 (**Open Dialogue**). *A structured interaction among a group of individuals aimed at fostering transparent communication, shared understanding, and collaborative problem-solving.*

It encourages the exchange of different perspectives in order to reach more informed and inclusive decisions.

Substitute and Complementary Goods

Definition 1.18 (**Substitute Goods**). *Products that can replace each other in satisfying the same need. An increase in the price of one good typically increases the demand for its substitute.*

Definition 1.19 (**Complementary Goods**). *Products that are used together. An increase in the price of one good typically reduces the demand for its complement.*

1.2 Introduction to Business

Definition 1.20 (**Business**)**.** *An organization (or economic system) that uses resources to satisfy customers' needs by providing goods or services demanded in the market.*

Goods and services are exchanged either directly (barter) or, more commonly, for money.

Definition 1.21 (**Investment**)**.** *The allocation of financial, physical, or human resources with the expectation of generating future returns.*

Every business requires:

- an initial investment;
- a sufficient customer base;
- a product or service that meets market demand.

Definition 1.22 (**Profit**)**.** *The positive difference between total revenues and total costs:*

$$\mathit{Profit} = \mathit{Revenues} - \mathit{Costs}$$

A business is **economically sustainable** only if revenues exceed costs over time.

1.2.1 Business Process

A business can be analyzed as a **process that transforms inputs into outputs**:

Input ⟶ **Output**

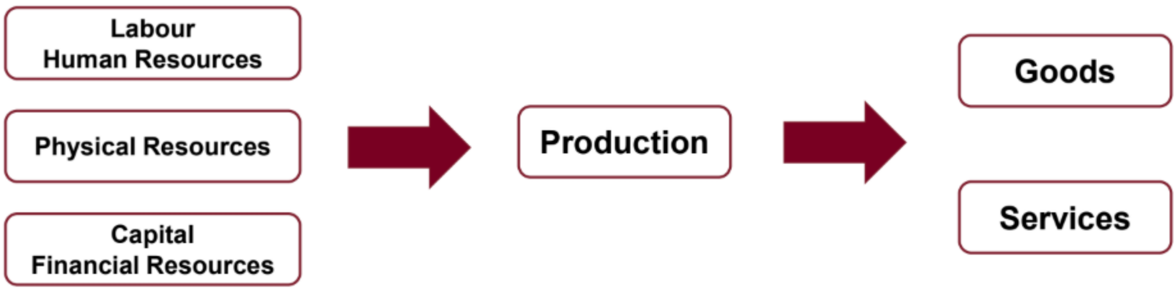
Resources **Products/Services**

Definition 1.23 (**Inputs**)**.** *The resources used by a firm to produce goods or services.*

Definition 1.24 (**Outputs**)**.** *Goods and services that satisfy identified customer needs and are typically sold in the market to generate revenues.*

Definition 1.25 (**Business Process**)**.** *The transformation of inputs into outputs in order to satisfy customer needs and generate profit.*

Organizations first identify **customer needs** and then acquire and combine resources (also called **production factors**) to produce one or more outputs.



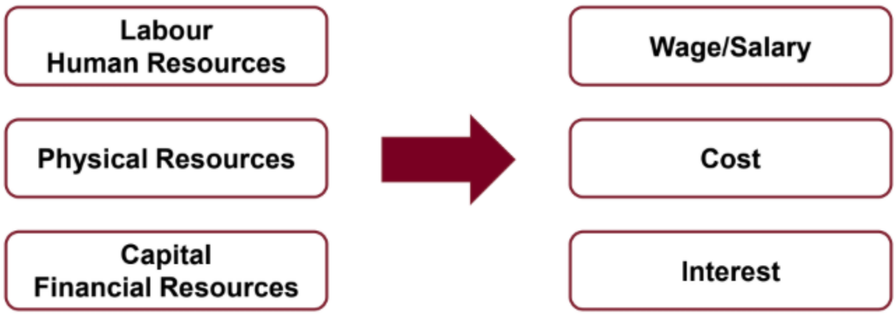
The main categories of inputs are:

- **Labour (human resources)**
- **Physical resources**
- **Capital (financial resources)**

The **production process** is usually distinguished into:

Definition 1.26 (**Capital-Intensive Production**). *A production process that relies primarily on physical and financial capital rather than human labour.*

Definition 1.27 (**Labour-Intensive Production**). *A production process that relies primarily on human labour rather than capital equipment.*



Each input used in production must be **supplied/purchased** and therefore **remunerated**.

Definition 1.28 (**Remuneration of Inputs**). *The payment made to resource owners in exchange for the use of production inputs.*

Tipically:

- Labour (human resources) is remunerated through **wages/salaries**.
- Physical resources generate a **cost** (e.g., purchase, maintenance, depreciation).
- Capital (financial resources) is remunerated through **interest**.

For what concerns outputs, they have almost opposite characteristics depending on whether we are talking about products (goods) or services.

Definition 1.29 (**Goods**)**.** *Tangible outputs that can be produced, stored, and transferred to consumers.*

Definition 1.30 (**Services**)**.** *Intangible outputs produced and consumed through an interaction between provider and customer.*

The typical characteristics of goods are the following:

- **Tangibility:** goods can be inventoried and displayed; their features can be communicated.
- **Standardization:** goods can be produced with a predefined and replicable set of features.
- **Separation:** production, purchase, and usage are typically distinct phases.
- **Durability:** goods can often be stored, returned, or resold.

While the typical characteristics of services are the following:

- **Intangibility:** services cannot be inventoried or displayed like physical products.
- **Heterogeneity:** delivery and perceived quality vary across providers, customers, and contexts.
- **Inseparability:** production and consumption occur simultaneously; customers often participate in delivery.
- **Perishability:** services cannot be stored, returned, or resold.

1.2.2 Business Size

Organizations can be classified according to different criteria. One of the most commonly adopted classifications is based on **business size**, typically measured by the number of employees.

Definition 1.31 (**Small and Medium Enterprises (SMEs)**)**.** *Firms with fewer than 250 employees.*

SMEs are usually divided into:

- **Micro Enterprises:** from 1 to 9 employees.
- **Small Enterprises:** from 10 to 49 employees.
- **Medium Enterprises:** from 50 to 249 employees.

Definition 1.32 (**Large Enterprises**)**.** *Firms with 250 or more employees.*

1.2.3 Business Functions

Definition 1.33 (Business Functions). *Specialized organizational areas responsible for specific sets of activities necessary for the firm to operate and create value.*

Organizations, independently of their size, are typically structured into **six main functional areas**:

Marketing and Sales

- Defines products and services.
- Sets pricing strategies.
- Manages distribution channels (place).
- Develops promotion strategies.
- Oversees packaging.

Finance and Administration

- Ensures the availability of financial resources (capital).
- Manages accounting and administrative tasks.

Human Resources

- Recruitment and training of employees.
- Compensation and incentive systems.

- Job assignment and workforce planning.

Logistics and Operations Management

- Management of production processes.
- Quality control.
- Management of logistics flows and warehouses.

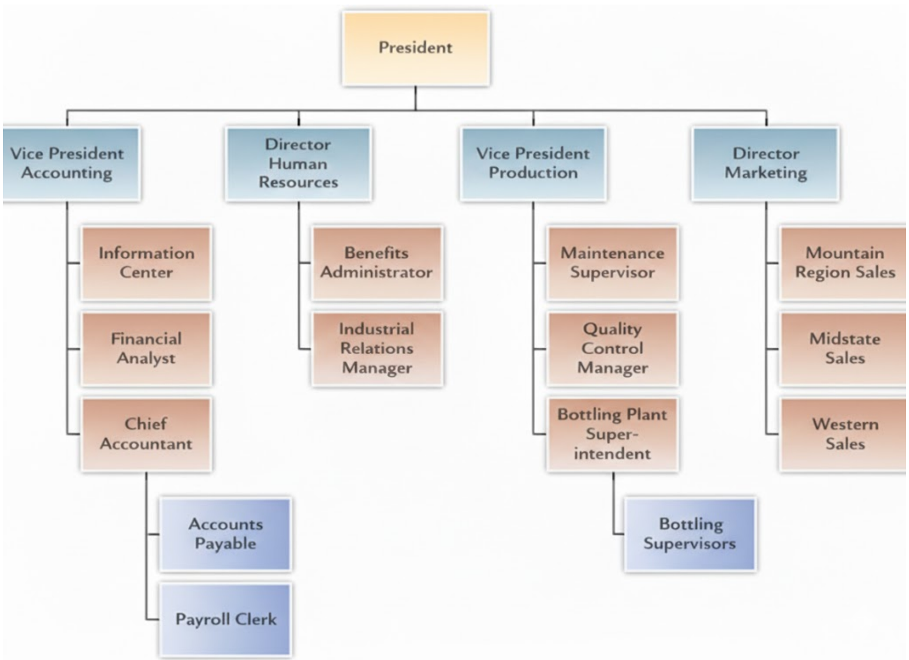
Research and Development (R&D)

- Research of new products/services.
- Development of new products/services.
- Improvement and upgrading of existing products.

Information Technology (IT)

- Management of IT infrastructures and digital systems.

1.2.4 Organizational Chart



Definition 1.34 (Organizational Chart). *A graphical representation of a company's internal structure, showing hierarchical relationships, roles, and lines of authority.*

In a typical hierarchical structure, authority flows from the top management (e.g., President) to functional managers (e.g., Accounting, Human Resources, Production, Marketing), and then to lower-level supervisors and operational staff.

This type of structure clarifies:

- who reports to whom;
- the division of responsibilities;
- the functional specialization within the organization.

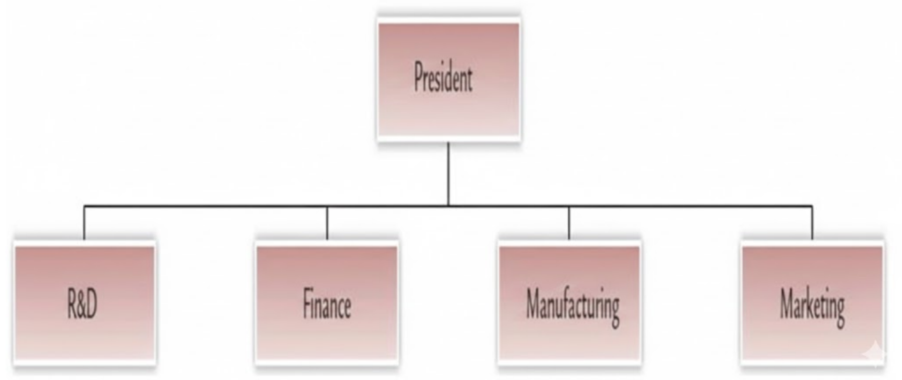
It is characteristic of a **functional organizational structure**, where activities are grouped by business functions.

1.2.5 Organizational Structure

Definition 1.35 (**Organizational Structure**). *It defines how tasks, responsibilities, authority, and communication are arranged within a firm.*

There are **three main types** of organizational structures.

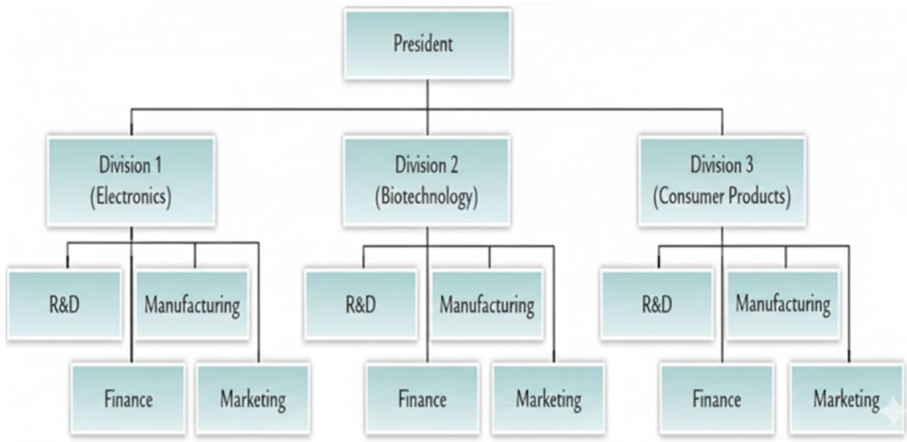
1.2.5.1 Functional Structure



Definition 1.36 (**Functional Structure**). *It groups activities according to business functions (e.g., R&D, Finance, Manufacturing, Marketing).*

This structure is generally the simplest organizational form. It centralizes activities by function and minimizes duplication of resources, leading to lower administrative costs. Since all departments report to top management, it promotes specialization and efficiency within each function but may reduce coordination across departments.

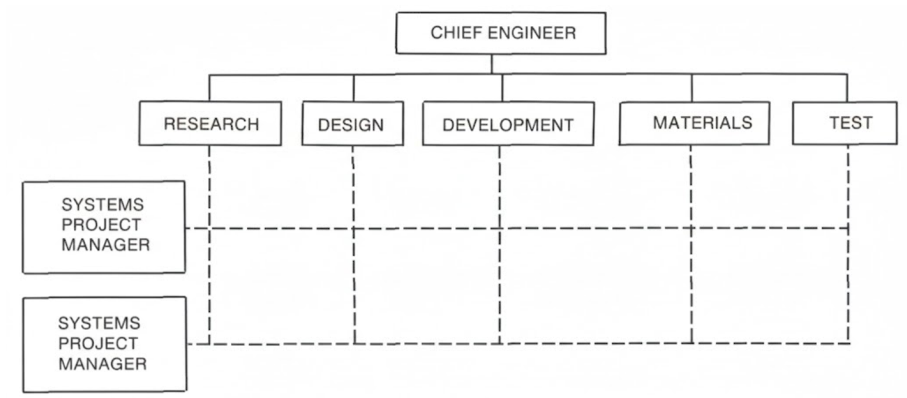
1.2.5.2 Divisional Structure



Definition 1.37 (**Divisional Structure**)**.** *It organizes the company into semi-autonomous divisions based on products, markets, or geographic areas.*

Each division contains its own functional units (e.g., R&D, Manufacturing, Finance, Marketing). It increases flexibility and market focus but may duplicate resources across divisions, leading to higher costs. A divisional structure is typically adopted when a firm operates in highly differentiated products or heterogeneous markets that require distinct strategies, competencies, and managerial focus.

1.2.5.3 Matrix Structure



Definition 1.38 (**Matrix Structure**)**.** *It combines functional and divisional structures, where employees report to both a functional manager and a project or product manager.*

This structure is more complex than both functional and divisional structures.
It introduces **bidirectional authority**, meaning that employees report simultaneously to:

- a functional manager (specialization dimension);

- a project/product manager (business or market dimension).

This structure combines high specialization with cross-functional coordination.

Definition 1.39 (**Strong Matrix**). *It gives greater authority to the project or product manager, while functional managers play a supporting role.*

Definition 1.40 (**Weak Matrix**). *It gives greater authority to functional managers, while project managers have limited coordination power.*

The matrix increases flexibility and integration but also raises coordination costs and potential conflicts due to dual reporting lines.

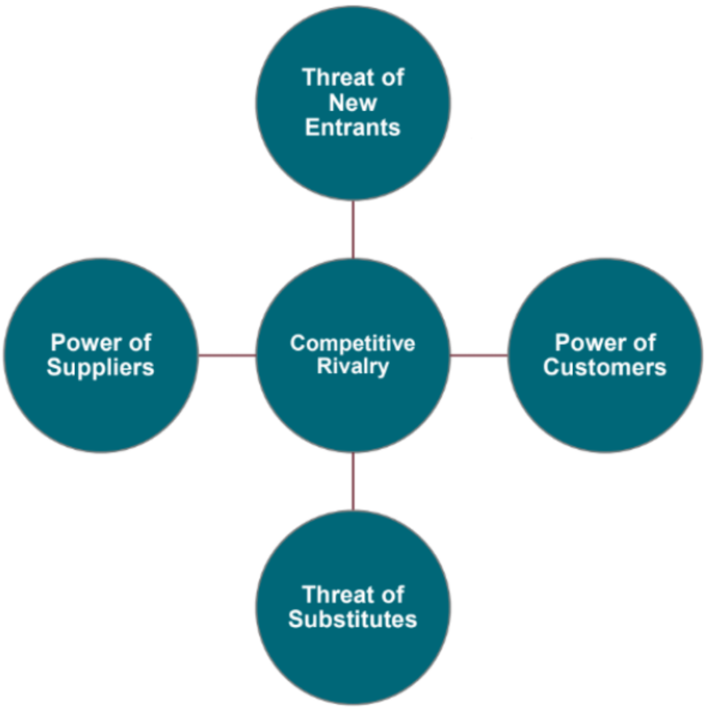
1.2.6 Competition

Definition 1.41 (**Industry (sector)**). *A group of firms operating in the same segment of the economy and sharing a similar type of business activity.*

Each industry includes one or more **markets**, where products and services are sold and where firms **compete** to achieve a favourable competitive position.

Definition 1.42 (**Market**). *The environment in which buyers and sellers interact to exchange goods or services.*

Porter's Five Forces Model (1985) is a framework used to analyze the competitive structure of an industry through **five key forces**.



- **Threat of New Entrants:** determined by entry barriers such as time, cost, specialized knowledge, and technology protection.
- **Power of Suppliers:** influenced by the number of suppliers, their size, uniqueness of inputs, and switching costs.
- **Power of Customers:** influenced by customer concentration, order size, price sensitivity, and switching costs.
- **Threat of Substitutes:** determined by the availability and performance of alternative products and switching costs.
- **Competitive Rivalry:** intensity of competition among existing firms, influenced by number of competitors, differentiation, and customer loyalty.

Definition 1.43 (**Competitive Strategy**). *The search for a favorable competitive position within an industry.*

Its objective is to achieve a profitable and sustainable position relative to competitors.

Definition 1.44 (**Competitive Advantage**). *An attribute, or a set of attributes, that enables a firm to outperform competitors and obtain a favorable competitive position.*

2
Noreen Ch01

2.1 Random concepts captured (25/02/2026)

Customer Engagement

Definition 2.1 (**Customer Engagement**). *The strategies and actions implemented by a firm to attract, involve, and retain customers by increasing their interaction and connection with the brand.*

Typical engagement tools include discounts, special offers, loyalty programs, and promotional campaigns aimed at stimulating customer participation and repeat purchases.

Work Capacity

Definition 2.2 (**Work Capacity**). *The amount of working time available to perform tasks within a given period.*

Before planning activities, it is necessary to verify the availability of working hours in order to allocate resources efficiently and avoid overload or underutilization.

Outsourcing

Definition 2.3 (**Outsourcing**). *The practice of contracting external firms or third parties to perform activities or services that were previously carried out internally.*

It is typically adopted to reduce costs, access specialized expertise, or increase organizational flexibility.

Strategy

Definition 2.4 (**Strategy**). *The long-term direction and scope of an organization, aimed at achieving competitive advantage and guiding future decisions.*

It defines how a firm allocates resources and positions itself in order to achieve its objectives over time.

Economics: Efficiency and Effectiveness

Definition 2.5 (**Economics**). *The study of how scarce resources are allocated, typically analyzed through quantitative measures and numerical evaluation.*

Economic analysis relies heavily on measurable variables such as costs, revenues, prices, and productivity.

Definition 2.6 (**Efficiency**). *The ability to achieve a given output using the minimum possible amount of resources.*

Definition 2.7 (**Effectiveness**). *The degree to which an objective or desired result is achieved.*

Efficiency concerns *how well* resources are used, while effectiveness concerns *whether* the intended goal is reached.

Financial vs. Economic Perspective

Definition 2.8 (**Financial Perspective**). *It concerns only monetary flows, particularly cash inflows and outflows (cash flow).*

It focuses on liquidity, revenues, costs, and the firm's ability to generate and manage money.

Definition 2.9 (**Economic Perspective**). *It concerns the creation of value through the efficient transformation of skills, resources, and services into outputs that generate utility and market value.*

While the financial dimension measures money movements, the economic dimension evaluates the firm's ability to create value from available resources.

Invention vs. Innovation

Definition 2.10 (**Invention**). *The creation of a new idea, concept, technology, or method.*

Definition 2.11 (**Innovation**). *The successful commercialization or practical application of an invention in the market.*

An invention becomes an innovation only when it generates economic value through market adoption.

Cost vs. Revenue

Definition 2.12 (**Cost**). *A monetary outflow (cash outflow) sustained by a firm for acquiring resources or conducting business activities.*

Definition 2.13 (**Revenue**). *A monetary inflow (cash inflow) generated from the sale of goods or services.*

Stakeholders vs. Shareholders

Definition 2.14 (**Stakeholders**). *All individuals or groups that have an interest in or are affected by the activities and performance of a firm.*

Stakeholders include shareholders, employees, customers, suppliers, creditors, governments, and the local community.

Definition 2.15 (**Shareholders**). *Individuals or entities that own shares of a company and therefore hold a financial ownership interest in it.*

All shareholders are stakeholders, but not all stakeholders are shareholders.

2.2 Managerial Accounting and Cost Concepts

Definition 2.16 (Financial Accounting). *Concerned with the preparation and reporting of financial information for external users, such as shareholders, creditors, regulators, and tax authorities.*

Its main objective is to provide standardized and reliable financial statements describing the economic performance and financial position of the firm.

Definition 2.17 (Managerial Accounting). *Concerned with providing financial and non-financial information to internal managers in order to support planning, control of operations, and decision-making.*

Unlike financial accounting, managerial accounting is internally oriented and not subject to strict external reporting standards.

Definition 2.18 (Commercial Activity). *It consists of purchasing finished goods and reselling them to customers without transforming them.*

The firm acts as an intermediary between producers and consumers.

Definition 2.19 (Manufacturing Activity). *It consists of purchasing raw materials or semi-finished goods, transforming them through a production process, and selling the resulting finished products.*

In this case, value is created through the transformation process.

2.2.1 Cost Classification

Costs are classified for five main purposes:

1. assigning costs to cost objects;
2. accounting for costs in manufacturing companies;
3. preparing financial statements;
4. predicting cost behavior in response to changes in activity;
5. supporting managerial decision-making.

The following cost classifications arise from **purpose 1**:

Definition 2.20 (Direct Cost). *A cost that can be easily and economically traced to a specific cost object.*

Definition 2.21 (**Indirect Cost**). *A cost that cannot be easily traced to a specific cost object and must be allocated.*

The following cost classifications arise from **purpose 2**:

Definition 2.22 (**Manufacturing Costs**). *Costs directly associated with the production process.*

They include:

- direct materials;
- direct labor;
- manufacturing overhead.

Definition 2.23 (**Nonmanufacturing Costs**). *Costs not directly related to production.*

They include:

- selling costs;
- administrative costs.

The following cost classifications arise from **purpose 3**:

Definition 2.24 (**Product Costs (inventoriable costs)**). *Costs assigned to products and included in inventory until the product is sold.*

Definition 2.25 (**Period Costs**). *Costs expensed in the period in which they are incurred.*

The following cost classifications arise from **purpose 4**:

Definition 2.26 (**Variable Cost**). *Cost that changes proportionally with the level of activity. A variable cost per unit is constant.*

Definition 2.27 (**Fixed Cost**). *Cost that remains constant in total within the relevant range of activity. If expressed on a per-unit basis, the average cost per unit varies inversely with the level of activity.*

Definition 2.28 (**Mixed Cost**). *Cost that contains both fixed and variable components.*

The following cost classifications arise from **purpose 5**:

Definition 2.29 (**Differential Cost**). *Cost that differs between alternative decisions.*

Definition 2.30 (**Sunk Cost**). *Cost that has already been incurred and cannot be recovered; it should not influence future decisions.*

Definition 2.31 (**Opportunity Cost**). *The benefit foregone by choosing one alternative over another.*

2.2.2 LO1[1]: Assigning Costs to Cost Objects

Definition 2.32 (**Cost Object**)**.** *Anything for which cost data are desired, such as a product, service, customer, project, or department.*

Costs assigned to cost objects can be classified as **direct** (2.20) or **indirect** (2.21). Typical examples of direct costs include:

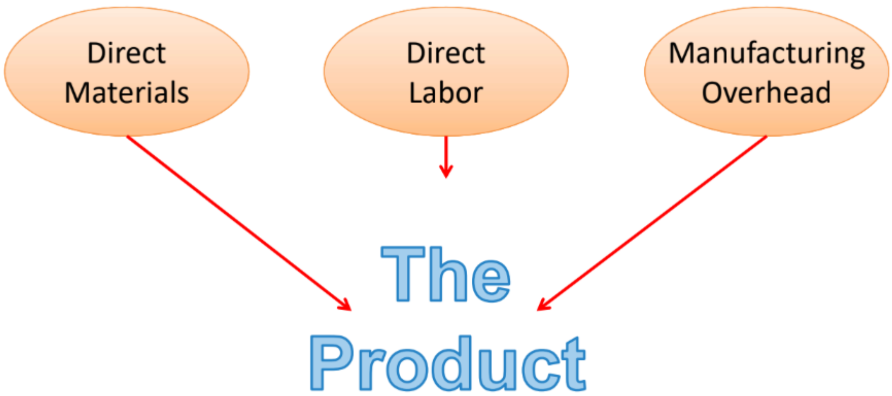
- direct materials;
- direct labor.

Typical examples of indirect costs include:

- manufacturing overhead;
- common costs.

Definition 2.33 (Common Cost)**.** *Indirect cost incurred to support multiple cost objects and cannot be traced to any single cost object.*

2.2.3 LO2: Classifications of Manufacturing Costs



Manufacturing costs (2.22) are classified into three main categories:

- direct materials;
- direct labor;
- manufacturing overhead.

These three elements together form the **total manufacturing cost** of a product.

[1]LO means Learning Objective.

Direct Materials and Direct Labor

Definition 2.34 (Direct Materials). *Raw materials that become an integral part of the finished product and can be conveniently traced to it.*

Example: A radio installed in an automobile.

Definition 2.35 (Direct Labor). *It consists of labor costs that can be easily and conveniently traced to individual units of product.*

Example: Wages paid to automobile assembly workers.

Manufacturing Overhead

Definition 2.36 (Manufacturing Overhead). *It includes all manufacturing costs except direct materials and direct labor. These costs cannot be directly traced to specific units of product.*

Manufacturing overhead includes:

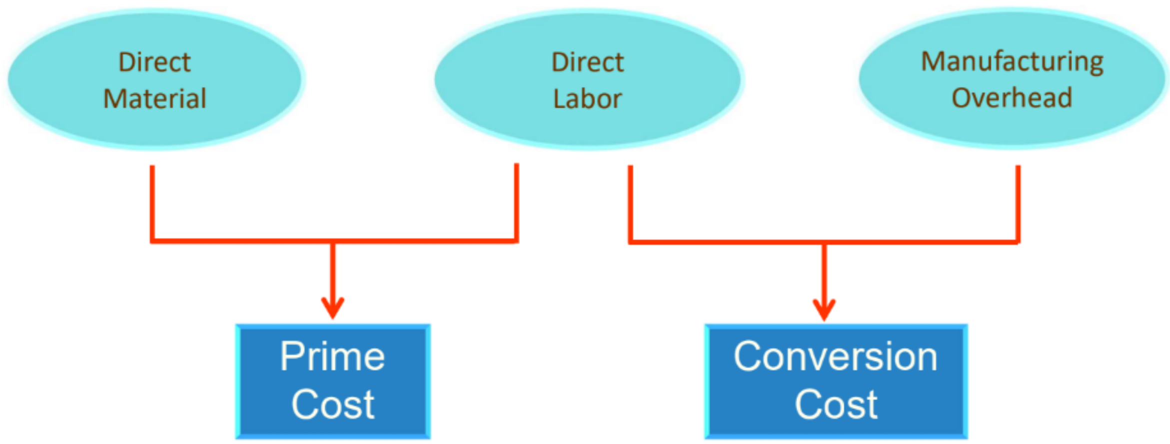
- indirect materials;
- indirect labor;
- other factory-related indirect costs.

Examples:

- lubricants and cleaning supplies used in the plant;
- maintenance workers and security guards;
- depreciation of manufacturing equipment;
- utility costs of the factory;
- property taxes and insurance on the manufacturing facility.

Only indirect costs associated with operating the factory are included in manufacturing overhead.

Prime and Conversion Costs



Definition 2.37 (Prime Cost). *The sum of direct materials and direct labor:*

$$\textit{Prime Cost} = \textit{Direct Materials} + \textit{Direct Labor}.$$

Definition 2.38 (Conversion Cost). *The sum of direct labor and manufacturing overhead:*

$$\textit{Conversion Cost} = \textit{Direct Labor} + \textit{Manufacturing Overhead}.$$

Nonmanufacturing Costs

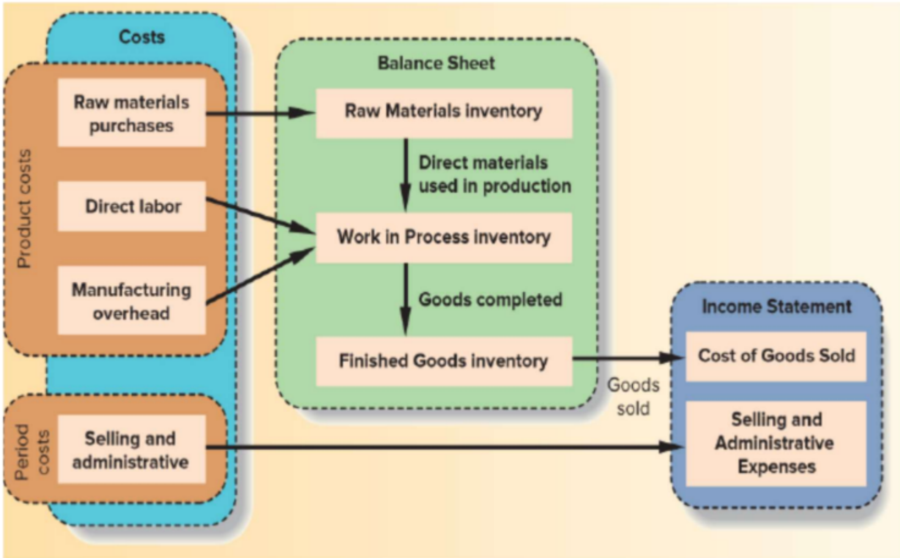
Nonmanufacturing costs (2.23) are classified into:

- selling costs (direct or indirect);
- administrative costs (direct or indirect);

Definition 2.39 (Selling Costs). *Costs incurred to secure customer orders and deliver the product.*

Definition 2.40 (Administrative Costs). *Costs that include executive, organizational, and clerical expenses necessary to manage the company.*

2.2.4 LO3: Preparing Financial Statements



Product costs (2.24) *attach* to units of product as they are purchased or manufactured and remain in inventory until the product is sold.

In manufacturing firms, product costs are composed of:

- raw material purchases (direct materials);
- direct labor;
- manufacturing overhead.

Period costs (2.25) tipically include selling (2.39) and administrative costs (2.40).

Definition 2.41 (**Balance Sheet**). *Financial statement that reports a company's assets, liabilities, and owner's equity at a specific point in time.*

Definition 2.42 (**Income Statement**). *Financial statement that reports revenues, expenses, and net income over a specific period of time.*

While the balance sheet represents the financial position at a given date, the income statement measures performance over a time interval. Both are based on the **fundamental accounting equation**.

Definition 2.43 (**Accounting Equation**). *Equation that expresses the relationship between a firm's resources and the claims on those resources:*

$$Assets = Liabilities + Equity.$$

Definition 2.44 (**Assets**). *Economic resources owned or controlled by a company that are expected to provide future economic benefits.*

Definition 2.45 (**Liabilities**). *Present obligations of a company arising from past events, which require the future transfer of assets or services to other parties.*

Definition 2.46 (**Equity (Owner's Equity)**). *It represents the residual interest in the assets of a company after deducting liabilities.*

- Product costs are recorded as **inventory** on the **balance sheet** and become an expense only when the goods are sold (COGS) on the **income statement**.
- Period costs are recorded directly as **expenses** on the **income statement**.

Definition 2.47 (**Raw Materials Inventory**). *It includes materials that will be used in production and become part of the final product.*

Definition 2.48 (**Work in Process (WIP) Inventory**). *It consists of units that are partially complete and will require further processing before sale.*

Definition 2.49 (**Finished Goods Inventory**). *It consists of completed units that have not yet been sold to customers.*

In a manufacturing company, product costs flow through inventory accounts as production progresses:

- When direct materials are used in production, their costs move from **Raw Materials** to **Work in Process**.
- Direct labor and manufacturing overhead are added to **Work in Process** to convert materials into finished goods.
- When units are completed, costs are transferred from **Work in Process** to **Finished Goods**.
- When finished goods are sold, costs are transferred from **Finished Goods** to **Cost of Goods Sold**.

Definition 2.50 (**Cost of Goods Sold (COGS)**). *The cost of product units that have been sold during the period; it is recognized as an expense in the income statement.*

EXAMPLE (Balance Sheet)

Example Company
Balance Sheet
December 31, 2016

ASSETS		LIABILITIES	
Current assets		Current liabilities	
Cash	$ 2,100	Notes payable	$ 5,000
Petty cash	100	Accounts payable	35,900
Temporary investments	10,000	Wages payable	8,500
Accounts receivable - net	40,500	Interest payable	2,000
Inventory	31,000	Taxes payable	6,000
Supplies	3,800	Warranty liability	1,100
Prepaid insurance	1,500	Unearned revenues	1,500
Total current assets	89,000	Total current liabilities	61,000
Investments	36,000	Long-term liabilities	
		Notes payable	20,000
Property, plant & equipment		Bonds payable	400,000
Land	5,500	Total long-term liabilities	420,000
Land improvements	6,500		
Buildings	180,000		
Equipment	201,000	Total liabilities	481,000
Less: accum depreciation	(56,000)		
Prop, plant & equip - net	337,000		
Intangible assets		STOCKHOLDERS' EQUITY	
Goodwill	105,000	Common stock	110,000
Trade names	200,000	Retained earnings	220,000
Total intangible assets	305,000	Accum other comprehensive income	9,000
		Less: Treasury stock	(50,000)
Other assets	3,000	Total stockholders' equity	289,000
Total assets	$ 770,000	Total liabilities & stockholders' equity	$ 770,000

The notes to the sample balance sheet have been omitted.

EXAMPLE (Income Statement)

Example Corporation
Income Statement
For the year ended December 31, 2016

Sales (all on credit)	$500,000
Cost of goods sold	380,000
Gross profit	120,000
Operating expenses	
Selling expenses	35,000
Administrative expenses	45,000
Total operating expenses	80,000
Operating income	40,000
Interest expense	12,000
Income before taxes	28,000
Income tax expense	5,000
Net income after taxes	$ 23,000

2.2.5 LO4: Predicting Cost Behavior

Definition 2.51 (**Cost Behavior**). *The way in which a cost reacts to changes in the level of activity.*

The cost classifications involved in this concept are:

- variable costs (2.26);
- fixed costs (2.27);
- mixed costs (2.28).

Variable Costs

Definition 2.52 (**Activity Base (Cost Driver)**). *A measure of what causes the incurrence of a variable cost.*

Examples of activity bases:

- units produced;
- machine hours;
- labor hours;
- miles driven.

Fixed Costs

There are two types of fixed costs:

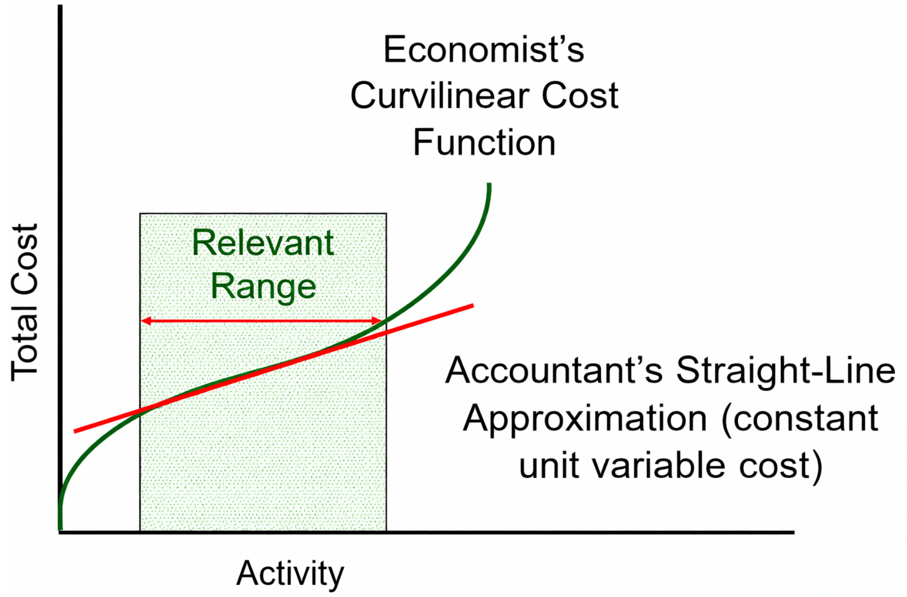
Definition 2.53 (**Committed Fixed Cost**). *Long-term fixed cost that cannot be significantly reduced in the short term.*

Definition 2.54 (**Discretionary Fixed Cost**). *Fixed cost that may be altered in the short term by current managerial decisions.*

Linearity Assumption and Relevant Range

Definition 2.55 (**Relevant Range**). *The range of activity within which cost behavior assumptions are valid.*

Observation 2.1 (**Linearity Assumption**). *Within the relevant range, total cost can be approximated by a linear function of the activity level.*

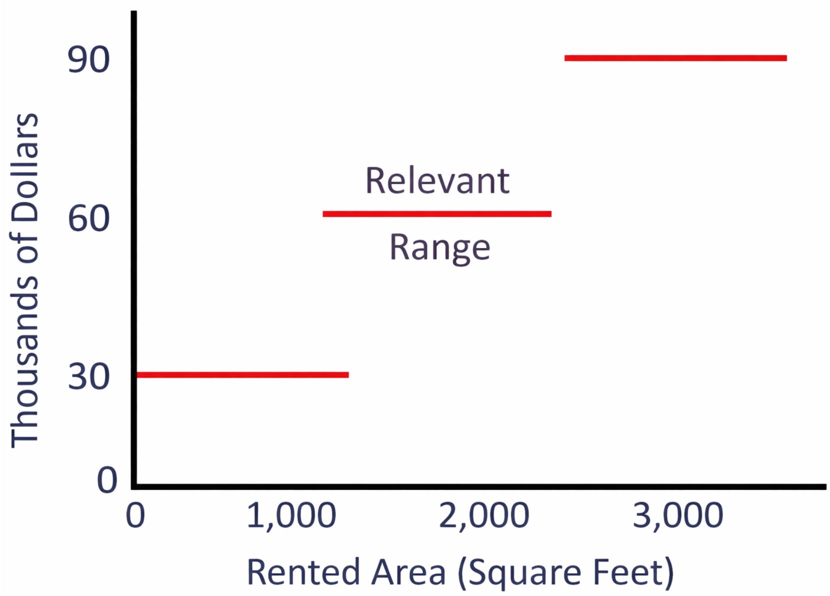


From a theoretical standpoint, total cost is typically **curvilinear**: as the level of activity increases, costs may initially increase at a decreasing rate and later at an increasing rate due to inefficiencies, congestion, overtime, or capacity constraints.

In managerial accounting, total cost is approximated using a **linear function**, assuming that:

- variable cost per unit is constant;
- total fixed cost remains constant.

Within the relevant range, the straight-line approximation closely follows the true (curvilinear) cost function. Outside the relevant range, the linear model may no longer provide an accurate representation of cost behavior.



Sometimes fixed costs increase in **steps** rather than continuously. For example, suppose office space costs \$30,000 per year for each additional 1,000 square feet rented. In this case, total fixed cost increases in a step fashion:

- From 0 to 1,000 sq ft → \$30,000
- From 1,001 to 2,000 sq ft → \$60,000
- From 2,001 to 3,000 sq ft → \$90,000

The cost is **fixed within each interval** (horizontal segment), but it increases in a **step pattern** when activity exceeds the current capacity. Therefore:

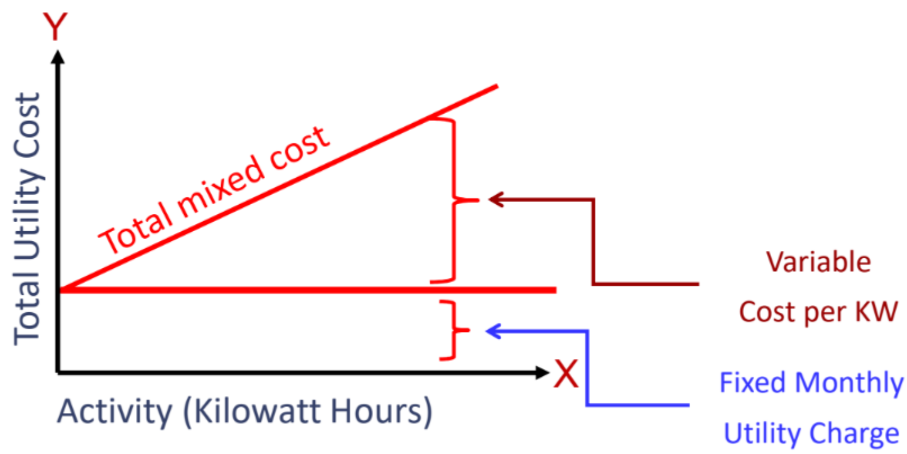
- within each step, the fixed cost is constant;
- when activity moves beyond the current capacity, total fixed cost jumps to a higher level.

The **relevant range** is the specific activity interval over which the fixed cost remains constant (i.e., where the graph is flat).

Comparison of Cost Behavior (Within the Relevant Range)

- **Variable Cost**
  - In total: changes proportionally with activity.
  - Per unit: remains constant.
- **Fixed Cost**
  - In total: remains constant.
  - Per unit: decreases as activity increases and increases as activity decreases.

Mixed Costs



A mixed cost has:

- A **fixed component** (incurred even if activity is zero).
- A **variable component** (changes proportionally with activity).

In the graph:

- The vertical intercept represents the **fixed cost**.
- The slope of the line represents the **variable cost per unit**.
- The total line represents the **total mixed cost**.

The total mixed cost can be expressed through a linear function:

$$Y = a + bX$$

Where:

- Y = total cost
- a = total fixed cost (vertical intercept)
- b = variable cost per unit (slope)
- X = level of activity

For example, suppose:

- Fixed monthly utility charge = \$40
- Variable cost per kilowatt hour = \$0.03
- Monthly activity level = 2,000 kilowatt hours

Then:

$$Y = 40 + (0.03 \times 2{,}000)$$

$$Y = 40 + 60$$

$$Y = \$100$$

Therefore, the total monthly utility bill is \$100.

2.3 Quick Checks

1. Which of the following costs would be considered a **period** rather than a **product** cost in a manufacturing company?

A. Manufacturing equipment depreciation.

B. Property taxes on corporate headquarters.

C. Direct materials costs.

D. Electrical costs to light the production facility.

E. Sales commissions.

Correct answer(s): B, E.

2. Which of the following costs would be **variable** with respect to the number of ice cream cones sold at a Baskins & Robbins shop? (There may be more than one correct answer.)

A. The cost of lighting the store.

B. The wages of the store manager.

C. The cost of ice cream.

D. The cost of napkins for customers.

Correct answer(s): C, D.

3
Noreen Ch02

3.1 Random concepts captured (03/03/2026)

Economist Perspective

Observation 3.1. *An economist focuses on analyzing the* ***percentage composition of costs****. In particular, the objective is to identify which cost components represent the largest share of the total cost in order to better understand cost structure and economic behavior.*

Saturation Level

Definition 3.1 (Saturation Level). *The degree to which the available production capacity is utilized.*

When demand increases, firms may increase production by using machinery more intensively or by installing additional machines. In these situations, variable costs increase because more resources (such as labor, energy, and materials) are consumed.
Production capacity is not always fully usable. For example, machines may require maintenance, during which production must stop.
If a machine is theoretically available for 24 hours per day but is used for 18 hours, the saturation level corresponds to those 18 hours of effective utilization.
When demand exceeds the available production capacity, firms may also resort to **outsourcing**, delegating part of the production to external suppliers due to the lack of internal resources or machinery.

Relevance of The Information

Definition 3.2 (Relevant Information). *Information that influences economic decisions because it differs between alternative courses of action.*

Definition 3.3 (Irrelevant Information). *Information that does not affect the outcome of a decision and therefore should not influence economic analysis.*

In everyday contexts, the distinction between relevant and irrelevant information can be subjective. In economics and managerial accounting, however, the distinction is more objective: information is relevant only if it affects the comparison between alternative decisions.

3.2 Cost-Volume-Profit (CVP) Relationships

To simplify CVP calculations, managers typically adopt several assumptions regarding prices, costs, and sales mix.

- **Selling price is constant**: the price of a product or service does not change as the sales volume changes.
- **Costs are linear**: costs can be divided into **variable** and **fixed** components. Variable costs remain constant **per unit**, while fixed costs remain constant **in total** within the relevant range.
- **Constant sales mix**: in multiproduct companies, the proportion in which different products are sold remains constant.

3.2.1 LO1: Changes in Activity, Contribution Margin, and Net Operating Income

Definition 3.4 (**Contribution Margin (CM)**). *The amount remaining from sales revenue after variable expenses have been deducted.*

The contribution margin is used first to cover **fixed expenses**. Any remaining contribution margin becomes **net operating income** (profit).

Using formulas:

$$\textbf{Contribution Margin} = \text{Sales} - \text{Variable Expenses}$$

$$\textbf{Net Operating Income} = \text{Contribution Margin} - \text{Fixed Expenses}$$

Where:

$$\textbf{Sales} = \text{Selling Price Per Unit } (P) \times \text{Quantity Sold } (Q)$$

$$\textbf{Variable Expenses} = \text{Variable Expenses Per Unit } (V) \times \text{Quantity Sold } (Q)$$

Therefore:

$$\textbf{Profit} = (P \times Q - V \times Q) - \text{Fixed Expenses}$$

Factoring Q:

$$\textbf{Profit} = (P - V) \times Q - \text{Fixed Expenses}$$

Where:

$$\textbf{Unit Contribution Margin} = P - V$$

Using this definition, the profit equation becomes:

$$\textbf{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

EXAMPLE

Consider the following contribution income statement:

Racing Bicycle Company Contribution Income Statement For the Month of June		
Sales (500 bicycles)	$	250,000
Less: Variable expenses		150,000
Contribution margin		100,000
Less: Fixed expenses		80,000
Net operating income	$	20,000

It is often useful to express values on a **per-unit basis** to highlight its contribution toward covering fixed expenses and generating profit:

Racing Bicycle Company Contribution Income Statemens For the Month of June				
	Total		For Unit	
Sales (500 bicycles)	$	250,000	$	500
Less: Variable expenses		150,000		300
Contribution margin		100,000	$	200
Less: Fixed expenses		80,000		
Net operating income	$	20,000		

This means that each additional unit sold contributes 200 $.

Definition 3.5 (Break-Even Point). *The level of sales at which profit is zero.*

The break-even point is reached when Contribution Margin = Fixed Expenses:

Racing Bicycle Company Contribution Income Statement, For the Month of June				
		Total		Per Unit
Sales (400 bicycles)	$	200,000	$	500
Less: Variable expenses		120,000		300
Contribution margin		80,000	$	200
Less: Fixed expenses		80,000		
Net operating income	$	-		

In our example, the break-even point is reached when the company sells 400 units:

$$400 \times 200\ \$ = 80.000\ \$$$

which exactly covers the fixed expenses (Profit = 0 $).

Racing Bicycle Company Contribution Income Statement For the Month of June				
		Total		Por Unit
Sales (401 bicycles)	$	200,500	$	500
Less: Variable expenses		120,300		300
Contribution margin		80,200	$	200
Less: Fixed expenses		80,000		
Net operating income	$	200		

If the company sells one more unit for a total of 401 units, the additional contribution margin is 200 $. Therefore:

$$\text{Net operating income} = 200\ \$$$

We would have gotten the same result using the formulas previously introduced:

$$\textbf{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

$$\textbf{Profit} = (500\ \$ - 300\ \$) \times 401 - 80.000\ \$ = 200\ \$ \times 401 - 80.000\ \$ = 80.200\ \$ - 80.000\ \$ = 200\ \$$$

3.2.2 LO2: CVP Graph & Profit Graph

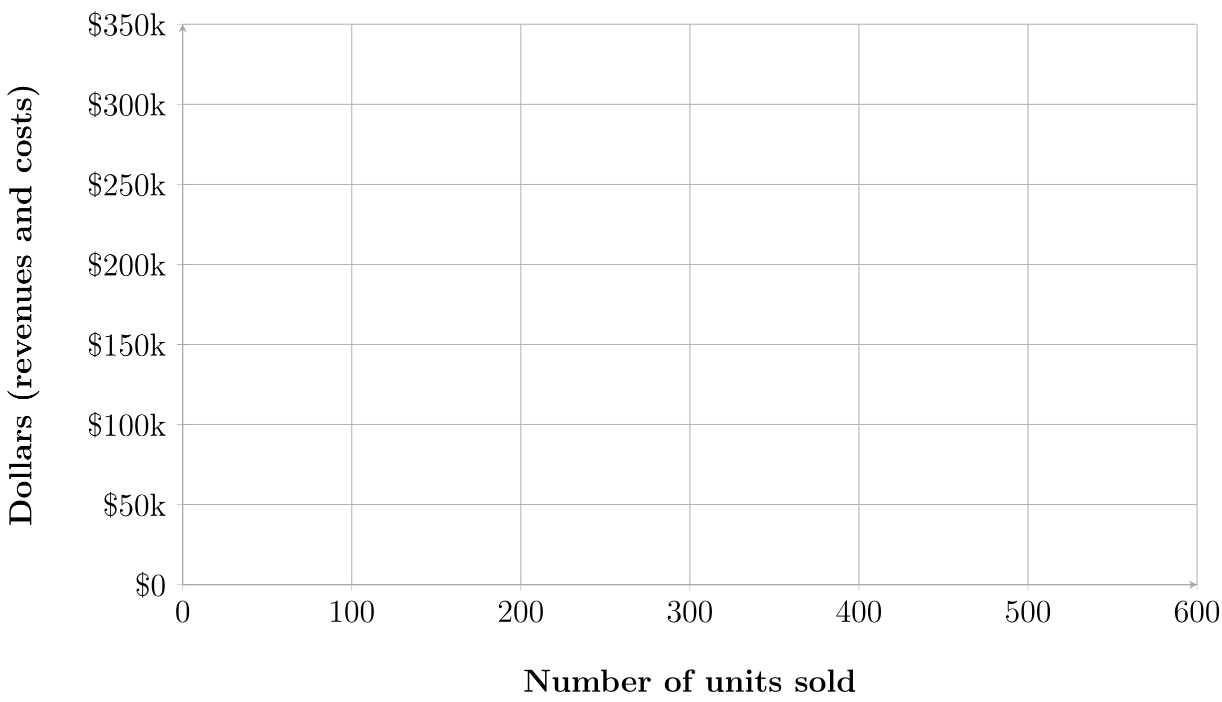
3.2.2.1 CVP Graph

The relationships among revenue, cost, profit, and volume can also be represented graphically using a **CVP graph**. Let's consider the following income statement prepared at 0, 200, 400 and 600 units sold:

	Units Sold			
	0	200	400	600
Sales	$ -	$ 100,000	$ 200,000	$ 300,000
Total variable expenses	-	60,000	120,000	180,000
Contribution margin	-	40,000	80,000	120,000
Fixed expenses	80,000	80,000	80,000	80,000
Net operating income (loss)	$ (80,000)	$ (40,000)	$ -	$ 40,000

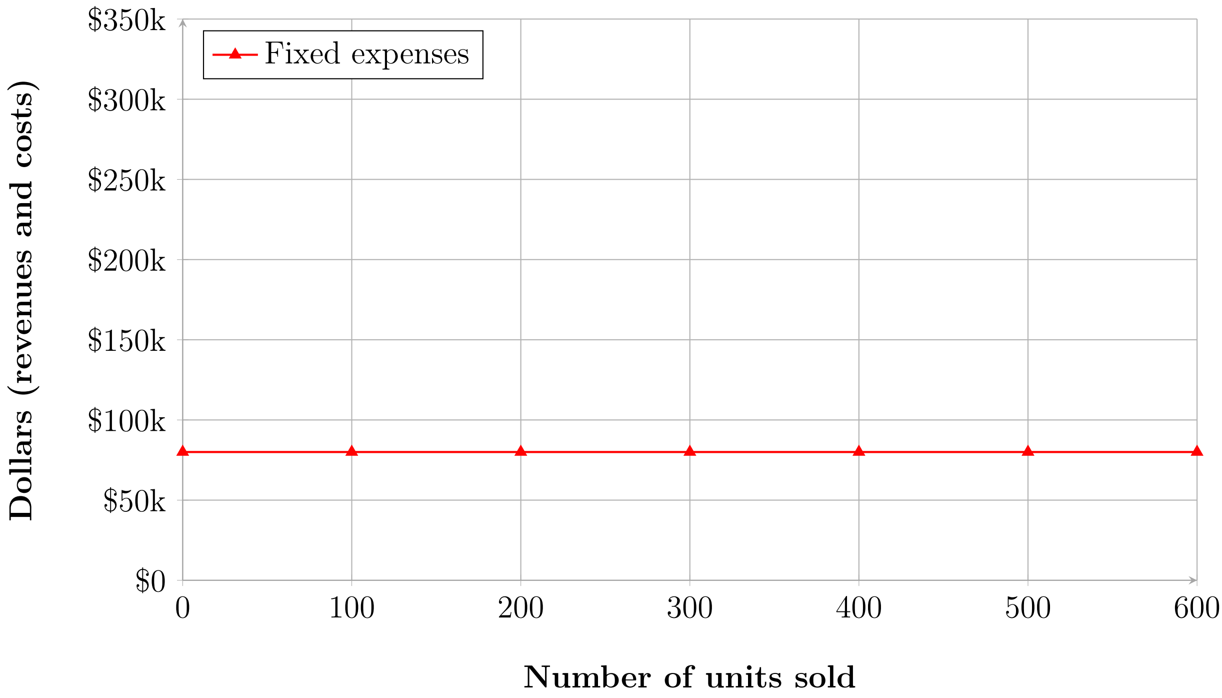
In a CVP graph:

- The **horizontal axis (X)** represents the number of units sold.
- The **vertical axis (Y)** represents dollars (revenues and costs).



Step 1: Plot Fixed Costs

We first draw a horizontal line parallel to the horizontal axis representing total **fixed expenses** (we assume 80.000):



Since fixed costs do not change with the number of units sold, this line remains constant.

Step 2: Plot Total Expenses

We choose a level of activity (we assume 400 units) and compute total expenses:

$$\text{Total expenses} = \text{Fixed expenses} + \text{Variable expenses}$$

If:

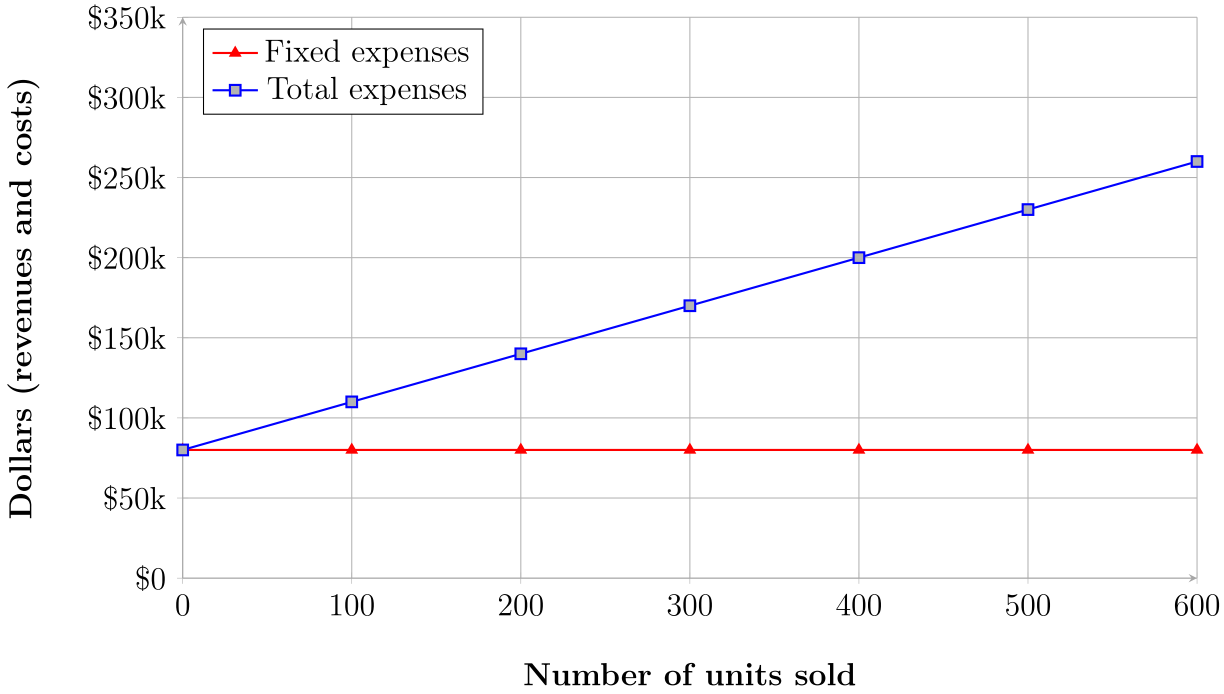
$$\text{Variable cost per unit} = 300$$

then:

$$\text{Variable expenses} = 400 \times 300 = 120.000$$

$$\text{Total expenses} = 80.000 + 120.000 = 200.000$$

So we plot the point and draw the **total expenses line** starting from the fixed-cost intercept:

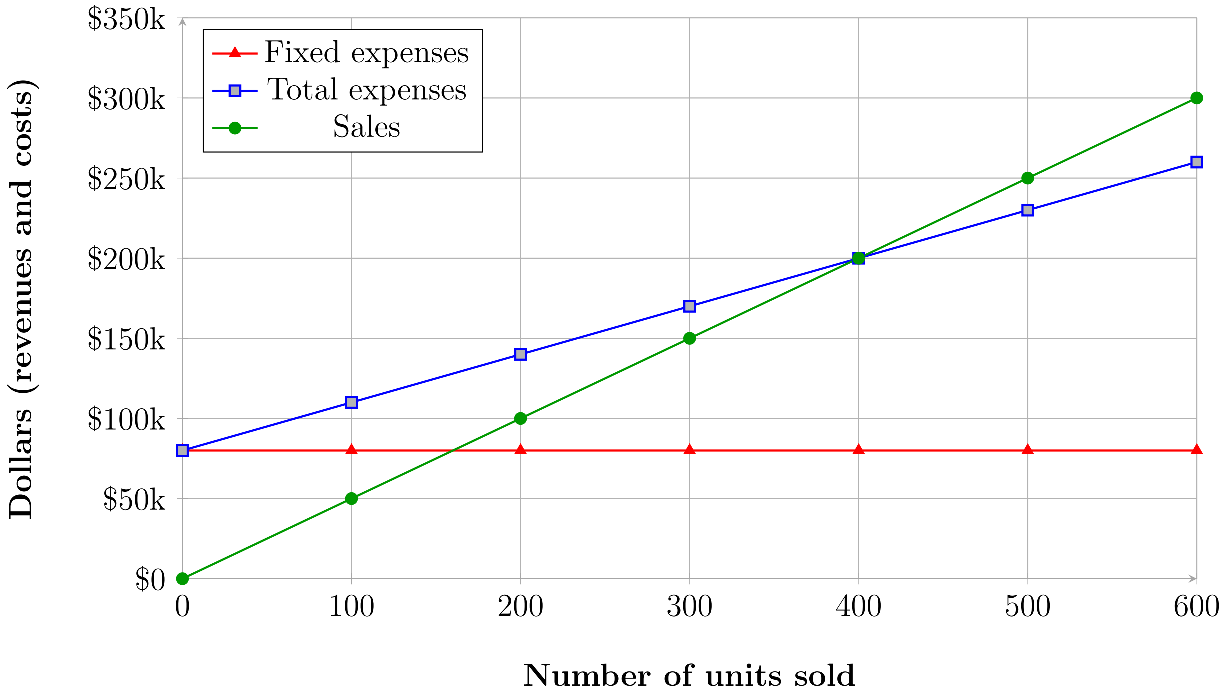


Step 3: Plot the Sales Line

We then compute sales for the same level of activity:

$$\text{Selling price per unit} = 500 \Rightarrow \text{Sales} = 400 \times 500 = 200.000$$

We plot this point and draw the **sales line** starting from the origin:

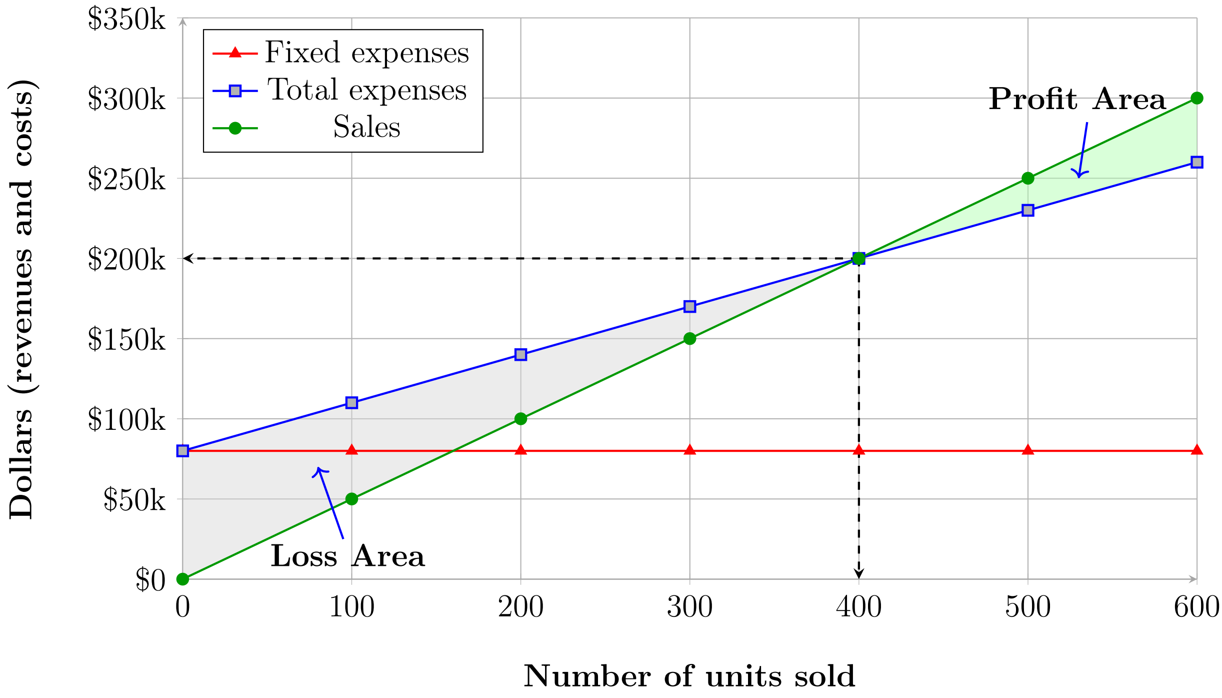


Step 4: Find the Break-Even Point

The **break-even point** occurs where the sales line intersects the total expenses line:

$$\text{Sales} = \text{Total expenses}$$

$$\text{Profit} = 0$$



In this example:

$$\text{Break-even} = 400 \text{ units} \quad \text{or} \quad \$200.000 \text{ in sales}$$

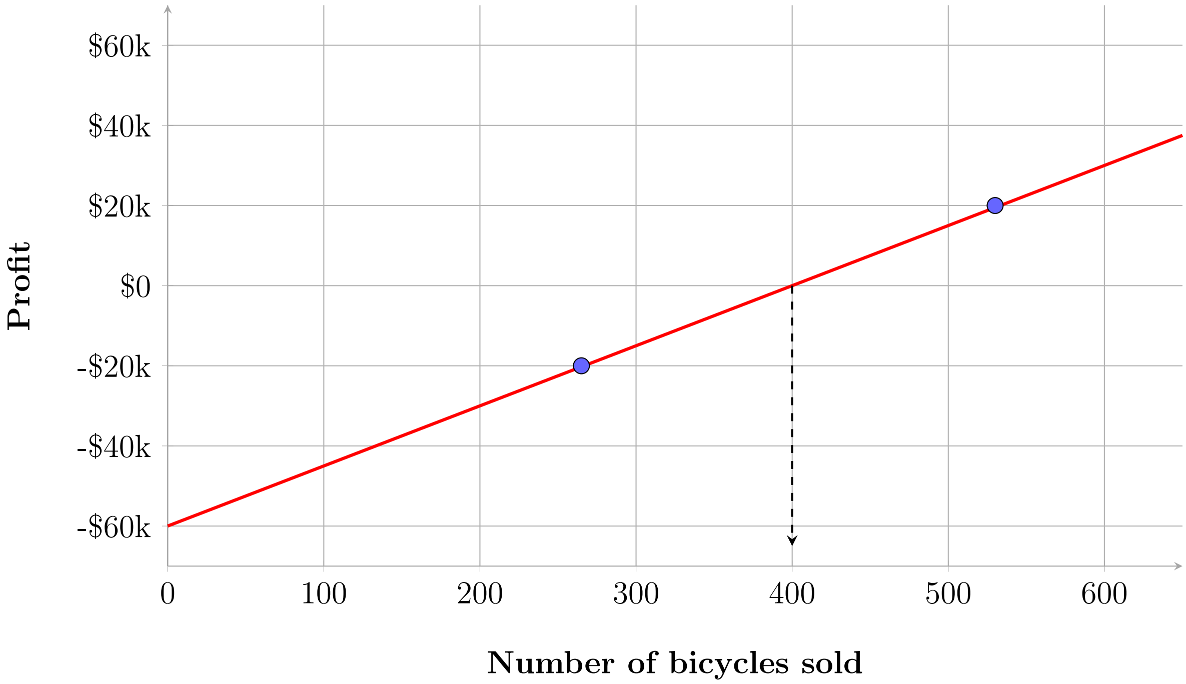
We can also see that:

- To the **left of the break-even point**: the company operates at a **loss**.
- To the **right of the break-even point**: the company generates a **profit**.

3.2.2.2 Profit Graph

A simpler representation of CVP relationships is the **profit graph**, which shows the relationship between profit and the number of units sold.
In a profit graph:

- The horizontal axis represents units sold.
- The vertical axis represents profit.



The profit equation is:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed costs}$$

where:

$$\text{Unit CM} = P - V$$

For example:

$$\text{Unit CM} = 500 - 350 = 150$$

$$\text{Profit} = 150Q - 60{,}000$$

When:

$$Q = 400$$

$$\text{Profit} = 0$$

which corresponds to the **break-even point**.

3.2.3 LO3: Using the CM/VE Ratio

Definition 3.6 (**Contribution Margin Ratio**). *The contribution margin expressed as a percentage of sales.*

$$\text{CM Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

Alternatively, the CM ratio can also be computed using per-unit values:

$$\text{CM Ratio} = \frac{\text{Contribution Margin per Unit}}{\text{Selling Price per Unit}}$$

Racing Bicycle Company
Contribution Income Statement,
For the Month of June

	Total		Per Unit	
Sales (400 bicycles)	$	200,000	$	500
Less: Variable expenses		120,000		300
Contribution margin		80,000	$	200
Less: Fixed expenses		80,000		
Net operating income	$	-		

$$\text{CM Ratio} = \frac{\$80.000}{\$200.000} = 40\% \quad \text{or} \quad \text{CM Ratio} = \frac{\$200}{\$500} = 40\%$$

The CM ratio indicates how much contribution margin is generated for each dollar of sales:

$$\text{For each \$1 of sales} \rightarrow +\$0.40 \text{ contribution margin}$$

Definition 3.7 (Variable Expense Ratio). *The variable expenses expressed as a percentage of sales.*

$$\text{Variable Expense Ratio} = \frac{\text{Variable Expenses}}{\text{Sales}}$$

Alternatively, the VE ratio can also be computed using per-unit values:

$$\text{VE Ratio} = \frac{\text{Variable Expenses per Unit}}{\text{Selling Price per Unit}}$$

Racing Bicycle Company
Contribution Income Statement,
For the Month of June

	Total	Per Unit
Sales (400 bicycles)	$ 200,000	$ 500
Less: Variable expenses	120,000	300
Contribution margin	80,000	$ 200
Less: Fixed expenses	80,000	
Net operating income	$ -	

$$\text{VE Ratio} = \frac{\$120.000}{\$200.000} = 60\% \quad \text{or} \quad \text{VE Ratio} = \frac{\$300}{\$500} = 60\%$$

The CM Ratio and VE Ratio are mathematically related to one another:

$$\text{CM Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}} = \frac{\text{Sales} - \text{Variable Expenses}}{\text{Sales}} = 1 - \text{VE Ratio}$$

$$\text{CM Ratio} = 1 - \text{VE Ratio} = 1 - 0.60 = 0.40 = 40\%$$

The change in contribution margin due to a change in sales can be computed as:

$$\Delta\text{CM} = \text{CM Ratio} \times \Delta\text{Sales}$$

For example, if sales increase by \$50.000:

Racing Bicycle Company
Contribution Income Statemens
For the Month of June

	Total		For Unit	
Sales (500 bicycles)	$	250,000	$	500
Less: Variable expenses		150,000		300
Contribution margin		100,000	$	200
Less: Fixed expenses		80,000		
Net operating income	$	20,000		

$$\Delta\text{CM} = 0.40 \times \$50.000 = \$20.000$$

Since fixed costs remain unchanged:

$$\Delta\text{Profit} = \Delta\text{CM} = \$20.000$$

The profit equation:

$$\textbf{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

can also be expressed in terms of CM Ratio (%) and sales ($):

$$\textbf{Profit} = \text{CM Ratio} \times \text{Sales} - \text{Fixed Expenses}$$

$$\textbf{Profit} = 0.40 \times \$250.000 - \$80.000 = \$20.000$$

3.2.4 LO4: Effects on Net Operating Income

The net operating income depends on four main factors:

- Selling price
- Variable costs
- Fixed costs
- Sales volume

Any change in one or more of these variables will affect profit. To analyze changes in profit, it is sufficient to focus only on the variations (**incremental analysis**), instead of recomputing the entire income statement.

Case 1: Change in Fixed Costs and Volume

- Increasing fixed costs increases the break-even point
- Increasing volume increases contribution margin

$$\Delta\text{Profit} = \Delta CM - \Delta F$$

Or similarly, but only if variable expenses are **constant**:

$$\Delta\text{Profit} = (\Delta\text{Sales} \times CM\ Ratio) - \Delta\text{F} = (\Delta Q \times Unit\ CM) - \Delta F$$

For example:

	500 units	540 units
Sales	$ 250,000	$ 270,000
Less: Variable expenses	150,000	162,000
Contribution margin	100,000	108,000
Less: Fixed expenses	80,000	90,000
Net operating income	$ 20,000	$ 18,000

- Sales increase from 500 to 540 units
- Fixed costs increase by $10.000 (advertising)

$$\Delta\text{CM} = 0.40 \times \$20.000 = 40 \times \$200 = \$8.000$$

$$\Delta\text{Profit} = \$8.000 - \$10.000 = -\$2.000$$

Observation 3.2. *An increase in fixed costs can more than offset the additional contribution margin generated by higher sales, leading to a decrease in profit.*

Case 2: Change in Variable Cost and Volume

- Increasing variable cost reduces the contribution margin
- Higher volume may compensate this reduction

For example:

	500 units	580 units
Sales	$ 250,000	$ 290,000
Less: Variable expenses	150,000	179,800
Contribution margin	100,000	110,200
Less: Fixed expenses	80,000	80,000
Net operating income	$ 20,000	$ 30,200

- Sales increase from 500 to 580 units
- Variable cost per unit increases from \$300 to \$310
- Contribution margin per unit decreases from \$200 to \$190

Since fixed costs do not change:

$$\Delta\text{Profit} = \Delta\text{CM} = \$10.200$$

Observation 3.3. *An increase in variable cost reduces the contribution margin per unit, but a sufficiently large increase in sales volume can still lead to higher profit.*

Case 3: Change in Selling Price, Fixed Costs, and Volume

- Decreasing price reduces CM
- Increasing fixed costs reduces profit
- Increasing volume increases total CM

For example:

	500 units	650 units
Sales	\$ 250,000	\$ 312,000
Less: Variable expenses	150,000	195,000
Contribution margin	100,000	117,000
Less: Fixed expenses	80,000	95,000
Net operating income	\$ 20,000	\$ 22,000

- Sales increase from 500 to 650 units
- Selling price decreases from \$500 to \$480
- Contribution margin per unit decreases from \$200 to \$180
- Fixed costs increase by \$15.000

$$\Delta\text{Profit} = \Delta\text{CM} - \Delta\text{F} = \$17.000 - \$15.000 = \$2.000$$

Observation 3.4. *When multiple factors change simultaneously, the effect on profit depends on the balance between an increase in sales volume, a decrease in contribution margin and an increase in fixed costs.*

Case 4: Shift from Fixed to Variable Costs

- Fixed costs decrease
- Variable costs increase
- Risk structure changes:
  - Lower fixed costs → lower risk
  - Higher variable costs → lower contribution margin per unit

For example:

	500 units	575 units
Sales	$ 250,000	$ 287,500
Less: Variable expenses	150,000	181,125
Contribution margin	100,000	106,375
Less: Fixed expenses	80,000	74,000
Net operating income	$ 20,000	$ 32,375

- Sales increase from 500 to 575 units
- Fixed costs decrease (removal of salaries) by \$6.000
- Variable cost per unit increases (sales commission) from \$300 to \$315
- Contribution amrgin per unit decreases from \$200 to \$185

$$\Delta\text{Profit} = \Delta\text{CM} - \Delta\text{F} = \$6.375 - (-\$6.000) = \$12.375$$

Observation 3.5. *Shifting from fixed costs to variable costs reduces fixed risk but also reduces the contribution margin per unit.*

Case 5: Change in Selling Price (Special Order)

- Target increase in profit
- Additional units to sell

For example:

- 150 units to sell (special order)
- Target profit: \$3.000
- Variable cost per unit: \$300

- Fixed costs remain constant

In order to reach the target increase in profit, the firm must increase the price of the product:

$$\text{Required CM per unit} = \frac{\text{Target Profit}}{\Delta Q} = \frac{\$3.000}{150} = \$20$$

Required selling price = Variable Cost per Unit + Required CM per Unit = \$300+\$20 = \$320

In fact:

$$\text{Sales} = P \times Q = \$320 \times 150 = \$48.000$$

$$\text{Variable Expenses} = \text{Variable Cost per Unit } \times Q = \$300 \times 150 = \$45.000$$

And since fixed costs do not change:

$$\Delta\text{Profit} = \Delta\text{CM} = \text{Required CM per unit} \times Q = \$20 \times 150 = \$3.000$$

Observation 3.6. *When fixed costs do not change, the required selling price must cover the variable cost per unit and the desired contribution margin per unit.*

3.2.5 LO5: Determining the Break-Even Point

Break-even analysis can be used to determine:

- the number of units that must be sold to earn zero profit,
- the dollar sales required to earn zero profit.

Break-even analysis shows the minimum sales level required to avoid a loss.

Break-Even in Unit Sales

For a single-product company, the break-even point can be found using the basic profit equation:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed expenses}$$

At break-even:

$$\text{Profit} = 0$$

Therefore:

$$0 = \text{Unit CM} \times Q - \text{Fixed expenses}$$

For example:

Racing Bicycle Company Contribution Income Statement For the Month of June			
	Total	Per Unit	CM Ratio
Sales (500 bicycles)	$ 250,000	$ 500	100%
Less: Variable expenses	150,000	300	60%
Contribution margin	100,000	$ 200	40%
Less: Fixed expenses	80,000		
Net operating income	$ 20,000		

$$0 = \$200 \times Q - \$80.000$$

$$200Q = \$80.000$$

$$Q = \frac{\$80.000}{\$200} = 400$$

Thus, the break-even point in unit sales is 400.
Alternatively, we can use the formula:

$$\text{Break-Even Unit Sales} = \frac{\text{Fixed Expenses}}{\text{Unit CM}} = \frac{\$80.000}{\$200} = 400$$

Break-Even in Dollar Sales

Break-even can also be expressed in terms of sales revenue.
Using the equation method:

$$\text{Profit} = \text{CM Ratio} \times \text{Sales} - \text{Fixed Expenses}$$

At break-even:

$$0 = \text{CM Ratio} \times \text{Sales} - \text{Fixed Expenses}$$

For example:

Racing Bicycle Company Contribution Income Statement For the Month of June			
	Total	Per Unit	CM Ratio
Sales (500 bicycles)	$ 250,000	$ 500	100%
Less: Variable expenses	150,000	300	60%
Contribution margin	100,000	$ 200	40%
Less: Fixed expenses	80,000		
Net operating income	$ 20,000		

$$0 = 0.40 \times \text{Sales} - \$80.000$$

$$0.40 \times \text{Sales} = \$80.000$$

$$\text{Sales} = \frac{\$80.000}{0.40} = \$200.000$$

Thus, the break-even point in dollar sales is \$200.000.
Alternatively, we can use the formula:

$$\text{Break-Even Dollar Sales} = \frac{\text{Fixed Expenses}}{\text{CM Ratio}} = \frac{\$80.000}{0.40} = \$200.000$$

In general, in our example:

- If sales are below 400 units, the company incurs a loss.
- If sales are exactly 400 units, profit is zero.
- If sales exceed 400 units, the company earns a profit.

3.2.6 LO6: Target Profit Analysis

Definition 3.8 (**Target Profit Analysis**). *An analysis that determines the level of sales needed to achieve a desired profit.*

Target Profit in Unit Sales

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

Solve for Q:

$$Q = \frac{\text{Target profit} + \text{Fixed Expenses}}{\text{Unit CM}}$$

For example, suppose a certain firm wants to know how many units must be sold to earn a target profit of \$100.000:

$$Q = \frac{\$100.000 + \$80.000}{\$200} = 900 \text{ Units}$$

Alternatively, we can use the formula:

$$\text{Unit Sales for Target Profit} = \frac{\text{Target Profit} + \text{Fixed Expenses}}{\text{Unit CM}} = 900$$

Target Profit in Dollar Sales

$$\text{Profit} = \text{CM ratio} \times \text{Sales} - \text{Fixed Expenses}$$

Solve for Sales:

$$\text{Sales} = \frac{\text{Target profit} + \text{Fixed expenses}}{\text{CM ratio}}$$

For example, suppose a certain firm wants to know the sales volume that must be generated to earn a target profit of \$100.000:

$$\text{Sales} = \frac{\$100.000 + \$80.000}{0.40} = \$450.000$$

Alternatively, we can use the formula:

$$\text{Dollar sales for Target Profit} = \frac{\text{Target Profit} + \text{Fixed Expenses}}{\text{CM ratio}} = \$450.000$$

3.2.7 LO7: Margin of Safety and Cost Structure

Definition 3.9 (**Margin of Safety**). *The excess of actual (or budgeted) sales over break-even sales. It represents how much sales can drop before the company incurs a loss.*

Observation 3.7. *A higher margin of safety indicates lower risk, since sales can decline more before reaching the break-even point.*

Definition 3.10 (**Cost Structure**). *The relative proportion of fixed and variable costs in a company.*

Margin of Safety in Dollars

$$\text{Margin of Safety} = \text{Total Sales} - \text{Break-Even Sales}$$

For example:

	Break-even sales 400 units	Actual sales 500 units
Sales	$ 200,000	$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net operating income	$ -	$ 20,000

$$\text{Margin of safety} = \$250.000 - \$200.000 = \$50.000$$

Margin of Safety Percentage

$$\text{Margin of safety \%} = \frac{\text{Margin of Safety}}{\text{Total Sales}}$$

In reference to the previous example:

$$\text{Margin of safety \%} = \frac{\$50.000}{\$250.000} = 20\%$$

Margin of Safety in Units

$$\text{Margin of Safety (units)} = \frac{\text{Margin of Safety (dollars)}}{\text{Selling Price per Unit}}$$

In reference to the previous example:

$$\text{Margin of Safety (units)} = \frac{\$50.000}{\$500} = 100 \text{ units}$$

High vs Low Fixed Cost Structures

- High fixed costs (low variable costs):
  - Higher profits in good years

  - Lower profits (or losses) in bad years

- Low fixed costs (high variable costs):
  - Lower profits in good years
  - More stable income across different levels of sales

Observation 3.8. *Companies with lower fixed costs have less risk and more stable income, while companies with higher fixed costs have higher risk but greater profit potential.*

3.2.8 LO8: Operating Leverage

Definition 3.11 (Operating Leverage). *Measure of how sensitive net operating income is to percentage changes in sales.*

$$\text{Degree of Operating Leverage (DOL)} = \frac{\text{Contribution Margin}}{\text{Net Operating Income}}$$

For example:

	Actual sales 500 Bikes
Sales	$ 250,000
Less: variable expenses	150,000
Contribution margin	100,000
Less: fixed expenses	80,000
Net income	$ 20,000

$$\text{DOL} = \frac{\$100.000}{\$20.000} = 5$$

Observation 3.9. *A higher degree of operating leverage means that a small percentage change in sales results in a larger percentage change in profit.*

$$\%\Delta\text{Profit} = \text{DOL} \times \%\Delta\text{Sales}$$

In reference to the previous example:

$$\%\Delta\text{Profit} = 5 \times 10\% = 50\%$$

This means that if the firm increases its sales by 10%, net operating income will increase by 50%. We can verify this:

	Actual sales (500)	Increased sales (550)
Sales	$ 250,000	$ 275,000
Less variable expenses	150,000	165,000
Contribution margin	100,000	110,000
Less fixed expenses	80,000	80,000
Net operating income	$ 20,000	$ 30,000

Using the percentage change formula:

$$\%\Delta\text{Profit} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} = \frac{\$30.000 - \$20.000}{\$20.000} = 50\%$$

Observation 3.10. *Operating leverage is driven by cost structure:*

- *High fixed costs* $\Rightarrow$ *high operating leverage* $\Rightarrow$ *higher risk and higher potential returns*
- *Low fixed costs* $\Rightarrow$ *low operating leverage* $\Rightarrow$ *lower risk and more stable profits*

Sales Commissions and Incentives

Sales commissions can be based on:

- Selling price (sales revenue)
- Contribution margin

For example, consider a firm that produces two types of surfboards:

- XR7:
$$\text{Price} = \$100, \quad \text{CM per unit} = \$25$$
- Turbo:
$$\text{Price} = \$150, \quad \text{CM per unit} = \$18$$

Observation 3.11 (Bad Approach). *If commissions are based on sales revenue, salespeople may prefer products with higher prices rather than higher contribution margins.*

Observation 3.12 (Correct Approach). *Aligning incentives with contribution margin ensures that sales decisions maximize company profit.*

3.2.9 LO9: Multiproduct Break-Even and Sales Mix

Definition 3.12 (Sales Mix). *The relative proportion in which a company's different products are sold.*

In a multiproduct company, break-even analysis is more complex because different products have different:

- selling prices
- variable cost structures
- contribution margins

For break-even analysis, it is usually assumed that the sales mix remains constant. For example, suppose a firm sells two products with the following sales mix:

- bicycles (45%)
- carts (55%)

	Bicycle		Carts		Total	
Sales	$ 250,000	100%	$ 300,000	100%	$ 550,000	100.0%
Variable expenses	150,000	60%	135,000	45%	285,000	51.8%
Contribution margin	100,000	40.0%	165,000	55%	265,000	48.2%
Fixed expenses					170,000	
Net operating income					$ 95,000	
Sales mix	$ 250,000	45%	$ 300,000	55%	$ 550,000	100%

The total contribution margin ratio is computed using total values:

$$\text{CM ratio} = \frac{\$265.000}{\$550.000} \approx 48.2\%$$

The break-even formula in dollar sales is still:

$$\text{Break-Even Dollar Sales} = \frac{\text{Fixed expenses}}{\text{CM ratio}}$$

Using the multiproduct data:

$$\text{Break-Even Dollar Sales} = \frac{\$170.000}{48.2\%} = \$352.697$$

This means that if the sales mix remains constant at 45% bicycles and 55% carts, the firm must generate approximately \$352.697 in total sales revenue to break even. These break-even sales are then allocated according to the sales mix:

$$\text{Bicycles} = 45\% \times \$352.697 \approx \$158.714$$

$$\text{Carts} = 55\% \times \$352.697 \approx \$193.983$$

Observation 3.13. *A shift in the sales mix affects the overall CM ratio:*

- *more sales of high-CM products ⇒ higher overall CM ratio ⇒ lower break-even point*
- *more sales of low-CM products ⇒ lower overall CM ratio ⇒ higher break-even point*

3.3 Quick Checks

1. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. An average of 2.100 cups are sold each month. What is the CM ratio for Coffee Klatch?

A. 1.319

B. 0.758

C. 0.242

D. 4.139

Correct answer(s): B.

Explanation:

$$\text{CM Ratio} = \frac{\$1.49 - \$0.36}{\$1.49} = \frac{\$1.13}{\$1.49} = 0.758$$

2. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. An average of 2.100 cups are sold each month. What is the break-even sales in dollars?

A. \$1.300

B. \$1.715

C. \$1.788

D. \$3.129

Correct answer(s): B.

Explanation:

$$\text{Break-Even Dollar Sales} = \frac{\$1.300}{0.758} = \$1.715$$

3. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. An average of 2.100 cups are sold each month. What is the break-even point in units?

A. 872 cups

B. 3.611 cups

C. 1.200 cups

D. 1.150 cups

Correct answer(s): D.

Explanation:

$$\text{Break-Even Unit Sales} = \frac{\$1.300}{\$1.13} = 1.150 \text{ cups}$$

4. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. Use the formula method to determine how many cups must be sold to attain a target profit of \$2.500.

A. 3.363 cups

B. 2.212 cups

C. 1.150 cups

D. 4.200 cups

Correct answer(s): A.

Explanation:

$$Q = \frac{\$2.500 + \$1.300}{\$1.13} = \frac{\$3.800}{\$1.13} = 3.363$$

5. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. Use the formula method to determine the sales dollars required to attain a target profit of \$2.500.

A. \$2.550

B. \$5.013

C. \$8.458

D. \$10.555

Correct answer(s): B.

Explanation:

$$\text{Dollar Sales} = \frac{\$3.800}{0.758} = \$5.013$$

6. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. An average of 2.100 cups are sold each month. What is the margin of safety expressed in cups?

A. 3.250 cups

B. 950 cups

C. 1.150 cups

D. 2.100 cups

Correct answer(s): B.

Explanation:

$$\text{Margin of safety in units} = 2.100 - 1.150 = 950 \text{ cups}$$

7. Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is \$1.49 and the average variable expense per cup is \$0.36. The average fixed expense per month is \$1.300. An average of 2.100 cups are sold each month. What is the degree of operating leverage?

A. 2.21

B. 0.45

C. 0.34

D. 2.92

Correct answer(s): A.

Explanation:

$$\text{Degree of operating leverage} = \frac{\$2.373}{\$1.073} = 2.21$$

8. At Coffee Klatch the average selling price of a cup of coffee is \$1.49, the average variable expense per cup is \$0.36, the average fixed expense per month is \$1.300, and an average of 2.100 cups are sold each month. If sales increase by 20%, by how much should net operating income increase?

A. 30.0%

B. 20.0%

C. 22.1%

D. 44.2%

Correct answer(s): D.

Explanation:

$$\%\Delta\text{Net operating income} = 2.21 \times 20\% = 44.2\%$$

4 Noreen Ch03

4.1 Random concepts captured (04/03/2026)

Difference Between Risk and Return

Definition 4.1 (**Risk**). *The uncertainty about future outcomes. It reflects the possibility that actual results will differ from expected results.*

Definition 4.2 (**Return**). *The gain or loss generated from an investment over a period of time, usually expressed as a percentage of the initial investment.*

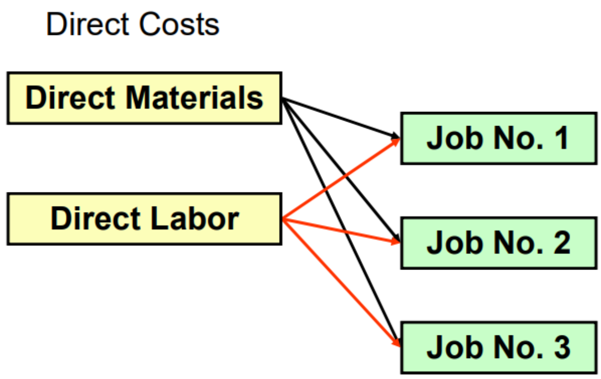
- **Risk** measures uncertainty and variability of outcomes.
- **Return** measures the actual or expected reward from an investment.

Observation 4.1. *There is a fundamental trade-off between risk and return: higher expected returns are typically associated with higher levels of risk.*

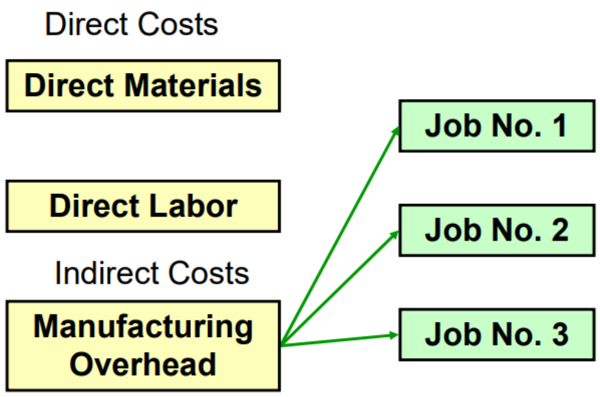
4.2 Job-Order Costing

Definition 4.3 (**Job-Order Costing System**). *A system used to assign costs to specific jobs or batches of products.*

- It is used when **many different products** are produced.
- Production is typically **customized** (made to order).
- Each job is unique, so costs must be **traced** (direct costs) or **allocated** (indirect costs).
- A separate **cost record** is maintained for each job.



For what concerns direct costs, **direct materials** and **direct labor** are **assigned directly** to specific jobs as work is performed.



For what concerns indirect costs, **manufacturing overhead**, including **indirect materials** and **indirect labor**, are allocated to all jobs rather than directly traced to each job.

Definition 4.4 (**Job Cost Sheet**). *A document used to record and accumulate all costs associated with a specific job.*

PearCo Job Cost Sheet

Job Number A - 143 Date Initiated 3-4-17

Date Completed

Department B3 Units Completed

Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount

Cost Summary	Units Shipped		
Direct Materials	Date	Number	Balance
Direct Labor			
Manufacturing Overhead			
Total Cost			
Unit Product Cost			

Definition 4.5 (**Materials Requisition Form**). *A document used to track the materials taken from inventory and assigned to a job.*

PearCo Materials Requisition Form

Requisition No. X7 - 6890 Date 3-4-17

Job No. A - 143

Department B3

Description	Quantity	Unit Cost	Total Cost
2 x 4, 12 feet	12	$ 3.00	$ 36.00
1 x 6, 12 feet	20	4.00	80.00
			$ 116.00

Authorized Signature *Will E. Delite*

Materials are:

- Withdrawn from inventory
- Assigned to a specific job (e.g., Job A-143)
- Recorded in the job cost sheet

PearCo Job Cost Sheet

Job Number A - 143
Date Initiated 3-4-17
Date Completed
Department B3
Units Completed
Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116						

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor				
Manufacturing Overhead				
Total Cost				
Unit Product Cost				

Definition 4.6 (Employee Time Ticket). *A document used to record the amount of time an employee spends on each job.*

PearCo Employee Time Ticket

Time Ticket No. 36
Date 3-5-17
Employee I. M. Skilled
Station 42

Starting Time	Ending Time	Hours Completed	Hourly Rate	Amount	Job No.
0800	1600	8.00	$ 15.00	$ 120.00	A-143
Totals		8.00	$ 15.00	$ 120.00	A-143

Supervisor C. M. Workman

Labor cost is:

- Traced to the job
- Added to the job cost sheet

4.2.1 LO1: Predetermined Overhead Rate

Definition 4.7 (**Allocation Base**)**.** *A measure such as direct labor hours, direct labor cost, or machine hours used to assign manufacturing overhead to jobs.*

An allocation base is used because:

- manufacturing overhead is difficult or impossible to trace directly to specific jobs;
- manufacturing overhead includes many different indirect cost items;
- many overhead costs are fixed, even though production volume changes during the period.

Definition 4.8 (**Predetermined Overhead Rate (POHR)**)**.** *A rate used to apply manufacturing overhead to jobs before the period ends.*

The POHR is computed as:

$$\text{POHR} = \frac{\text{Estimated total manufacturing overhead cost for the coming period}}{\text{Estimated total units in the allocation base for the coming period}}$$

Definition 4.9 (**Cost Driver**)**.** *A factor that causes overhead costs.*

Ideally, the allocation base should be a **cost driver**.

Estimated data are used because:

- actual manufacturing overhead for the period is not known until the end of the period;
- actual overhead costs may fluctuate seasonally and may therefore distort job costs during the period.

The predetermined overhead rate is computed before the period begins through the following steps:

1. Estimate the total amount of the allocation base that will be required for the coming period.
2. Estimate the total fixed manufacturing overhead cost and the variable manufacturing overhead cost per unit of the allocation base.
3. Estimate total manufacturing overhead using:

$$Y = a + bX$$

where:

- Y = estimated total manufacturing overhead cost
- a = estimated total fixed manufacturing overhead cost
- b = estimated variable manufacturing overhead cost per unit of the allocation base
- X = estimated total amount of the allocation base

4. Compute the predetermined overhead rate:

$$\text{POHR} = \frac{Y}{X}$$

4.2.2 LO2: Applying Overhead to Jobs

Once the **predetermined overhead rate (POHR)** has been computed, it is used to assign overhead to jobs:

$$\text{Applied Overhead} = \text{POHR} \times \text{Actual amount of allocation base used}$$

EXAMPLE

PearCo estimates that it will require 160.000 direct labor-hours to meet the coming period's estimated production level. In addition, the company estimates total fixed manufacturing overhead at \$200.000, and variable manufacturing overhead costs of \$2.75 per direct labor hour.

Using:

$$Y = a + bX = \$200.000 + (\$2.75 \times 160.000) = \$640.000$$

We calculate the POHR as follows:

$$\text{POHR} = \frac{\$640.000}{160.000} = \$4 \text{ per direct labor-hour}$$

If, for example, a job uses 8 direct labor hours:

$$\text{Applied Overhead} = 8 \times \$4 = \$32$$

This amount is added to the job cost sheet under manufacturing overhead:

PearCo Job Cost Sheet

Job Number A - 143

Date Initiated 3-4-17

Date Completed 3-5-17

Department B3

Units Completed 2

Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead	$ 32			
Total Cost				
Unit Product Cost				

4.2.3 LO3: Total Cost and Unit Product Cost

$$\boxed{\text{Total Cost} = \text{Direct Materials} + \text{Direct Labor} + \text{Manufacturing Overhead}}$$

PearCo Job Cost Sheet

Job Number A - 143 | Date Initiated 3-4-17
Date Completed 3-5-17
Department B3 | Units Completed 2
Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead	$ 32			
Total Cost	$ 268			
Unit Product Cost				

$$\boxed{\text{Unit Cost} = \frac{\text{Total Job Cost}}{\text{Units Produced}}}$$

PearCo Job Cost Sheet

Job Number A - 143 | Date Initiated 3-4-17
Date Completed 3-5-17
Department B3 | Units Completed 2
Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead	$ 32			
Total Cost	$ 268			
Unit Product Cost	$ 134			

Direct materials and direct labor are usually **accurately traced**, while manufacturing overhead is **more difficult to allocate correctly**. Inaccurate assignment of manufacturing costs can negatively affect managerial decisions.

The allocation base should reflect how jobs **actually consume resources**. Many firms use a single **plantwide overhead rate**, but this could be overly simplistic and inaccurate.

For this reason, using **multiple predetermined overhead rates** (multiple cost drivers) leads to more accurate job costing.

4.2.4 LO4: Multiple POHRs

In more complex settings, companies compute **separate predetermined overhead rates for each department**.

EXAMPLE

Dickson Company has two production departments, Milling and Assembly. The company uses a job-order costing system and computes a predetermined overhead rate in each production department. The predetermined overhead rate in the Milling Department is based on machine-hours and in the Assembly Department it is based on direct labor-hours. The company uses cost-plus pricing (and a markup percentage of 75% of total manufacturing cost) to establish selling prices for all of its jobs. At the beginning of the year, the company made the following estimates:

	Department	
	Milling	Assembly
Machine-hours	60,000	3,000
Direct labor-hours	8,000	80,000
Total fixed manufacturing overhead cost	$390,000	$500,000
Variable manufacturing overhead per machine-hour	$2.00	—
Variable manufacturing overhead per direct labor-hour	—	$3.75

From now on:

- DLH(s) = Direct Labor-Hour(s)
- MH(s) = Machine-Hour(s)

Step 1: Estimate Total Overhead for Each Department

$$Y = a + bX$$

Milling Department:

$$Y = \$390.000 + (\$2.00 \text{ per MH} \times 60.000 \text{ MHs}) = \$510.000$$

Assembly Department:

$$Y = \$500.000 + (\$3.75 \text{ per DLH} \times 80.000 \text{ DLHs}) = \$800.000$$

Step 2: Compute POHR for Each Department

$$\text{POHR} = \frac{\text{Estimated overhead}}{\text{Estimated allocation base}}$$

Milling Department:

$$\text{POHR} = \frac{\$510.000}{60.000} = \$8.50 \text{ per MH}$$

Assembly Department:

$$\text{POHR} = \frac{\$800.000}{80.000} = \$10.00 \text{ per DLH}$$

Step 3: Apply Overhead to a Job

Consider the following job:

	Department	
Job 407	Milling	Assembly
Machine-hours	90	4
Direct labor-hours	5	20
Direct materials	$800	$370
Direct labor cost	$70	$280

Milling Department:

$$\text{Applied Overhead} = 90 \text{ MHs} \times \$8.50 = \$765$$

Assembly Department:

$$\text{Applied Overhead} = 20 \text{ DLHs} \times \$10 = \$200$$

Step 4: Compute Total Job Cost

	Milling	*Assembly*	*Total*
Direct materials	$800	$370	$1,170
Direct labor	$ 70	$280	350
Manufacturing overhead applied	$765	$200	965
Total cost of Job 407			**$2,485**

$$\text{Total Cost} = \$1.170 + \$350 + \$965 = \$2.485$$

Step 5: Compute Selling Price (Cost-Plus)

Assuming a 75% markup:

Total cost of Job 407	$2,485.00
Markup ($2,485 × 75%)	1,863.75
Selling price of Job 407	$4,348.75

$$\text{Markup} = \$2.485 \times 75\% = \$1.863.75$$

$$\text{Selling Price} = \$2.485 + \$1.863.75 = \$4.348.75$$

An Alternative: Activity-Based Approach

Definition 4.10 (**Activity-Based Costing (ABC)**). *Approach that assigns overhead based on the activities that generate costs.*

The ABC is an alternative approach to developing multiple predetermined overhead rates:

- It uses multiple cost drivers
- It provides more accurate cost allocation

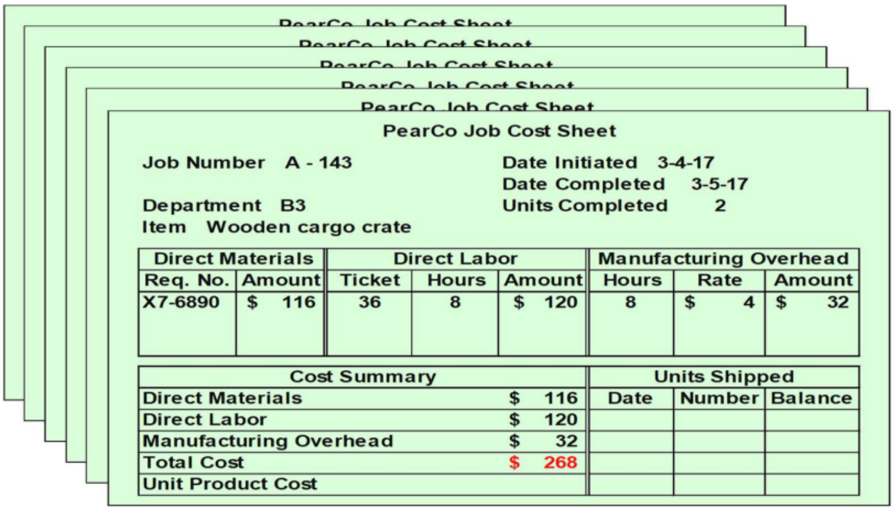
Underapplied and Overapplied Overhead

The estimated amount of overhead almost always differs from the actual amount incurred in a specific period. There are two possible cases:

- **Underapplied overhead:**
  - Applied < Actual
  - **Increases** cost of goods sold
  - **Decreases** net operating income
- **Overapplied overhead:**
  - Applied > Actual
  - **Decreases** cost of goods sold
  - **Increases** net operating income

4.2.5 Subsidiary Ledger

Definition 4.11 (**Subsidiary Ledger**). *The collection of all of a company's job cost sheets.*



PearCo Job Cost Sheet

Job Number A - 143
Date Initiated 3-4-17
Date Completed 3-5-17
Department B3
Units Completed 2
Item Wooden cargo crate

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
X7-6890	$ 116	36	8	$ 120	8	$ 4	$ 32

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	$ 116			
Direct Labor	$ 120			
Manufacturing Overhead	$ 32			
Total Cost	$ 268			
Unit Product Cost				

Job cost sheets provide the detailed support for the amounts reported in the **balance sheet**. Specifically:

- **Work in Process (WIP)** includes jobs that are not yet completed
- **Finished Goods** includes completed jobs that have not yet been sold

The balance sheet reports aggregated amounts, while job cost sheets show the **individual jobs** that make up those totals.

Job cost sheets also provide the detailed support for the **Cost of Goods Sold (COGS)** reported in the **income statement**. When a job is sold, its cost moves from Finished Goods to COGS. The income statement reports total COGS, while job cost sheets show **which specific jobs** make up that cost.

4.3 Quick Checks

1. Job WR53 at NW Fab, Inc. required \$200 of direct materials and 10 direct labor hours at \$15 per hour. Estimated total overhead for the year was \$760.000 and estimated direct labor hours were 20.000. What would be recorded as the cost of job WR53?

A. \$200

B. \$350

C. \$380

D. \$730

Correct answer: D.

Explanation:
First, compute the predetermined overhead rate (POHR):

$$\text{POHR} = \frac{\$760.000}{20.000} = \$38 \text{ per DLH}$$

Now compute the cost components of the job:

- Direct materials:
$$\$200$$
- Direct labor:
$$10 \times \$15 = \$150$$
- Manufacturing overhead:
$$10 \times \$38 = \$380$$

Total job cost:

$$\$200 + \$150 + \$380 = \$730$$

5 Exercises Ch01-03

5.1 Random concepts captured (10/03/2026)

Gold, Labor, and Oil as Fundamental Economic Inputs

Gold, labor, and oil can be interpreted as three fundamental elements in economic systems, each representing a different dimension of value creation and wealth:

- **Gold** represents **stored value and financial wealth**. Historically, it has been used as a medium of exchange and a store of value, symbolizing capital and financial stability.
- **Labor** represents **human effort and productivity**. It is the primary source of value creation in most economic activities.
- **Oil** represents **natural resources and energy**. It is a key input for industrial production and transportation, and therefore a driver of economic growth.

Price Elasticity of Demand

Definition 5.1 (Price Elasticity of Demand). *Measure of how sensitive the quantity demanded of a good is to a change in its price.*

The price elasticity of demand is defined as:

$$E_d = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}}$$

5.2 Exercise 1

Otsego Inc. is a merchandiser that provided the following information:

Number of units sold	12,000
Selling price per unit	$25
Variable selling expense per unit	$2.50
Variable administrative expense per unit	$2
Total fixed selling expense	$16,000
Total fixed administrative expense	$17,000
Merchandise inventory, beginning balance	$25,000
Merchandise inventory, ending balance	$18,000
Merchandise purchases	$101,000

Required:

1. Define net operating income.
2. Define contribution margin per unit.
3. Calculate the number of units sold if the contribution margin ratio is equal to 50%.

Solution (1)

We start from the fundamental relationships:

$$\text{Net Operating Income} = \text{Contribution Margin} - \text{Fixed Expenses}$$

$$\text{Contribution Margin} = \text{Sales} - \text{Variable Expenses}$$

First, compute **total sales**:

$$\text{Sales} = \text{Selling Price Per Unit} \times \text{Quantity Sold} = \$25 \times 12.000 = \$300.000$$

Next, compute total **variable expenses**. These include:

- Variable selling expenses
- Variable administrative expenses

- Cost of Goods Sold (COGS):
  - Merchandise inventory, beginning balance
  - Merchandise inventory, ending balance
  - Merchandise purchases

$$\begin{aligned} \text{COGS} &= \text{Beginning Inventory} + \text{Purchases} - \text{Ending Inventory} = \\ &= \$25.000 + \$101.000 - \$18.000 = \\ &= \$108.000 \end{aligned}$$

$$\begin{aligned} \text{Variable Expenses} &= \text{Variable Expenses Per Unit} \times \text{Quantity Sold} + \text{Cost of Goods Sold} = \\ &= (\$2{,}50 + \$2) \times 12.000 + \$108.000 = \\ &= \$54.000 + \$108.000 = \$162.000 \end{aligned}$$

$$\text{Contribution Margin} = \$300.000 - \$162.000 = \$138.000$$

Now, compute total **fixed expenses**. These include:

- Total fixed selling expenses
- Total fixed administrative expenses

$$\text{Fixed Expenses} = \$16.000 + \$17.000 = \$33.000$$

Finally, compute **net operating income**:

$$\text{Net Operating Income} = \$138.000 - \$33.000 = \$105.000$$

Solution (2)

$$\text{CM per Unit} = \frac{\text{CM}}{\text{Quantity Sold}} = \frac{\$138.000}{12.000} = \$11{,}5 \text{ per unit}$$

Solution (3)

$$\text{CM Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

$$0.5 = \frac{\text{Sales - Variable Expenses}}{\text{Sales}}$$

$$0.5 = \frac{(\text{Selling Price per Unit} \times \text{Q}) - (\text{Variable Expenses Per Unit} \times \text{Q} + \text{COGS})}{\text{Selling Price per Unit} \times \text{Q}}$$

$$0.5 = \frac{(\$25 \times \text{Q}) - (\$4{,}5 \times \text{Q} + \$108.000)}{\$25 \times \text{Q}}$$

$$\$12{,}5 \times Q = \$20{,}5 \times \text{Q} - \$108.000$$

$$Q = \frac{\$108.000}{\$8} = 13.500$$

5.3 Exercise 2

Allwill Products distributes a single product, a decorative plate whose selling price is \$10 and whose variable cost is \$6 per unit. The company's monthly fixed expense is \$7.500.

Required:

1. Calculate the company's break-even point in unit sales.
2. Calculate the company's break-even point in dollar sales.
3. If the company's fixed expenses increase by \$500, what would become the new break-even point in unit sales? In dollar sales?

Solution (1)

Let's consider the profit equation:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

Compute unit CM:

$$\text{Unit CM} = \text{Selling Price per Unit} - \text{Variable Expenses per Unit} = \$10 - \$6 = \$4$$

At break-even, Profit = 0:

$$0 = \$4 \times Q - \$7.500$$

$$Q = \frac{\$7.500}{\$4} = 1.875$$

Solution (2)

$$\text{Break-Even Dollar Sales} = \text{Break-Even Unit Sales} \times \text{Selling Price per Unit} =$$

$$= 1.875 \times \$10 = \$18.750$$

Solution (3)

$$\text{Break-Even Unit Sales} = \frac{\$8.000}{\$4} = \text{2.000}$$

$$\text{Break-Even Dollar Sales} = 2.000 \times \$10 = \text{\$20.000}$$

5.4 Exercise 3

Stepman Corporation has a single product whose selling price is \$200 and whose variable expense is \$150 per unit. The company's monthly fixed expense is \$75.000.

Required:

1. Calculate the unit sales needed to attain a target profit of \$9.000.
2. Calculate the dollar sales needed to attain a target profit of \$10.000.

Solution (1)

Let's consider the profit equation:

$$\text{Profit} = \text{Unit CM} \times Q - \text{Fixed Expenses}$$

Solve for Q:

$$Q = \frac{\text{Target Profit} + \text{Fixed Expenses}}{\text{Unit CM}}$$

First, we calculate the CM per unit:

$$\text{Unit CM} = \text{Selling Price per Unit} - \text{Variable Expenses per Unit} = \$200 - \$150 = \$50$$

Since the target profit is \$9.000:

$$Q = \frac{\$9.000 + \$75.000}{\$50} = 1.680 \ \text{units}$$

Solution (2)

Let's consider the profit equation:

$$\text{Profit} = \text{CM Ratio} \times \text{Sales} - \text{Fixed Expenses}$$

Solve for sales:

$$\text{Sales} = \frac{\text{Target Profit} + \text{Fixed Expenses}}{\text{CM Ratio}}$$

First, we calculate the CM ratio:

$$\text{CM Ratio} = \frac{\text{Contribution Margin per Unit}}{\text{Selling Price per Unit}} = \frac{\$50}{\$200} = 25\%$$

Since the target profit is \$10.000:

$$\text{Sales} = \frac{\$10.000 + \$75.000}{0.25} = \$340.000$$

Another way would be to calculate the unit sales to attain a target profit of \$10.000 and then multiply it for the selling price:

$$Q = \frac{\$10.000 + \$75.000}{\$50} = 1.700 \text{ units}$$

$$\text{Sales} = 1.700 \times \$200 = \$340.000$$

5.5 Exercise 5

Miller Company's contribution format income statement for the most recent month is shown below:

	Total	Per Unit
Sales (25,000 units)	$450,000	$18.00
Variable expenses	250,000	10.00
Contribution margin	$200,000	$ 8.00
Fixed expenses	85,000	
Net operating income	$115,000	

Required: (Consider each case independently)

1. What is the revised net operating income if unit sales increase by 20%?
2. What is the revised net operating income if the selling price decreases by $2.00 per unit and the number of units sold increases by 15%?
3. What is the revised net operating income if the selling price increases by $2.00 per unit, fixed expenses increase by $15,000, and the number of units sold decreases by 4%?
4. What is the revised net operating income if the selling price per unit increases by 10%, variable expenses increase by 80 cents per unit, and the number of units sold decreases by 8%?

Solution (1)

An increase in sales of 20% means:

$$\begin{aligned} \text{Quantity Sold} &= \text{Current Quantity Sold} + 20\% \times \text{Current Quantity Sold} = \\ &= 25.000 + 5.000 = 30.000 \end{aligned}$$

$$\text{Sales} = \text{Quantity Sold} \times \text{Selling Price Per Unit} = 30.000 \times \$18 = \$540.000$$

Consequently:

$$\text{Variable Expenses} = \text{Variable Expenses per Unit} \times \text{Quantity Sold} = \$10 \times 30.000 = \$300.000$$

$$\text{Contribution Margin} = \text{Sales} - \text{Variable Expenses} = \$540.000 - \$300.000 = \$240.000$$

$$\text{Net Operating Income} = \text{CM} - \text{Fixed Expenses} = \$240.000 - \$85.000 = \$155.000$$

Solution (2)

A decrease in selling price of $2 per unit means:

$$\text{Sales per Unit} = \text{Current Sales per Unit} - \$2 = \$18 - \$2 = \$16$$

An increase in units sold of 15% means:

$$\begin{aligned} \text{Quantity Sold} &= \text{Current Quantity Sold} + 15\% \times \text{Current Quantity Sold} = \\ &= 25.000 + 3.750 = 28.750 \end{aligned}$$

$$\text{Sales} = \text{Sales per Unit} \times \text{Quantity Sold} = \$16 \times 28.750 = \$460.000$$

Consequently:

$$\text{Variable Expenses} = \text{Variable Expenses per Unit} \times \text{Quantity Sold} = \$10 \times 28.750 = \$287.500$$

$$\text{CM} = \text{Sales} - \text{Variable Expenses} = \$460.000 - \$287.500 = \$172.500$$

$$\text{NOI} = \text{CM} - \text{Fixed Expenses} = \$172.500 - \$85.000 = \$87.500$$

Solution (3)

An increase in selling price of \$2 per unit means:

$$\text{Sales per Unit} = \text{Current Sales per Unit} + \$2 = \$18 + \$2 = \$20$$

An increase in fixed expenses of \$15.000 means:

$$\text{Fixed Expenses} = \text{Current Fixed Expenses} + \$15.000 = \$85.000 + \$15.000 = \$100.000$$

A decrease in units sold of 4% means:

$$\begin{aligned}\text{Quantity Sold} &= \text{Current Quantity Sold} - 4\% \times \text{Current Quantity Sold} = \\ &= 25.000 - 1.000 = 24.000\end{aligned}$$

$$\text{Sales} = \text{Sales per Unit} \times \text{Quantity Sold} = \$20 \times 24.000 = \$480.000$$

Consequently:

$$\text{Variable Expenses} = \text{Variable Expenses per Unit} \times \text{Quantity Sold} = \$10 \times 24.000 = \$240.000$$

$$\text{CM} = \text{Sales} - \text{Variable Expenses} = \$480.000 - \$240.000 = \$240.000$$

$$\text{NOI} = \text{CM} - \text{Fixed Expenses} = \$240.000 - \$100.000 = \$140.000$$

Solution (4)

An increase in selling price of 10% per unit means:

$$\begin{aligned}\text{Sales per Unit} &= \text{Current Sales per Unit} + 10\% \times \text{Current Sales per Unit} = \\ &= \$18 + \$1{,}8 = \$19{,}8\end{aligned}$$

An increase in variable expenses of 80 cents per unit means:

$$\begin{aligned}\text{Variable Expenses per Unit} &= \text{Current Variable Expenses per Unit} + \$0{,}80 = \\ &= \$10 + \$0{,}80 = \$10{,}80\end{aligned}$$

A decrease in units sold of 8% means:

$$\begin{aligned}\text{Quantity Sold} &= \text{Current Quantity Sold} - 8\% \times \text{Current Quantity Sold} = \\ &= 25.000 - 2.000 = 23.000\end{aligned}$$

$$\text{Sales} = \text{Sales per Unit} \times \text{Quantity Sold} = \$19{,}8 \times 23.000 = \$455.400$$

Consequently:

$$\text{Variable Expenses} = \text{Variable Expenses per Unit} \times \text{Quantity Sold} = \$10{,}80 \times 23.000 = \$248.400$$

$$\text{CM} = \text{Sales} - \text{Variable Expenses} = \$455.400 - \$248.400 = \$207.000$$

$$\text{NOI} = \text{CM} - \text{Fixed Expenses} = \$207.000 - \$85.000 = \$122.000$$

5.6 Exercise 6

Weaver Company's plantwide predetermined overhead rate is $21.00 per direct labor-hour and its direct labor wage rate is $14.00 per hour. The following information pertains to Job A-200:

- **Direct materials:** $290
- **Direct labor:** $210

Required:

1. What is the total manufacturing cost assigned to Job A-200?
2. If Job A-200 consists of 50 units, what is the average cost assigned to each unit in the job?

Solution (1)

$$\begin{aligned}\text{Total Manufacturing Cost} &= \text{Direct Materials Cost} + \\ &\quad \text{Direct Labor Cost} + \\ &\quad \text{Manufacturing Overhead}\end{aligned}$$

$$\text{Manufacturing Overhead} = \text{POHR} \times \text{Amount of Allocation Base}$$

We calculate the amount of allocation base used:

$$\text{Amount of Allocation Base} = \frac{\text{Direct Labor Cost}}{\text{Direct Labor Wage Rate per Hour}} = \frac{\$210}{\$14} = 15 \text{ DLHs}$$

Consequently:

$$\text{Manufacturing Overhead} = \$21 \text{ per DLH} \times 15 \text{ DLHs} = \$315$$

$$\text{Total Manufacturing Cost} = \$290 + \$210 + \$315 = \$815$$

Solution (2)

$$\text{Average Cost per Unit} = \frac{\text{Total Manufacturing Cost}}{\text{Number of Units in the Job}} = \frac{\$815}{50} = \$16{,}30$$

5.7 Exercise 7

Lionheart Company has two manufacturing departments—Molding and Firing. The predetermined departmental overhead rates in Molding and Firing are \$23.00 per direct labor-hour and 150% of direct materials cost, respectively. The company's direct labor wage rate is \$18.00 per hour. The following information pertains to Job HC-916:

	Molding	**Firing**
Direct materials	\$290	\$340
Direct labor	\$198	\$72

Required:

1. What is the total manufacturing cost assigned to Job HC-916?
2. If Job HC-916 consists of 20 units, what is the average cost assigned to each unit in the job?
3. What is the revised total manufacturing cost assigned to Job HC-916 considering a reduction of 10% of all direct materials?

Solution (1)

$$\begin{aligned}\text{Total Manufacturing Cost} = {} & \text{Direct Materials Cost} + \\ & \text{Direct Labor Cost} + \text{Manufacturing Overhead}\end{aligned}$$

$$\text{Manufacturing Overhead} = \text{POHR} \times \text{Amount of Allocation Base}$$

We calculate the amount of allocation base used:

Molding:

$$\text{Amount of Allocation Base} = \frac{\text{Direct Labor Cost}}{\text{Direct Labor Wage Rate per Hour}} = \frac{\$198}{\$18} = 11\text{ DLHs}$$

Firing:

$$\text{Amount of Allocation Base} = \frac{\text{Direct Labor Cost}}{\text{Direct Labor Wage Rate per Hour}} = \frac{\$72}{\$18} = 4\text{ DLHs}$$

Consequently:

Molding:

$$\text{Manufacturing Overhead} = \$23 \text{ per DLH} \times 11 \text{ DLHs} = \$253$$

$$\text{Total Manufacturing Cost} = \$290 + \$198 + \$253 = \$741$$

Firing:

$$\text{Manufacturing Overhead} = 150\% \times \$340 = 1,5 \times \$340 = \$510$$

$$\text{Total Manufacturing Cost} = \$340 + \$72 + \$510 = \$922$$

The total manufacturing cost is the sum of those of the molding and firing departments:

$$\text{Total Manufacturing Cost} = \$741 + \$922 = \$1.663$$

Solution (2)

$$\text{Average Cost per Unit} = \frac{\text{Total Manufacturing Cost}}{\text{Number of Units in the Job}} = \frac{\$1.663}{20} = \$83{,}15$$

Solution (3)

Molding:

$$\text{Manufacturing Overhead} = \$23 \text{ per DLH} \times 11 \text{ DLHs} = \$253$$

$$\text{Total Manufacturing Cost} = \$261 + \$198 + \$253 = \$712$$

Firing:

$$\text{Manufacturing Overhead} = 150\% \times \$306 = \$459$$

$$\text{Total Manufacturing Cost} = \$306 + \$72 + \$459 = \$837$$

The total manufacturing cost is the sum of those of the molding and firing departments:

$$\text{Total Manufacturing Cost} = \$712 + \$837 = \$1.549$$

5.8 Exercise 9

Vence Corporation is currently operating at 40% of its available manufacturing capacity. It uses a job-order costing system with a plantwide predetermined overhead rate based on machine-hours. At the beginning of the year, the company made the following estimates:

Item	**Amount**
Machine-hours required to support estimated production	40,000
Fixed manufacturing overhead cost	\$792,000
Variable manufacturing overhead cost per machine-hour	\$1.50

Required:

1. Compute the plantwide predetermined overhead rate.
2. During the year, Job 2K17 was started, completed, and sold to the customer for \$4,000. The following information was available with respect to this job:

Item	**Amount**
Direct materials	\$2,100
Direct labor cost	\$1,265
Machine-hours used	90

   Compute the total manufacturing cost assigned to Job 2K17 and net operating income.
3. Calculate net operating income considering that the firm works at 100% of its available manufacturing capacity.

Solution (1)

$$\text{POHR} = \frac{\text{Estimated total manufacturing overhead cost}}{\text{Estimated total units in the allocation base}} = \frac{\$792.000 + \$1{,}50 \times 40.000 \text{ MHs}}{40.000 \text{ MHs}} =$$

$$= \frac{\$852.000}{40.000 \text{ MHs}} = \boxed{\$21{,}30 \text{ per MH}}$$

Solution (2)

$$\begin{aligned}\text{Total Manufacturing Cost} &= \text{Direct Materials Cost} + \\ &\quad \text{Direct Labor Cost} + \\ &\quad \text{Manufacturing Overhead}\end{aligned}$$

$$\begin{aligned}\text{Manufacturing Overhead} &= \text{POHR} \times \text{Amount of Allocation Base} = \\ &= \$21{,}30 \text{ per MH} \times 90 \text{ MHs} = \$1.917\end{aligned}$$

Consequently:

$$\text{Total Manufacturing Cost} = \$2.100 + \$1.265 + \$1.917 = \boxed{\$5.282}$$

$$\begin{aligned}\text{Net Operating Income} &= \text{Sales} - \text{Total Manufacturing Cost} = \\ &= \$4.000 - \$5.282 = \boxed{\text{-}\$1.282}\end{aligned}$$

This means that Job 2K17 produced a loss.

Solution (3)

If 40% of the firm's available manufacturing capacity corresponds to 40.000 machine-hours required to support estimated production, then 100% of the capacity corresponds to 100.000 machine-hours. Consequently:

$$\text{POHR} = \frac{\text{Estimated total manufacturing overhead cost}}{\text{Estimated total units in the allocation base}} = \frac{\$792.000 + \$1{,}50 \times 100.000 \text{ MHs}}{100.000 \text{ MHs}} =$$

$$= \frac{\$942.000}{100.000 \text{ MHs}} = \$9{,}42 \text{ per MH}$$

Then:

$$\begin{aligned}\text{Manufacturing Overhead} &= \text{POHR} \times \text{Amount of Allocation Base} = \\ &= \$9{,}42 \text{ per MH} \times 90 \text{ MHs} = \$847{,}8\end{aligned}$$

$$\text{Total Manufacturing Cost} = \$2.100 + \$1.265 + \$847{,}8 = \$4.212{,}8$$

$$\begin{aligned}\text{Net Operating Income} &= \text{Sales} - \text{Total Manufacturing Cost} = \\ &= \$4.000 - \$4.212{,}8 = \text{-\$212,8}\end{aligned}$$

Job 2K17 still produces a loss, even if smaller.

6
Decision-Making Methods (11/03/2026)

6.1 Decision-Making Methods

Definition 6.1 (Decision-Making Methods). *Structured approaches or techniques used to evaluate alternatives and support the selection of the best possible option based on specific criteria, data, or preferences.*

These methods help reduce uncertainty and subjectivity by providing a systematic way to compare different choices and justify decisions.

6.1.1 Likert Scale

Definition 6.2 (Likert Scale). *A measurement instrument used to capture individuals' opinions, attitudes, or perceptions through a set of structured statements.*

Its main strength lies in its **simplicity** and ease of interpretation.

A Likert scale consists of a series of **statements (items)** related to a specific concept that the researcher wants to investigate. Each item is designed to measure the **same underlying construct**, making the scale **one-dimensional**.

Respondents are asked to indicate their level of agreement or disagreement with each statement by selecting one of several ordered response options:

- Strongly disagree
- Disagree
- Neutral / Uncertain
- Agree
- Strongly agree

Observation 6.1. *The presence of a* ***neutral option*** *allows respondents to take no clear position, which can be useful but may also reduce the decisiveness of the results.*

Each response is typically associated with a **numerical score**, for example:

- 1 = totally disagree
- 2 = disagree
- 3 = neither agree nor disagree
- 4 = agree
- 5 = completely agree

Using numerical values is generally preferred because:

- it allows the data to be processed using **analytical tools and software**;
- it enables the computation of **averages and aggregate indicators**.

Each response contributes to a **total score**. The **sum or average** of all responses across items represents the respondent's overall position with respect to the investigated concept. For this reason, the Likert scale is considered an **additive scale**.

The **quality** of a Likert scale depends heavily on the design of the questionnaire.
Questions can be defined based on:

- Literature data
- Interaction with experts
- Company or empirical data

Respondents can be selected through:

- Online channels (web surveys)
- Personal knowledge and networks
- Experts in the field, possibly belonging to different categories of **stakeholders**

6.1.2 Statistical Methods

The choice of the appropriate statistical method depends on the type of data and the number of groups being compared.

Likert Scale (Ordinal Data)

- **Two groups:** Mann–Whitney test statskingdom.com/170median_mann_whitney
- **More than two groups:** Kruskal–Wallis test statskingdom.com/kruskal-wallis-calculator

Numerical Data

- **Two groups:** t-test statskingdom.com/140MeanT2eq
  socscistatistics.com/tests/studentttest
- **More than two groups:** ANOVA (with Tukey post-hoc test) statskingdom.com/180Anova1way

Observation 6.2. *Non-parametric tests (such as Mann–Whitney and Kruskal–Wallis) are typically used for ordinal data like Likert scales, while parametric tests (such as t-test and ANOVA) are used for numerical data under appropriate assumptions.*

6.1.3 Multicriteria Decision Analysis (MCDA)

Definition 6.3 (**Multicriteria Decision Analysis (MCDA)**). *A decision-making framework used to evaluate and compare multiple alternatives based on several (often heterogeneous) criteria.*

MCDA is particularly useful when:

- data are incomplete or not directly comparable;
- multiple conflicting criteria must be considered;
- both qualitative and quantitative aspects are involved.

Observation 6.3. *The goal is to identify the* ***best alternative****, i.e., the one with the highest overall evaluation obtained by aggregating the values associated with each criterion.*

The value of each alternative is obtained through the combination (typically a weighted aggregation) of:

- a set of **criteria**;
- the **values** assigned to each criterion;

- the **weights** associated with those criteria.

The final score of an alternative is obtained by aggregating values and weights.

Observation 6.4. *The **weights of criteria are independent of the alternatives** and represent only their relative importance in the decision-making process.*

Observation 6.5. *All **weights are mutually independent**, meaning that the importance assigned to one criterion does not depend on the others.*

When applying MCDA, several fundamental questions arise:

- What categories of stakeholders should be considered?
- How many criteria should be included?
- How should criteria be selected?
- How should weights be assigned to criteria?
- How should values be assigned to criteria?
- How should weights and values be aggregated?

6.1.4 Analytic Hierarchy Process (AHP)

Definition 6.4 (**Analytic Hierarchy Process (AHP)**)**.** *A multicriteria decision-making method that structures a problem into a hierarchy of criteria and alternatives and uses pairwise comparisons to derive priorities.*

AHP is one of the most used methods within the MCDA framework due to its relative **simplicity of application**.

Main features:

- Information is structured in a **hierarchy** of criteria and alternatives.
- Information is aggregated to determine the **relative importance** of alternatives.
- Both **qualitative and quantitative criteria** can be compared.
- Weights and priorities are derived from **pairwise comparisons**.

6.1.4.1 Categories of Stakeholders

According to the UNEP–SETAC classification, stakeholders can be grouped into:

- workers
- consumers
- general society
- local community
- value chain actors

6.1.4.2 Number of Criteria

The number of criteria should be limited to ensure consistency and usability. Typically, the number of criteria follows the range:

$$7 \pm 2$$

In practice, it is recommended not to exceed **10 criteria**.

If the number of criteria is large:

- criteria can be grouped into **categories**;
- **local priorities** (within each category) and **global priorities** (overall importance) can be used:

$$\text{Global Priority} = \text{Local Priority} \times \text{Category Priority}$$

- some criteria may be repeated across categories;
- the number of criteria should be **balanced across categories**.

6.1.4.3 Selection of Criteria

Alternatives must be **comprehensively compared**, therefore criteria should be chosen carefully.
Criteria can be selected based on:

- literature data
- company data
- statistical data
- survey data

6.1.4.4 Assignment of Weights

Weights can be assigned in different ways:

- **Equal weights** (simplest approach):
  - all criteria are considered equally important;
  - easy to implement but often **not realistic**.
- **Eigenvalue method**:
  - Each criterion is associated with a **priority level**.
  - Two types of priorities are defined:
    * **Local priority**: within a category
    * **Global priority**: overall importance

Observation 6.6. *Weights are* ***independent of alternatives*** *and represent only the importance of criteria.*

In AHP, comparisons are made pairwise (e.g., criterion A vs B, A vs C, etc.). Since these judgments are subjective, inconsistencies may arise.

Definition 6.5 (**Consistency Ratio (CR)**)**.** *Measure of how consistent the judgments are in a pairwise comparison matrix.*

The matrix is considered **consistent** if:

$$CR < 0.10$$

If $CR \geq 0.10$, the judgments should be revised.

The consistency ratio is defined as:

$$CR = \frac{CI}{RI}$$

where:

$$CI = \text{Consistency Index} = \frac{\lambda_{\max} - n}{n - 1}$$

- $\lambda_{\max}$ = maximum eigenvalue of the comparison matrix
- n = number of criteria
- RI = Random Inconsistency Index

The value RI is a benchmark value that represents the consistency of a randomly generated comparison matrix. It depends only on the number of criteria n and is obtained from standard tables:

Random Inconsistency (RI) values for different numbers of factors (n).

n	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

***Note:** These values are used to calculate the Consistency Ratio (CR) in Saaty's Analytic Hierarchy Process (AHP), typically as the divisor.

The value $\lambda_{\max}$ is obtained from the comparison matrix using the **eigenvalue method**. In practice:

- compute the **priority vector** (eigenvector);
- compute the **weighted sum vector**;
- divide each component of the weighted sum vector by the corresponding element of the priority vector;
- take the **average** of these values.

This average gives $\lambda_{\max}$.

Observation 6.7. *The closer $\lambda_{\max}$ is to n, the more consistent the matrix is.*

In AHP, comparisons are performed using a scale from 1 to 9:

Numerical rating	Verbal judgements of preferences
1	Equally preferred
2	Equally to moderately
3	Moderately preferred
4	Moderately to strongly
5	Strongly preferred
6	Strongly to very strongly
7	Very strongly preferred
8	Very strongly to extremely
9	Extremely preferred

6.1.4.5 Assignment of Values to Criteria

Values associated with criteria can be obtained from:

- company data
- laboratory data

- literature data

If data are not available:

- a panel of experts can assign values;
- typically using a scale from **1 (minimum)** to **10 (maximum)**.

6.1.5 Other MCDA Methods

Other commonly used MCDA methods are:

- **TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution)**: method used to rank alternatives based on their distance from an ideal and a non-ideal solution. In particular, it is based on proximity to the **positive ideal solution** and distance from the **negative ideal solution**.
- **VIKOR**: method that focuses on ranking and selecting alternatives based on compromise solutions.

Observation 6.8. *AHP is based on **weighted aggregation**, while TOPSIS and VIKOR are based on **distance measures** from ideal solutions.*

6.1.6 Example: Weight Assignment (AHP)

This example illustrates the computation of weights using AHP.

AHP Example - Weight Assignment

Categories Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS
WORKERS	1,00	0,50	0,50	0,50	0,50
CONSUMERS	2,00	1,00	0,50	0,50	0,50
GENERAL SOCIETY	2,00	2,00	1,00	2,00	0,50
LOCAL COMMUNITY	2,00	2,00	0,50	1,00	0,50
VALUE CHAIN ACTORS	2,00	2,00	2,00	2,00	1,00
Total	9,00	7,50	4,50	6,00	3,00

Normalized Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS	Total	Average
WORKERS	0,11	0,07	0,11	0,08	0,17	0,54	0,11
CONSUMERS	0,22	0,13	0,11	0,08	0,17	0,72	0,14
GENERAL SOCIETY	0,22	0,27	0,22	0,33	0,17	1,21	0,24
LOCAL COMMUNITY	0,22	0,27	0,11	0,17	0,17	0,93	0,19
VALUE CHAIN ACTORS	0,22	0,27	0,44	0,33	0,33	1,60	0,32
Total	1,00	1,00	1,00	1,00	1,00	5,00	1,00

n	Lambda max	CI	RI	CR	Consistency Check
5,00	5,22	0,05	1,12	0,05	OK

Editable inputs are highlighted in yellow. Upper-triangular cells are calculated automatically as reciprocals of the corresponding lower-triangular inputs.

If CR < 0.10, the pairwise comparison matrix is considered acceptably consistent.

n	RI
1,00	0,00
2,00	0,00
3,00	0,58
4,00	0,90
5,00	1,12
6,00	1,24
7,00	1,32
8,00	1,41
9,00	1,45
10,00	1,49

Download AHP Excel File

Step 1: Pairwise Comparison Matrix

AHP Example - Weight Assignment

Categories Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS
WORKERS	1,00	0,50	0,50	0,50	0,50
CONSUMERS	2,00	1,00	0,50	0,50	0,50
GENERAL SOCIETY	2,00	2,00	1,00	2,00	0,50
LOCAL COMMUNITY	2,00	2,00	0,50	1,00	0,50
VALUE CHAIN ACTORS	2,00	2,00	2,00	2,00	1,00
Total	**9,00**	**7,50**	**4,50**	**6,00**	**3,00**

Normalized Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS	Total	Average
WORKERS	0,11	0,07	0,11	0,08	0,17	0,54	0,11
CONSUMERS	0,22	0,13	0,11	0,08	0,17	0,72	0,14
GENERAL SOCIETY	0,22	0,27	0,22	0,33	0,17	1,21	0,24
LOCAL COMMUNITY	0,22	0,27	0,11	0,17	0,17	0,93	0,19
VALUE CHAIN ACTORS	0,22	0,27	0,44	0,33	0,33	1,60	0,32
Total	**1,00**	**1,00**	**1,00**	**1,00**	**1,00**	**5,00**	**1,00**

n	Lambda max	CI	RI	CR	Consistency Check
5,00	5,22	0,05	1,12	**0,05**	**OK**

Editable inputs are highlighted in yellow. Upper-triangular cells are calculated automatically as reciprocals of the corresponding lower-triangular inputs.

If CR < 0.10, the pairwise comparison matrix is considered acceptably consistent.

n	RI
1,00	0,00
2,00	0,00
3,00	0,58
4,00	0,90
5,00	1,12
6,00	1,24
7,00	1,32
8,00	1,41
9,00	1,45
10,00	1,49

A pairwise comparison matrix is constructed by comparing each category with the others.

- 1 = equal importance
- > 1 = more important
- < 1 = less important (reciprocal values)

Step 2: Column Normalization

AHP Example - Weight Assignment

Categories Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS
WORKERS	1,00 ÷	0,50	0,50	0,50	0,50
CONSUMERS	2,00 ÷	1,00	0,50	0,50	0,50
GENERAL SOCIETY	2,00 ÷	2,00	1,00	2,00	0,50
LOCAL COMMUNITY	2,00 ÷	2,00	0,50	1,00	0,50
VALUE CHAIN ACTORS	2,00 ÷	2,00	2,00	2,00	1,00
Total	9,00	7,50	4,50	6,00	3,00

Normalized Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS	Total	Average
WORKERS	0,11	0,07	0,11	0,08	0,17	0,54	0,11
CONSUMERS	0,22	0,13	0,11	0,08	0,17	0,72	0,14
GENERAL SOCIETY	0,22	0,27	0,22	0,33	0,17	1,21	0,24
LOCAL COMMUNITY	0,22	0,27	0,11	0,17	0,17	0,93	0,19
VALUE CHAIN ACTORS	0,22	0,27	0,44	0,33	0,33	1,60	0,32
Total	1,00	1,00	1,00	1,00	1,00	5,00	1,00

n	Lambda max	CI	RI	CR	Consistency Check
5,00	5,22	0,05	1,12	0,05	OK

Editable inputs are highlighted in yellow. Upper-triangular cells are calculated automatically as reciprocals of the corresponding lower-triangular inputs.

If CR < 0.10, the pairwise comparison matrix is considered acceptably consistent.

n	RI
1,00	0,00
2,00	0,00
3,00	0,58
4,00	0,90
5,00	1,12
6,00	1,24
7,00	1,32
8,00	1,41
9,00	1,45
10,00	1,49

Each element of the matrix is divided by the sum of its column:

$$\text{Normalized value} = \frac{\text{element}}{\text{column sum}}$$

Step 3: Priority Vector (Weights)

AHP Example - Weight Assignment

Categories Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS
WORKERS	1,00	0,50	0,50	0,50	0,50
CONSUMERS	2,00	1,00	0,50	0,50	0,50
GENERAL SOCIETY	2,00	2,00	1,00	2,00	0,50
LOCAL COMMUNITY	2,00	2,00	0,50	1,00	0,50
VALUE CHAIN ACTORS	2,00	2,00	2,00	2,00	1,00
Total	9,00	7,50	4,50	6,00	3,00

Normalized Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS	Total	Average
WORKERS	0,11	0,07	0,11	0,08	0,17	0,54	0,11
CONSUMERS	0,22	0,13	0,11	0,08	0,17	0,72	0,14
GENERAL SOCIETY	0,22	0,27	0,22	0,33	0,17	1,21	0,24
LOCAL COMMUNITY	0,22	0,27	0,11	0,17	0,17	0,93	0,19
VALUE CHAIN ACTORS	0,22	0,27	0,44	0,33	0,33	1,60	0,32
Total	1,00	1,00	1,00	1,00	1,00	5,00	1,00

n	Lambda max	CI	RI	CR	Consistency Check
5,00	5,22	0,05	1,12	0,05	OK

Editable inputs are highlighted in yellow. Upper-triangular cells are calculated automatically as reciprocals of the corresponding lower-triangular inputs.

If CR < 0.10, the pairwise comparison matrix is considered acceptably consistent.

n	RI
1,00	0,00
2,00	0,00
3,00	0,58
4,00	0,90
5,00	1,12
6,00	1,24
7,00	1,32
8,00	1,41
9,00	1,45
10,00	1,49

The weight of each criterion is obtained by averaging the normalized values across each row.

Step 4: Maximum Eigenvalue

AHP Example - Weight Assignment

Categories Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS
WORKERS	1,00	0,50	0,50	0,50	0,50
CONSUMERS	2,00	1,00	0,50	0,50	0,50
GENERAL SOCIETY	2,00	2,00	1,00	2,00	0,50
LOCAL COMMUNITY	2,00	2,00	0,50	1,00	0,50
VALUE CHAIN ACTORS	2,00	2,00	2,00	2,00	1,00
Total	9,00	7,50	4,50	6,00	3,00

Normalized Matrix	WORKERS	CONSUMERS	GENERAL SOCIETY	LOCAL COMMUNITY	VALUE CHAIN ACTORS	Total	Average
WORKERS	0,11	0,07	0,11	0,08	0,17	0,54	0,11
CONSUMERS	0,22	0,13	0,11	0,08	0,17	0,72	0,14
GENERAL SOCIETY	0,22	0,27	0,22	0,33	0,17	1,21	0,24
LOCAL COMMUNITY	0,22	0,27	0,11	0,17	0,17	0,93	0,19
VALUE CHAIN ACTORS	0,22	0,27	0,44	0,33	0,33	1,60	0,32
Total	1,00	1,00	1,00	1,00	1,00	5,00	1,00

n	Lambda max	CI	RI	CR	Consistency Check
5,00	5,22	0,05	1,12	0,05	OK

Editable inputs are highlighted in yellow. Upper-triangular cells are calculated automatically as reciprocals of the corresponding lower-triangular inputs.

If CR < 0.10, the pairwise comparison matrix is considered acceptably consistent.

n	RI
1,00	0,00
2,00	0,00
3,00	0,58
4,00	0,90
5,00	1,12
6,00	1,24
7,00	1,32
8,00	1,41
9,00	1,45
10,00	1,49

The maximum eigenvalue is computed as:

$$\lambda_{\max} = \sum(\text{column sum} \times \text{weight})$$

Step 5: Consistency Index

$$CI = \frac{\lambda_{\max} - n}{n-1}$$

Step 6: Consistency Ratio

$$CR = \frac{CI}{RI}$$

where RI is the random inconsistency index (see the table).

7
Noreen Ch06

7.1 Random concepts captured (17/03/2026)

Product Line vs Business Line

Definition 7.1 (Product Line). *The set of products developed using a specific technology or technological capability.*

- It is mainly related to the **technology used** to design and produce a product.
- It focuses on **how a product is realized**.
- Different products may belong to the same product line if they share the same technological base.

Definition 7.2 (Business Line). *The way a company uses its technologies and resources to create value and generate revenue in a specific market.*

- It is related to **how the technology is applied** to satisfy a market need.
- It focuses on **what value is delivered and to whom**.
- It includes aspects such as customers, market segments, and revenue generation.

Observation 7.1. *A product line is technology-oriented, while a business line is market-oriented. The same technology (product line) can support multiple business lines.*

Backward and Forward Analysis

Definition 7.3 (Backward Analysis). *An approach that starts from a desired goal or outcome and works backward to identify the necessary steps, resources, and conditions required to achieve it.*

- Focuses on **starting from the objective**.

- Identifies the **requirements and constraints** needed to reach the goal.
- Useful for **planning and strategic design**.

Definition 7.4 (**Forward Analysis**). *An approach that starts from the current situation (available resources, technologies, and constraints) and moves forward to determine possible outcomes or opportunities.*

- Focuses on **starting from what is available**.
- Explores **feasible developments and opportunities**.
- Useful for **innovation and opportunity discovery**.

Observation 7.2. *Backward analysis is goal-driven, while forward analysis is resource-driven. In practice, effective decision-making often combines both approaches.*

7.2 Decision-Making

7.2.1 LO1: Relevant/Irrelevant Costs and Benefits

7.2.1.1 Step 1: Definition of Alternatives

Observation 7.3. *Every decision involves choosing among at least two alternatives.*

Therefore, the first step in decision-making is to **define the alternatives** being considered.

7.2.1.2 Step 2: Identification of Criteria

Once the alternatives have been defined, it is necessary to identify the criteria for choosing among them. In general:

- **Relevant costs** and **relevant benefits** should be considered.
- **Irrelevant costs** and **irrelevant benefits** should be ignored.

7.2.1.3 Step 3: Differential Analysis

Definition 7.5 (**Differential Analysis**). *An analysis that focuses on the future costs and benefits that differ between alternatives.*

Observation 7.4. *Everything that does not differ between alternatives is irrelevant and should be ignored.*

Definition 7.6 (**Differential Cost**). *A future cost that differs between alternatives.*

Definition 7.7 (**Differential Revenue**). *A future revenue that differs between alternatives.*

Definition 7.8 (**Incremental Cost**). *An increase in cost between two alternatives.*

Definition 7.9 (**Avoidable Cost**). *A cost that can be eliminated by choosing one alternative over another.*

7.2.1.4 Step 4: Identification of Sunk Costs

Definition 7.10 (**Sunk Cost**). *A cost that has already been incurred and cannot be changed regardless of future decisions.*

Sunk costs are always **irrelevant** when choosing among alternatives.

7.2.1.5 Step 5: Identification of Irrelevant Future Costs

Future costs and benefits that do not differ between alternatives are **irrelevant** to the decision-making process.

7.2.1.6 Step 6: Identification of Opportunity Costs

Definition 7.11 (**Opportunity Cost**). *The potential benefit that is given up when one alternative is selected over another.*

Opportunity costs must always be considered when making decisions, even if they are not explicitly recorded in accounting systems.

EXAMPLE (Identifying Relevant Costs)

Cynthia, a Boston student, is considering visiting her friend in New York. She has two alternatives:

- Drive her car
- Take the train

The distance is 230 miles (one way), and she wants to determine which alternative is less expensive.

Automobile Costs (based on 10,000 miles driven per year)			
		Annual Cost of Fixed Items	Cost per Mile
1	Annual straight-line depreciation on car	$ 2,800	$ 0.280
2	Cost of gasoline		0.100
3	Annual cost of auto insurance and license	1,380	0.138
4	Maintenance and repairs		0.065
5	Parking fees at school	360	0.036
6	Total average cost		$ 0.619

	Additional Information	
7	Reduction in resale value of car per mile of wear	$ 0.026
8	Round-tip train fare	$ 104
9	Benefits of relaxing on train trip	????
10	Cost of putting dog in kennel while gone	$ 40
11	Benefit of having car in New York	????
12	Hassle of parking car in New York	????
13	Per day cost of parking car in New York	$ 25

Among the irrelevant costs:

- **Cost of the car (purchase price):** sunk cost → not relevant
- **Insurance and license:** does not change between alternatives
- **Parking fees at school:** must be paid regardless of the decision
- **Kennel cost:** incurred in both alternatives

Among the relevant costs:

- **Gasoline:** only incurred if she drives
- **Maintenance and repairs:** depend on miles driven
- **Reduction in resale value:** additional wear from driving
- **Parking in New York:** avoidable if she takes the train
- **Train ticket:** only incurred if she takes the train

There are also some non-financial benefits to consider:

- Relaxing on the train
- Having a car available in New York
- Difficulty of finding parking

Observation 7.5. *Some benefits are relevant even if they cannot be easily expressed in monetary terms.*

If we compare the relevant costs of driving and taking the train:

Relevant Financial Cost of Driving	
Gasoline (460 @ $0.100 per mile)	$ 46.00
Maintenance (460 @ $0.065 per mile)	29.90
Reduction in resale (460 @ $0.026 per mile)	11.96
Parking in New York (2 days @ $25 per day)	50.00
Total	$ 137.86

Relevant Financial Cost of Taking the Train	
Round-trip ticket	$ 104.00

From a purely financial standpoint, Cynthia should take the train because it is less expensive. However, non-financial factors (comfort, convenience, flexibility) may influence the final decision.

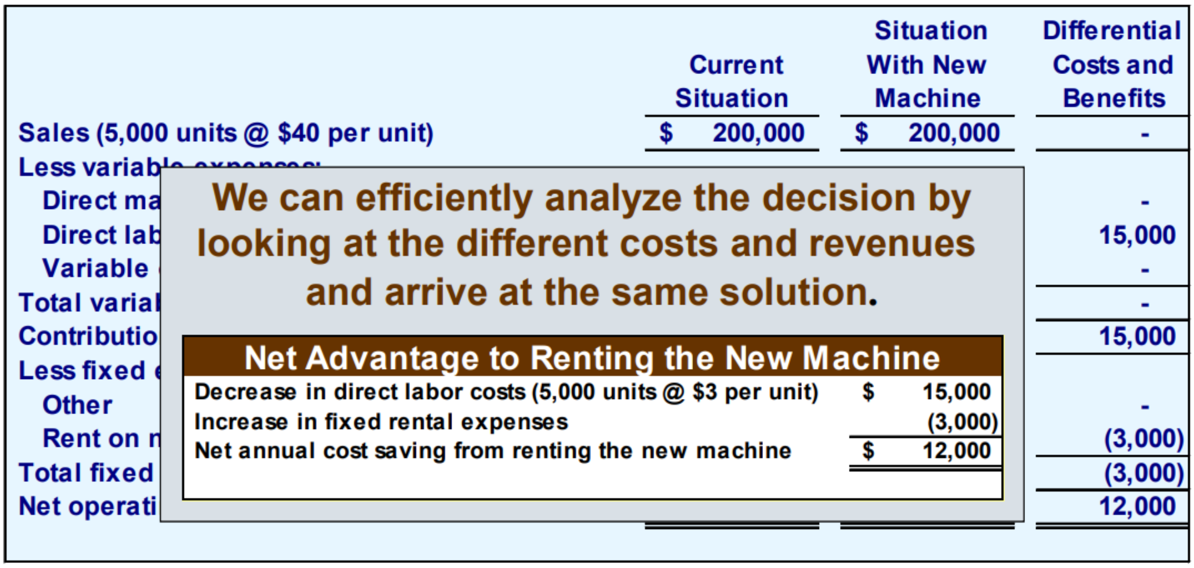
EXAMPLE (Total vs. Differential Cost Approaches)

The management of a company is considering a new labor saving machine that rents for $3,000 per year. Data about the company's annual sales and costs with and without the new machine are the following:

	Current Situation	Situation With New Machine	Differential Costs and Benefits
Sales (5,000 units @ $40 per unit)	$ 200,000	$ 200,000	-
Less variable expenses:			
Direct materials (5,000 units @ $14 per unit)	70,000	70,000	-
Direct labor (5,000 units @ $8 and $5 per unit)	40,000	25,000	15,000
Variable overhead (5,000 units @ $2 per unit)	10,000	10,000	-
Total variable expenses	120,000	105,000	-
Contribution margin	80,000	95,000	15,000
Less fixed expense:			
Other	62,000	62,000	-
Rent on new machine	-	3,000	(3,000)
Total fixed expenses	62,000	65,000	(3,000)
Net operating income	$ 18,000	$ 30,000	12,000

The total approach compares complete income statements for each alternative, including all revenues and costs. This approach certainly leads to the correct decision, but it may include many irrelevant items.

On the contrary, if we use the differential approach only differential costs and benefits affect the decision:



It is often **impractical** to prepare full income statements for all alternatives. Including irrelevant costs can create confusion and distract from critical information. The differential approach is more efficient and focuses directly on what matters for decision-making.

7.2.2 LO2: Adding/Dropping Business Segments

One of the most important decisions managers make is whether to add or drop a business segment (a product line, division, or department). The decision is primarily based on its **financial impact**.

Observation 7.6. *To properly assess the impact, it is necessary to carefully analyze the costs.*

EXAMPLE (Contribution Margin Approach)

Segment Income Statement Digital Watches		
Sales		$ 500,000
Less: variable expenses		
Variable manufacturing costs	$ 120,000	
Variable shipping costs	5,000	
Commissions	75,000	200,000
Contribution margin		$ 300,000
Less: fixed expenses		
General factory overhead	$ 60,000	
Salary of line manager	90,000	
Depreciation of equipment	50,000	
Advertising - direct	100,000	
Rent - factory space	70,000	
General admin. expenses	30,000	400,000
Net operating loss		$ (100,000)

Lovell Company is considering whether to drop its digital watch product line, which has not been profitable for several years.

Observation 7.7 (Decision Rule). *A segment should be dropped only if overall company profit increases.*

The analysis is based on comparing:

- The **contribution margin lost** if the segment is dropped
- The **fixed costs avoided** if the segment is dropped

$$\text{Net Effect} = \text{Avoided Fixed Costs} - \text{Lost Contribution Margin}$$

The theory says that:

- If avoided fixed costs > lost contribution margin → **drop the segment**
- If avoided fixed costs < lost contribution margin → **keep the segment**

So, in order to make the best decision, we have to distinguish between:

- **Avoidable fixed costs:** relevant (they disappear if the segment is dropped)
- **Unavoidable fixed costs:** irrelevant (they remain and are reallocated)

Observation 7.8. *Allocated fixed costs are often misleading because they may not actually be eliminated.*

From the income statement we learn that:

- Some fixed costs (e.g., general overhead, administrative expenses) **do not change** and are therefore irrelevant. These costs are simply **reallocated** to other segments.
- The equipment used to manufacture digital watches has **no resale value or alternative use**, therefore it is not relevant to the decision.

Contribution Margin Solution

Contribution margin lost if digital watches are dropped		$ (300,000)
Less fixed costs that can be avoided		
Salary of the line manager	$ 90,000	
Advertising - direct	100,000	
Rent - factory space	70,000	260,000
Financial disadvantage of dropping the digital watches product line		$ (40,000)

Since:

$$\text{Net Effect} = \$260.000 - \$300.000 = -\$40.000$$

We learn that it is not convenient for the company to drop the digital watch product line.

Observation 7.9. *Even if a segment reports an accounting loss, it should not necessarily be dropped. The correct decision depends only on the* ***incremental effect on overall profit****.*

EXAMPLE (Comparative Income Approach)

	Keep Digital Watches	Drop Digital Watches	Difference
Sales	$ 500,000	$ -	$ (500,000)
Less variable expenses:			
Manufacturing expenses	120,000	-	120,000
Shipping	5,000	-	5,000
Commissions	75,000	-	75,000
Total variable expenses	200,000	-	200,000
Contribution margin	300,000	-	(300,000)
Less fixed expenses:			
General factory overhead	60,000	60,000	-
Salary of line manager	90,000	-	90,000
Depreciation	50,000	50,000	-
Advertising - direct	100,000	-	100,000
Rent - factory space	70,000	-	70,000
General admin. expenses	30,000	30,000	-
Total fixed expenses	400,000	140,000	260,000
Net operating income	$ (100,000)	$ (140,000)	$ (40,000)

The **comparative income approach** consists in preparing two income statements:

- one assuming the segment is **kept**
- one assuming the segment is **dropped**

The decision is based on the **difference in net operating income** between the two alternatives.

We remember that:

- If dropping the segment **increases profit** ⇒ drop it
- If dropping the segment **decreases profit** ⇒ keep it

Only **costs and revenues that differ between the alternatives** are relevant.

Among the relevant costs:

- Contribution margin (lost if the segment is dropped)
- Avoidable fixed costs

Among the irrelevant costs:

- Allocated common fixed costs
- Sunk costs (e.g., depreciation if no resale value)

Even if the segment shows an accounting loss, it may still be contributing to cover common fixed costs.

Dropping the segment causes:

- Loss of contribution margin
- Saving only of **avoidable** fixed costs

If:

$$\text{Loss of contribution margin} > \text{Avoidable fixed costs}$$

then:

$$\text{Net operating income decreases} \Rightarrow \text{Do NOT drop the segment}$$

Allocated fixed costs can distort the decision:

- These costs **do not disappear** if the segment is dropped
- They are simply **reallocated** to other segments

Therefore, allocated common fixed costs are irrelevant for the decision.

The correct question is not:

"Is the segment profitable?"

but:

"What is the effect on total profit if the segment is dropped?"

7.2.3 LO3: Make or Buy Analysis

A **make or buy decision** arises when a company must decide whether to:

- produce a component **internally** (make)
- purchase it from an **external supplier** (buy)

When a firm performs multiple activities along the value chain, it is said to be **vertically integrated**.

Among the advantages:

- Smoother flow of parts and materials
- Better quality control
- Ability to realize profits internally

Among the disadvantages:

- Loss of economies of scale from specialized suppliers
- Risk of inefficiencies compared to external providers

The decision must be based only on **relevant (avoidable) costs**, and must follow the following rule:

Make if Cost to make < Cost to buy

Buy if Cost to buy < Cost to make

EXAMPLE

Essex Company manufactures part 4A that is used in one of its products. The unit product cost of this part is:

Direct materials	$ 9
Direct labor	5
Variable overhead	1
Depreciation of special equip.	3
Supervisor's salary	2
General factory overhead	10
Unit product cost	$ 30

The $30 unit product cost is based on 20,000 parts produced each year. An outside supplier has offered to provide the 20,000 parts at a cost of $25 per part. Should the company stop making part 4A and buy it from the outside supplier?

Even if the unit product cost is $30, not all of this cost is relevant.

Among the relevant (avoidable) costs:

- Direct materials
- Direct labor
- Variable overhead
- Any fixed costs that can be avoided (e.g., supervisor salary)

Among the irrelevant costs:

- Sunk costs (e.g., depreciation of equipment already purchased)

- Allocated fixed overhead (unchanged regardless of the decision)

Only the **avoidable portion** must be compared with the purchase price:

	Cost Per Unit	Cost of 20,000 Units	
		Make	Buy
Outside purchase price	$ 25		$ 500,000
Direct materials (20,000 units)	$ 9	180,000	
Direct labor	5	100,000	
Variable overhead	1	20,000	
Depreciation of equip.	3	-	
Supervisor's salary	2	40,000	
General factory overhead	10	-	
Total cost	$ 30	$ 340,000	$ 500,000

Since:

$$\text{Relevant cost to make} < \text{Cost to buy}$$

the company should continue to make the part internally and save $160.000.

7.2.3.1 Opportunity Cost

Definition 7.12 (Opportunity Cost). *The potential benefit that is given up when one alternative is chosen over another.*

It is not an actual cash outflow and therefore it is not recorded in accounting systems, but it's still relevant for decision-making.

If a resource (e.g., space, machines) can be used in different ways, choosing one option implies giving up the benefit of the best alternative use:

$$\text{Opportunity cost} = \text{benefit from the best alternative use}$$

In other words, if the company uses its resources internally, it may lose the opportunity to use them elsewhere. This lost benefit must be **added** to the cost of making.

Therefore, the decision rule becomes:

$$\textbf{Make} \text{ if } (\text{Relevant cost} + \text{Opportunity cost}) < \text{Cost to buy}$$

$$\textbf{Buy} \text{ if } (\text{Relevant cost} + \text{Opportunity cost}) > \text{Cost to buy}$$

7.2.4 LO4: Special Orders

Definition 7.13 (Special Order). *A one-time order that is not considered part of the company's normal ongoing business.*

When analyzing a special order, only **incremental costs** and **incremental benefits** are relevant.

The decision rule is:

Accept the order if Incremental revenue > Incremental cost

Reject the order if Incremental revenue < Incremental cost

A special order is typically accepted when the company has **unused capacity**, since it can produce additional units without affecting normal sales. In this case, fixed costs remain unchanged and therefore are **irrelevant**.

Only the following are relevant:

- Additional revenues generated by the order
- Additional (variable) costs required to fulfill the order

EXAMPLE

Jet Inc. makes a single product whose normal selling price is \$20 per unit. A foreign distributor offers to purchase 3,000 units for \$10 per unit. This is a one-time order that would not affect the company's regular business. Annual capacity is 10,000 units, but Jet Inc. is currently producing and selling only 5,000 units. Should the company accept the offer?

Jet Inc.
Contribution Inc. Stmt., before considering a special order

Revenue (5,000 × \$20)		\$ 100,000
Variable costs:		
Direct materials	\$ 20,000	
Direct labor	5,000	
Manufacturing overhead	10,000	
Marketing costs	5,000	
Total variable costs		40,000
Contribution margin		60,000
Fixed costs:		
Manufacturing overhead	\$ 28,000	
Marketing costs	20,000	
Total fixed costs		48,000
Net operating income		\$ 12,000

Fixed costs are unavoidable, so we compare:

$$\text{Incremental revenue} = \text{Special price} \times \text{Units}$$

$$\text{Incremental cost} = \text{Variable cost per unit} \times \text{Units}$$

If incremental revenue exceeds incremental cost, the order should be accepted because it increases net operating income.

In the example:

Increase in revenue (3,000 × $10)	$30,000
Increase in costs (3,000 × $8 variable cost)	24,000
Financial advantage of accepting the order	$ 6,000

Since net operating income increases, the company should accept the special order. This proves that even if the special price is lower than the normal selling price, the order can still be convenient if it covers variable costs and contributes to fixed costs.

7.2.5 LO5: Utilization of a Constrained Resource

Companies are forced to make **volume trade-off decisions** when they do not have enough capacity to produce all the products and sales volumes demanded by their customers.

In these situations, companies must sacrifice the production of some products in favor of others in order to maximize profits.

Definition 7.14 (**Constraint**)**.** *A limited resource that restricts the company's ability to satisfy demand.*

Definition 7.15 (**Bottleneck**)**.** *The machine or process that limits overall output. It is the constrained resource.*

When a company operates under a constrained resource, fixed costs are usually unaffected. Therefore, the product mix that maximizes the company's **total contribution margin** should ordinarily be selected.

Observation 7.10. *A company should not necessarily emphasize the products with the highest unit contribution margins.*

Instead, it should emphasize the products that generate the highest contribution margin in relation to the constraining resource.

EXAMPLE

Ensign Company produces two products. The selected data are shown in the following figure:

	Product 1	Product 2
Selling price per unit	$ 60	$ 50
Less variable expenses per unit	36	35
Contribution margin per unit	$ 24	$ 15
Current demand per week (units)	2,000	2,200
Contribution margin ratio	40%	30%
Processing time required on machine A1 per unit	1.00 min.	0.50 min.

Machine A1 is the constrained resource and is being used at 100% of its capacity (2.400 minutes per week). There is excess capacity on all the other machines. Should Ensign focus its efforts on Product 1 or Product 2?

The key is to compute the **contribution margin per unit of the constrained resource**.

$$\text{Contribution margin per minute} = \frac{\text{Contribution margin per unit}}{\text{Minutes required per unit}}$$

For Product 1:

$$\frac{\$24}{1.00 \text{ min.}} = \$24 \text{ per minute}$$

For Product 2:

$$\frac{\$15}{0.50 \text{ min.}} = \$30 \text{ per minute}$$

Therefore, Ensign should emphasize **Product 2**, because it generates a higher contribution margin per minute of the constrained resource.

The company should first produce Product 2 to satisfy customer demand and then use any remaining capacity to produce Product 1.

If weekly demand for Product 2 is 2.200 units, the time required to produce it is:

$$2.200 \times 0{,}50 = 1.100 \text{ minutes}$$

Since total available time for machine A1 is 2.400 minutes, the remaining time for Product 1 is:

$$2.400 - 1.100 = 1.300 \text{ minutes}$$

Given that Product 1 requires 1 minute per unit, the company can produce 1.300 units of Product 1.

In summary, the optimal production plan is:

- 2.200 units of Product 2
- 1.300 units of Product 1

And the total contribution margin is therefore:

$$\text{CM} = (1.300 \times \$24) + (2.200 \times \$15) = \$31.200 + \$33.000 = \$64.200$$

7.2.6 LO6: Value of a Constrained Resource

In the presence of a constrained resource (bottleneck), increasing its capacity allows the company to increase production and sales.

The key question is:

How much should a company be willing to pay for an additional unit of the constrained resource?

The value of an additional unit of the constrained resource is equal to the **contribution margin that can be generated using that additional capacity**.

Observation 7.11. *The maximum price that should be paid for one additional unit of the constrained resource is equal to the* ***contribution margin per unit of the constrained resource****.*

EXAMPLE

From the previous example, Product 2 is prioritized because it generates the highest contribution margin per minute.

Therefore, any additional capacity will be used to produce more units of the next best alternative, which is Product 1.

Product 1 generates:

$$\$24 \text{ per minute}$$

Thus, the company should be willing to pay up to:

$$\$24 \text{ per minute}$$

for additional machine time.

Managing Constraints

It is often possible to increase the capacity of a bottleneck (i.e., relax the constraint) in several ways:

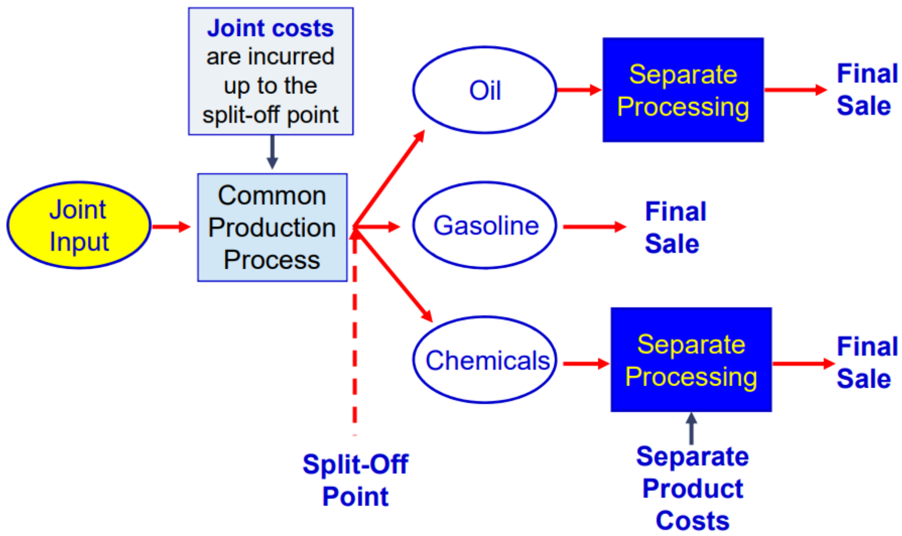
- Working overtime on the bottleneck
- Subcontracting part of the processing
- Investing in additional machines at the bottleneck
- Shifting workers from non-bottleneck processes
- Improving efficiency of the bottleneck process
- Reducing defective units processed through the bottleneck

7.2.7 LO7: Joint Products and Sell or Process Further Decisions

In some industries, two or more products are produced from a single raw material input. These products are called **joint products**.

Definition 7.16 (**Split-off point**). *The point in the production process where joint products can be recognized as separate products.*

Definition 7.17 (**Sell or Process Further Decision**). *A decision concerning whether a joint product should be sold at the split-off point or processed further before being sold.*



Definition 7.18 (**Joint Costs**). *Costs incurred up to the split-off point in the production of joint products.*

Joint costs are common to all products and cannot be traced to individual products before the split-off point.

Observation 7.12. *Joint costs are* ***irrelevant*** *for decisions made after the split-off point.*

Although joint costs are often allocated to products (e.g., based on relative sales value), such allocations are not useful for decision-making purposes.

Observation 7.13. *Allocating joint costs can be misleading and should not influence sell or process further decisions.*

The decision must be based only on **incremental (differential) analysis**:

$$\textbf{Process further} \text{ if Incremental revenue} > \text{Incremental processing cost}$$

$$\textbf{Sell at split-off} \text{ if Incremental revenue} < \text{Incremental processing cost}$$

Where:

$$\text{Incremental revenue} = \text{Sales value after further processing} - \text{Sales value at split-off}$$

EXAMPLE

Sawmill Inc. cuts logs from which unfinished lumber and sawdust are the immediate joint products. Unfinished lumber can be sold "as is" or processed further into finished lumber. Sawdust can also be sold "as is" to gardening wholesalers or processed further into "presto-logs."

Analysis of Sell or Process Further

	Per Log	
	Lumber	Sawdust
Final sales value after further processing	$ 270	$ 50
Sales value at the split-off point	140	40
Incremental revenue from further processing	130	10
Cost of further processing	50	20
Financial advantage (disadvantage) of further processing	$ 80	$ (10)

For **lumber**:

$$\begin{aligned}\text{Incremental revenue} &= \$270 - \$140 = \$130\\ \text{Incremental cost} &= \$50\\ \text{Net benefit} &= \$130 - \$50 = \$80\end{aligned}$$

Since the net benefit is positive, lumber should be **processed further**.

For **sawdust**:

$$\begin{aligned}\text{Incremental revenue} &= \$50 - \$40 = \$10\\ \text{Incremental cost} &= \$20\\ \text{Net benefit} &= \$10 - \$20 = -\$10\end{aligned}$$

Since the net benefit is negative, sawdust should be **sold at the split-off point**.

Activity-Based Costing and Relevant Costs

Activity-Based Costing (ABC) can help identify potentially relevant costs for decision-making.

Observation 7.14. *Not all traceable costs are avoidable. A cost is relevant only if it differs between alternatives.*

Managers must be careful not to assume that every traceable cost is relevant. Only **avoidable and differential costs** should be considered in the analysis.

8 Noreen Ch06A

8.1 Random concepts captured (24/03/2026)

Emission Trading Scheme

Definition 8.1 (Emission Trading Scheme). *A regulatory system in which firms must pay (or hold permits) when they produce pollution above a certain allowed level.*

Firms are assigned or must purchase **emission allowances**, each of which permits a firm to emit a certain amount of pollution. If a firm emits more than its allowances, it must **buy additional permits** or pay a penalty. If a firm emits less, it can **sell excess permits** to other firms.

Observation 8.1. *The emission trading scheme creates a **market price for pollution**, encouraging firms to reduce emissions when it is cost-effective.*

Bias

Definition 8.2 (Bias). *A systematic tendency to deviate from rational or objective decision-making due to psychological, cognitive, or organizational factors.*

Bias can lead managers to make **suboptimal decisions**, often arising from **personal beliefs, past experiences, or incentives**. It may cause individuals to **ignore relevant information** or give too much weight to irrelevant factors.

Observation 8.2. *Bias can distort decision-making by causing managers to focus on non-relevant costs or benefits instead of relying on proper differential analysis.*

8.2 Pricing Decisions

Many factors can influence how companies establish their selling prices. In particular, we focus on three main elements:

- **Customers**
- **Competitors**
- **Costs**

8.2.1 Customers and Elasticity

Customers usually possess two important elements that complicate the pricing process:

- **Attitude**
- **Private information**

Many companies face heterogeneous customers, meaning that:

- Different customers have different **willingness to pay**
- A single price may not be optimal for all customers

For this reason, firms are interested in measuring the **price elasticity of demand**.

Definition 8.3 (**Price Elasticity of Demand**). *The degree to which the unit sales of a product or service are affected by a change in its price.*

Demand can be:

- **Inelastic** if a change in price has little effect on the number of units sold. Luxury goods (e.g., designer perfumes) often exhibit relatively inelastic demand.
- **Elastic** if a change in price has a significant effect on the number of units sold. Products with many substitutes (e.g., gasoline across stations) tend to have more elastic demand.

Observation 8.3. *Pricing decisions depend heavily on demand elasticity.*

Definition 8.4 (**Markup**). *The difference between a product's selling price and its cost, usually expressed as a percentage of the cost.*

Usually:

- If demand is **inelastic** $\rightarrow$ firms can apply **higher markups**
- If demand is **elastic** $\rightarrow$ firms should apply **lower markups**

8.2.2 Competitors

Competitors influence pricing decisions because they provide **reference prices**. Customers often compare prices across competitors and this affects the **price elasticity of demand**.

Observation 8.4. *The presence of close substitutes increases demand elasticity.*

8.2.3 Costs and Cost-Plus Pricing

Definition 8.5 (Price Ceiling). *The maximum price that customers are willing to pay for a product, influenced by their perceived value and by competitors' prices.*

Definition 8.6 (Price Floor). *The minimum price a company is willing to accept for a product, determined by its incremental (relevant) costs.*

Observation 8.5. *Pricing above the price floor does not guarantee profitability.*

This is because revenues must cover not only incremental costs but also **fixed costs**.

A commonly used pricing method is **cost-plus pricing**, where firms apply a markup over cost.

$$\text{Selling Price} = (1 + \text{Markup }\%) \times \text{Cost}$$

For example, if a company applies a 20% markup and the cost is \$50:

$$\text{Selling Price} = (1 + 0.20) \times \$50 = \$60$$

8.2.4 LO8: Absorption Costing and Cost-Plus Pricing

To determine the selling price using the absorption costing approach, firms follow three steps:

1. Compute the **unit product cost**
2. Determine the **markup percentage**
3. Compute the **selling price**

Definition 8.7 (Absorption Costing Unit Product Cost). *The cost of a product including direct materials, direct labor, variable manufacturing overhead, and fixed manufacturing overhead.*

Observation 8.6. *Under cost-plus pricing, the cost base is the absorption costing unit product cost, not the variable cost.*

EXAMPLE

The following information is provided by the management of Ritter Company:

	Per Unit	Total
Direct materials	**$ 6**	
Direct labor	**4**	
Variable manufacturing overhead	**3**	
Fixed manufacturing overhead		**$ 70,000**
Variable S & A expenses	**2**	
Fixed S & A expenses		**60,000**

Management will use the absorption costing approach to cost-plus pricing to determine the selling price of the product in a three-step process.

Step 1: Unit Product Cost

Assuming 10.000 units:

$$\text{Fixed MOH} = \frac{\$70.000}{10.000} = \$7 \text{ per unit}$$

The unit product cost is computed as:

$$\begin{aligned}\text{Unit product cost} &= \text{DM} + \text{DL} + \text{Variable MOH} + \text{Fixed MOH} = \\ &= \$6 + \$4 + \$3 + \$7 = \$20\end{aligned}$$

Step 2: Markup Percentage

The markup must cover:

- Selling and administrative expenses (S&A)
- Desired return on investment (ROI)

Definition 8.8 (**Return on Investment (ROI)**). *A measure of profitability that indicates the return earned on an investment, calculated as the ratio of net income to the invested capital.*

Let's assume that Ritter must invest $100.000 in the product and market 10.000 units of product each year. The company requires a 20% ROI on all investments.

$$\text{Markup \%} = \frac{(\text{Required ROI} \times \text{Investment}) + \text{S\&A expenses}}{\text{Unit sales} \times \text{Unit product cost}} =$$

$$= \frac{(20\% \times \$100.000) + (\$2 \times 10.000 + \$60.000))}{10.000 \times \$20} =$$

$$= \frac{\$20.000 + \$80.000}{\$200.000} = 50\%$$

Observation 8.7. *The markup is not arbitrary: it is determined to ensure cost coverage and a target return.*

Step 3: Selling Price

Once the markup is known:

$$\text{Selling price} = (1 + \text{Markup \%}) \times \text{Unit product cost} = 1,5 \times \$20 = \$30$$

Observation 8.8. *The absorption costing approach ensures that both manufacturing costs and operating expenses are covered, while also providing a desired return.*

Problems with the Absorption Costing Approach

The absorption costing approach assumes that customers will purchase the forecasted number of units and accept the price set by the company. However, this assumption is often unrealistic because customers have alternatives and may not be willing to buy at that price. If actual sales are lower than expected, the company may not recover its fixed costs, leading to poor financial performance.

For example, suppose Ritter company sets the price of $30 based on expected sales of 10.000 units, but actual sales are only 7.000 units.

RITTER COMPANY Income Statement For the Year Ended December 31, 2017		
Sales (7,000 units × $30)	$	210,000
Cost of goods sold (7,000 units × $23)		161,000
Gross margin		49,000
SG&A expenses		74,000
Net operating loss	$	(25,000)

$$\text{Fixed MOH} = \frac{\$70.000}{7.000} = \$10 \text{ per unit}$$

$$\begin{aligned}\text{Unit product cost} &= \text{DM} + \text{DL} + \text{Variable MOH} + \text{Fixed MOH} = \\ &= \$6 + \$4 + \$3 + \$10 = \$23\end{aligned}$$

$$\text{S\&A expenses} = \$2 \times 7.000 + \$60.000 = \$74.000$$

As a consequence, the actual ROI is the following:

$$\text{ROI} = \frac{-\$25.000}{\$100.000} = -25\%$$

Therefore, absorption costing-based pricing is reliable only if actual demand meets or exceeds the forecasted demand.

8.2.5 LO9: Pricing and Customer Latitude

Observation 8.9. *Pricing decisions must consider not only costs, but also how customers react to price changes.*

Definition 8.9 (Customer Latitude). *The ability of customers to switch to alternative products or reallocate their spending when prices change.*

Customer latitude implies that:

- customers can **purchase from competitors**;
- customers can **shift their budget** to other goods;
- firms cannot assume that quantity sold remains constant when price changes.

EXAMPLE

Suppose a company is considering increasing the price of a product from \$5.00 to \$5.50.

- Current unit sales: 200.000 units
- Expected unit sales after increase: 170.000 units
- Variable cost per unit: \$2
- Fixed expenses: \$570.000

The profit function is:

$$\text{Profit} = (P - V) \times Q - \text{Fixed expenses}$$

Current situation:

$$\text{Profit} = (\$5 - \$2) \times 200.000 - \$570.000 = \$30.000$$

After price increase:

$$\text{Profit} = (\$5{,}50 - \$2) \times 170.000 - \$570.000 = \$25.000$$

Observation 8.10. *Even though the price increases, profit decreases because the reduction in sales volume more than offsets the higher margin per unit.*

A key managerial question is:

How many units must be sold at the new price to maintain the same profit?

Assuming fixed expenses remain unchanged, we solve for Q:

$$\text{Profit} = (P - V) \times Q - \text{Fixed expenses}$$

$$\$30.000 = (\$5{,}50 - \$2) \times Q - \$570.000$$

$$Q \approx 171.429 \text{ units}$$

Observation 8.11. *If unit sales remain above this threshold, the price increase improves profit. Otherwise, profit decreases.*

Optimal Pricing Model

Managers can analyze different price–volume combinations to identify the price that maximizes profit. The key idea is that increasing price means higher margin per unit but also lower sales volume.

Observation 8.12. *The optimal price is the one that balances margin and volume in order to maximize total profit.*

In practice, tools such as optimization models (e.g., spreadsheet solvers) can be used to evaluate different scenarios.

Assume the following data:

	A	B	C	D	E	F
1	**Optimal Pricing Model**					
2						
3	**Apple-Almond Shampoo:**					
4	**Current unit sales**	200.000				
5	**Current selling price per unit**	$5,00				
6	**Variable cost per unit**	$2,00				
7	**Traceable fixed costs**	$570.000				
8						
9	**Percentage change in selling price**	10%				
10	**Percentage change in unit sales**	-13%				
11	**Implied price elasticity**	-1,461				
12		**Per Unit**		**Unit Sales**		**Total**
13	**Sales**	$6,34	x	141.467	=	$896.481
14	**Variable expenses**	$2,00	x	141.467	=	$282.934
15	**Contribution margin**	$4,34	x	141.467	=	$613.547
16	**Traceable fixed expenses**					$570.000
17	**Net operating income**					$43.547
18						

Download OPM Excel File

The optimal price is higher than the initial price, even though unit sales decrease significantly. This is due to the following trade-off:

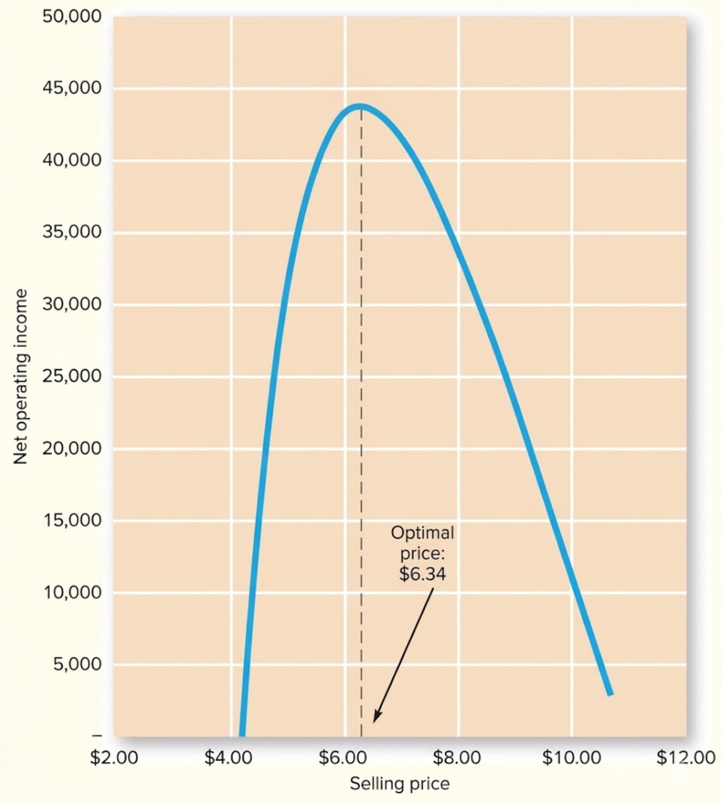
- Increasing price $\rightarrow$ increases the **contribution margin per unit**
- Increasing price $\rightarrow$ reduces the **number of units sold**

Observation 8.13. *Even though the company sells fewer units, the higher contribution margin per unit more than compensates for the reduction in volume, leading to a higher total profit.*

Observation 8.14. *The objective of pricing is not to maximize sales volume, but to maximize total profit.*

Observation 8.15. *The optimal price depends on customer sensitivity to price changes: ignoring this relationship can lead to suboptimal pricing decisions.*

The relationship between price and profit can be represented graphically:



Observation 8.16. *Profit is typically maximized at an intermediate price level, not at the lowest or highest possible price.*

8.2.6 LO10: Value-Based Pricing

Definition 8.10 (**Value-Based Pricing**). *A pricing approach in which the selling price is based on the* ***perceived economic value*** *of a product or service to the customer, rather than on its cost.*

Value-based pricing is an alternative to cost-plus pricing and focuses on **customer value** instead of production cost. Companies using value-based pricing set prices according to the **benefits** that customers receive from the product.

Definition 8.11 (**Economic Value to the Customer (EVC)**). *The maximum price that a customer would be willing to pay for a product, based on the value it provides compared to the best available alternative.*

The EVC is defined as:

$$\text{EVC} = \text{Reference value} + \text{Differentiation value}$$

Where:

- **Reference value**: price of the best alternative available to the customer.

- **Differentiation value**: additional value provided by the product compared to the alternative.

The selling price should lie between the reference value and the EVC:

$$\text{Reference value} \leq \text{Price} \leq \text{EVC}$$

- If the price is equal to the **reference value**, the customer is indifferent between the product and its alternative.
- If the price is equal to the **EVC**, the company captures all the value created.

In practice, firms choose a price **between these two values** to share value with customers.

EXAMPLE

Assume that the managers of Hike America magazine want to establish a selling price for a one-month full-page advertisement in their magazine. Hiking Trails magazine, their primary competitor, charges $5,000 per month for a full-page ad. The managers of Hike America believe that they can justify a higher selling price by quantifying the EVC. The chart calculates a differential value of $4,000.

	Hike America	Hiking Trails
Number of readers (a)	200,000	300,000
Percent of readers who buy advertised products each month (b)	.002	.001
Number of readers per month buying advertised products (a) × (b)	400	300
Monthly sales per reader who buys advertised products (a)	$100	$80
Contribution margin ratio (b)	25%	25%
Monthly contribution margin per reader who buys advertised products (a) × (b)	$25	$20
Number of readers per month buying advertised products (a)	400	300
Monthly contribution margin per reader who buys advertised products (b)	$25	$20
Contribution margin per month provided by a full-page ad (a) × (b)	$10,000	$6,000
Differentiation value		$4,000

Therefore:

$$\text{EVC} = \$5.000 + \$4.000 = \$9.000$$

The feasible price range is:

$$\$5.000 \leq \text{Price} \leq \$9.000$$

Value-based pricing requires understanding **customer behavior and benefits**, not only costs. It is closely related to **demand elasticity**: higher perceived value $\Rightarrow$ lower sensitivity to price. Unlike cost-plus pricing, this approach is **market-driven**.

Observation 8.17. *The key question is not:*

"What does the product cost?"

but:

"What is the product worth to the customer?"

8.2.7 LO11: Target Costing

Definition 8.12 (**Target Costing**)**.** *A process used to determine the* ***maximum allowable cost*** *of a product based on its expected selling price and desired profit.*

The fundamental equation is:

$$\text{Target Cost} = \text{Anticipated Selling Price} - \text{Desired Profit}$$

Once the target cost is determined, the product must be **designed** so that it can be produced at or below that cost. Two fundamental observations justify this approach:

- The **market determines the price** (supply and demand).
- Most of a product's cost is determined during the **design stage**.

Observation 8.18. *Traditional approaches try to reduce costs* ***after production begins****, while target costing focuses on cost control* ***before production****.*

EXAMPLE

Handy Appliance feels there is a niche for a hand mixer with special features. The marketing department believes that a price of \$30 would be about right and that about 40.000 mixers could be sold. An investment of \$2.000.000 is required to gear up for production. The company requires a 15% ROI on invested funds.

Step 1: Compute projected sales

$$\text{Sales} = 40.000 \times \$30 = \$1.200.000$$

Step 2: Compute desired profit

$$\text{Desired profit} = 15\% \times \$2.000.000 = \$300.000$$

Step 3: Compute total target cost

$$\text{Target cost} = \$1.200.000 - \$300.000 = \$900.000$$

Step 4: Compute target cost per unit

$$\text{Target cost per unit} = \frac{\$900.000}{40.000} = \$22{,}50$$

This means that:

- The product must be designed so that its cost is **at most $22,50 per unit**.
- If actual cost exceeds this level, the product is **not acceptable**.

9
Exercises Ch06

9.1 Random concepts captured (31/03/2026)

Green vs Grey Hydrogen

Definition 9.1 (**Grey Hydrogen**)**.** *Hydrogen produced from* ***fossil fuels****, typically natural gas, through processes such as steam methane reforming (SMR).*

- It is the **most common and cheapest** form of hydrogen production.
- It generates significant **CO_2 emissions**.
- It is **not environmentally sustainable**.

Definition 9.2 (**Green Hydrogen**)**.** *Hydrogen produced using* ***renewable energy sources*** *(e.g., solar, wind) through water electrolysis.*

- It produces **no direct CO_2 emissions**.
- It is considered **environmentally sustainable**.
- It is currently **more expensive** than grey hydrogen.

Observation 9.1. *The transition from grey to green hydrogen is driven by climate policies, carbon pricing, and technological improvements in renewable energy and electrolysis.*

9.2 Exercise 10

Country Curtains Company makes fashionable window curtains and valances. Data concerning the company's three product lines appear below.

	6' Curtains	3' Curtains	Valance
Selling price per unit	\$50	\$40	\$55
Variable cost per unit	\$26	\$16	\$25
Feet of fabric	6 feet	3 feet	4 feet
Sewing machine time	6 minutes	8 minutes	16 minutes

Required:

1. If we assume that the total fabric available is the constraint in the production process, how much contribution margin per foot of the constrained resource is earned by each product?
2. Which product offers the most profitable use of the fabric?
3. If we assume that a labor shortage has required the company to cut back its production so much that its new constraint has become the total available sewing machine time, how much contribution margin per minute of the constrained resource is earned by each product?
4. Which product offers the most profitable use of the sewing machine time?
5. Which product has the largest contribution margin per unit?
6. Assuming that the required number of units for all three variants is equal to 300, and that the total available fabric is equal to 1.800 feet, what is the optimal mix?
7. Assuming that the required number of units for all three variants is equal to 300, and that the total available sewing machine time is equal to 2.600 minutes, what is the optimal mix?

Solution (1)

$$\text{Contribution Margin per Foot} = \frac{\text{Contribution Margin per Unit}}{\text{Feet Required per Unit}}$$

6' Curtains:

$$\text{Contribution Margin per Foot} = \frac{\$50 - \$26}{6 \text{ feet}} = \frac{\$24}{6 \text{ feet}} = \$4 \text{ per foot}$$

3' Curtains:

$$\text{Contribution Margin per Foot} = \frac{\$40 - \$16}{3 \text{ feet}} = \frac{\$24}{3 \text{ feet}} = \$8 \text{ per foot}$$

Valance:

$$\text{Contribution Margin per Foot} = \frac{\$55 - \$25}{4 \text{ feet}} = \frac{\$30}{4 \text{ feet}} = \$7{,}50 \text{ per foot}$$

Solution (2)

The answer is the second product (**3' Curtains**), as it has the highest contribution margin per foot.

Solution (3)

$$\text{Contribution Margin per Minute} = \frac{\text{Contribution Margin per Unit}}{\text{Minutes Required per Unit}}$$

6' Curtains:

$$\text{Contribution Margin per Minute} = \frac{\$50 - \$26}{6 \text{ minutes}} = \frac{\$24}{6 \text{ minutes}} = \$4 \text{ per minute}$$

3' Curtains:

$$\text{Contribution Margin per Minute} = \frac{\$40 - \$16}{8 \text{ minutes}} = \frac{\$24}{8 \text{ minutes}} = \$3 \text{ per minute}$$

Valance:

$$\text{Contribution Margin per Minute} = \frac{\$55 - \$25}{16 \text{ minutes}} = \frac{\$30}{16 \text{ minutes}} = \$1{,}88 \text{ per minute}$$

Solution (4)

The answer is the first product (**6' Curtains**), as it has the highest contribution margin per minute.

Solution (5)

The answer is the third product (**Valance**) with $30 against $24 of the other two products.

Solution (6)

Since the constrained resource is the **fabric (feet)**, the optimal production plan is obtained by prioritizing products with the **highest contribution margin per foot**.

- 1st: Product 2
- 2nd: Product 3
- 3rd: Product 1

We start by producing Product 2 up to its maximum demand:

$$\text{Feet required} = 300 \times 3 = 900 \text{ feet}$$
$$\text{Contribution margin} = 300 \times \$24 = \$7.200$$

The total available fabric is 1.800 feet, therefore the remaining fabric is 900 feet.

Next, we allocate the remaining capacity to Product 3:

$$Q \times 4 = 900 \Rightarrow Q = \frac{900}{4} = 225 \text{ units}$$

$$\text{Contribution margin} = 225 \times \$30 = \$6.750$$

No capacity remains for Product 1.

Therefore, the total contribution margin is:

$$\text{Total CM} = \$7.200 + \$6.750 = \$13.950$$

Solution (7)

Since the constrained resource is the **sewing machine time (minutes)**, the optimal production plan is obtained by prioritizing products with the **highest contribution margin per minute**.

- 1st: Product 1
- 2nd: Product 2
- 3rd: Product 3

We start by producing Product 1 up to its maximum demand:

$$\text{Minutes required} = 300 \times 6 = 1.800 \text{ minutes}$$
$$\text{Contribution margin} = 300 \times \$24 = \$7.200$$

The total available time is 2.600 minutes, therefore the remaining time is 800 minutes.

Next, we allocate the remaining capacity to Product 2:

$$Q \times 8 = 800 \Rightarrow Q = \frac{800}{8} = 100 \text{ units}$$

$$\text{Contribution margin} = 100 \times \$24 = \$2.400$$

No capacity remains for Product 3.

Therefore, the total contribution margin is:

$$\text{Total CM} = \$7.200 + \$2.400 = \$9.600$$

9.3 Exercise 11

Thomas Ltd. makes fashionable handcrafted dining tables. The bottleneck in the production process is in the carving shop. Demand for these handcrafted tables exceeds the company's capacity. Information concerning the company's three product lines appears below:

	Ornate Table	**Classic Table**	**Modern Table**
Selling price per unit	\$2.000	\$1.900	\$2.200
Variable cost per unit	\$1.200	\$1.000	\$1.200
Carving shop hours per unit	8	5	5

Required:

1. More time could be made available in the carving shop by asking the employees who work in this shop to work overtime. Assuming that this extra time would be used to produce Ornate Tables, up to how much a premium should the company be willing to pay per hour to keep the carving shop open after normal working hours?

2. A small nearby woodworking company has offered to carve furniture for Thomas at a fixed charge of \$130 per hour. The management of Thomas is confident that this company's work is high quality and their craftsmen can work as quickly as Thomas's own craftsmen on the simpler carving jobs such as the Modern Table. How much additional contribution margin per hour can Thomas earn if it hires the nearby woodworking company to make Modern Tables?

Solution (1)

We first compute the **contribution margin per unit**:

$$\begin{aligned}\text{CM per Unit} &= \text{Selling Price per Unit} - \text{Variable Cost per Unit} = \\ &= \$2.000 - \$1.200 = \$800\end{aligned}$$

Since the **carving shop time** is the constrained resource and each unit requires 8 hours, we compute the **contribution margin per hour**:

$$\text{CM per Hour} = \frac{\text{CM per Unit}}{\text{Hours per Unit}} = \frac{\$800}{8} = \$100 \text{ per hour}$$

The maximum premium that the company should be willing to pay for one additional hour of carving time is equal to the contribution margin generated by that hour.

$$\text{Maximum Premium} = \$100 \text{ per hour}$$

Solution (2)

We first compute the **contribution margin per unit** for the Modern Table:

$$\begin{aligned}\text{CM per Unit} &= \text{Selling Price per Unit} - \text{Variable Cost per Unit} = \\ &= \$2.200 - \$1.200 = \$1.000\end{aligned}$$

Since each unit requires 5 hours of carving time, we compute the **contribution margin per hour**:

$$\text{CM per Hour} = \frac{\text{CM per Unit}}{\text{Hours per Unit}} = \frac{\$1.000}{5} = \$200 \text{ per hour}$$

If the external company is hired, the cost is $130 per hour. Therefore, the **additional contribution margin per hour** is:

$$\text{Additional CM} = \text{CM per Hour} - \text{Outsourcing Cost} = \$200 - \$130 = \$70 \text{ per hour}$$

9.4 Exercise 12

Porky Products Company manufactures three products from a pig in a joint processing operation. Joint processing costs up to the split-off point total $40.000 per year. The company allocates these costs to the joint products on the basis of their total sales value at the split-off point. Each product may be sold at the split-off point or processed further.

Additional processing requires no special facilities. The additional processing costs and the sales value after further processing for each product (on an annual basis) are shown below:

Product	**Additional Processing Costs**	**Sales Value after Further Processing**	**Sales Value at Split-off Point**
Ham	$20.000	$44.000	$20.000
Bacon	$30.000	$60.000	$35.000
Pork chops	$12.000	$50.000	$30.000

Required:
Which product or products should be sold at the split-off point, and which product or products should be processed further?

Solution

The decision is based on **incremental analysis**. Joint costs are irrelevant and therefore ignored.

Ham:

$$\begin{aligned}\text{Incremental Revenue} &= \$44.000 - \$20.000 = \$24.000\\ \text{Incremental Cost} &= \$20.000\\ \text{Net Benefit} &= \$24.000 - \$20.000 = \$4.000\end{aligned}$$

Since the net benefit is positive, Ham should be **processed further**.

Bacon:

$$\begin{aligned}\text{Incremental Revenue} &= \$60.000 - \$35.000 = \$25.000\\ \text{Incremental Cost} &= \$30.000\\ \text{Net Benefit} &= \$25.000 - \$30.000 = -\$5.000\end{aligned}$$

Since the net benefit is negative, Bacon should be **sold at the split-off point**.

Pork chops:

$$\begin{aligned}\text{Incremental Revenue} &= \$50.000 - \$30.000 = \$20.000\\ \text{Incremental Cost} &= \$12.000\\ \text{Net Benefit} &= \$20.000 - \$12.000 = \$8.000\end{aligned}$$

Since the net benefit is positive, Pork chops should be **processed further**.

9.5 Exercise 13

Delta Company produces a single product. The cost of producing and selling a single unit of this product at the company's normal activity level of 65.000 units per year is:

Direct materials	\$7,50
Direct labor	\$4,60
Variable manufacturing overhead	\$2,50
Fixed manufacturing overhead	\$6,90
Variable selling and administrative expense	\$2,75
Fixed selling and administrative expense	\$4,30

The normal selling price is \$40 per unit. The company's capacity is 90.000 units per year. An order has been received from a mail-order house for 25.000 units at a special price of \$24,00 per unit. This order would not affect regular sales or the company's total fixed costs.

Required:

1. What is the financial advantage (disadvantage) of accepting the special order?
2. As a separate matter from the special order, assume the company's inventory includes 800 units of this product that were produced last year and that are inferior to the current model. The units must be sold through regular channels at reduced prices. What unit cost is relevant for establishing a minimum selling price for these units?

Solution (1)

The decision is based on **incremental analysis**. Since the company has unused capacity and fixed costs are unaffected, only **variable costs** are relevant.

The decision rule is:

$$\textbf{Accept} \text{ if Incremental Revenue} > \text{Incremental Cost}$$

First, compute the **variable cost per unit**:

$$\text{Variable Cost per Unit} = 7{,}50 + 4{,}60 + 2{,}50 + 2{,}75 = \$17{,}35$$

Now compute the incremental values:

$$\text{Incremental Revenue} = 25.000 \times \$24 = \$600.000$$
$$\text{Incremental Cost} = 25.000 \times \$17{,}35 = \$433.750$$
$$\text{Financial Advantage} = \$600.000 - \$433.750 = \$166.250$$

Since the financial advantage is positive, the company should **accept the special order**.

Solution (2)

Since the units must be sold through regular channels, the only relevant cost for establishing a minimum selling price is the **variable selling and administrative expense**, because this cost will be incurred only if the units are sold.

$$\text{Relevant Unit Cost} = \$2{,}75$$

The manufacturing cost of the 800 units is irrelevant because it was incurred in the past and cannot be changed. Only future costs that differ between alternatives are relevant.

9.6 Exercise 14

Dianya Company uses the absorption costing approach to cost-plus pricing. It is considering the introduction of a new product. To determine a selling price, the company has gathered the following information:

Number of units to be produced and sold each year	20.000
Unit product cost	\$50
Estimated annual selling and administrative expenses	\$450.000
Estimated investment required by the company	\$1.000.000
Desired return on investment (ROI)	20%

Required:
Compute the selling price per unit.

Solution

$$\text{Markup }\% = \frac{(\text{Required ROI} \times \text{Investment}) + \text{S\&A expenses}}{\text{Unit sales} \times \text{Unit product cost}} =$$

$$= \frac{(20\% \times \$1.000.000) + \$450.000}{20.000 \times \$50} =$$

$$= \frac{\$200.000 + \$450.000}{\$1.000.000} = 65\%$$

Once the markup percentage has been determined, the selling price is computed as:

$$\text{Selling price} = (1 + \text{Markup }\%) \times \text{Unit product cost} = 1{,}65 \times \$50 = \$82{,}50$$

9.7 Exercise 15

Yazzie Company has developed a new industrial component called MB-50. The company is excited about MB-50 because it offers superior performance relative to the comparable component sold by Yazzie's primary competitor. The competing part sells for $1.500 and needs to be replaced after 3.000 hours of use. It also requires $350 of preventive maintenance during its useful life.

The MB-50's performance capabilities are similar to its competing product with two important exceptions—it needs to be replaced after 5.400 hours of use and it requires $500 of preventive maintenance during its useful life.

Required:
From a value-based pricing standpoint:

1. What is MB-50's economic value to the customer over its 5.400-hour life?
2. What range of possible prices should Yazzie consider when setting a price for MB-50?

Solution (1)

$$\text{EVC} = \text{Reference value} + \text{Differentiation value}$$

The reference value is the price of the competing product:

$$\text{Reference value} = \$1.500$$

The differentiation value captures the **economic advantage** of MB-50 compared to the competitor.

Preventive maintenance cost over 5.400 hours:

$$\text{Competing product} = \$350 \times \frac{5.400}{3.000} = \$630$$
$$\text{MB-50} = \$500$$
$$\text{Maintenance savings} = \$630 - \$500 = \$130$$

Capital cost savings (longer useful life):
The MB-50 lasts longer (5.400 hours vs 3.000 hours), so the customer avoids purchasing an additional fraction of the competing product:

$$\text{Additional usage} = 5.400 - 3.000 = 2.400 \text{ hours}$$

$$\text{Capital savings} = \$1.500 \times \frac{2.400}{3.000} = \$1.200$$

Total differentiation value:

$$\begin{aligned}\text{Differentiation value} &= \text{Maintenance savings} + \text{Capital savings} = \\ &= \$130 + \$1.200 = \$1.330\end{aligned}$$

Economic value to the customer:

$$\begin{aligned}\text{EVC} &= \text{Reference value} + \text{Differentiation value} = \\ &= \$1.500 + \$1.330 = \$2.830\end{aligned}$$

Solution (2)

The selling price should lie between the reference value and the EVC:

$$\text{Reference value} \leq \text{Price} \leq \text{EVC}$$

$$\$1.500 \leq \text{Price} \leq \$2.830$$

9.8 Exercise 16

Schmidt Products Corporation of Germany is anxious to enter the stapler market. Management believes that in order to be competitive in world markets, the price of the stapler that the company is developing cannot exceed \$7 U.S. Schmidt's required rate of return is 8% on all investments. An investment of \$2.000.000 would be required to purchase the equipment needed to produce the 80.000 staplers that management believes can be sold each year at the \$7 price.

Required:
Compute the target cost of one stapler.

Solution

First we compute the **projected sales**:

$$\text{Sales} = 80.000 \times \$7 = \$560.000$$

Then we compute **desired profit**:

$$\text{Desired profit} = 8\% \times \$2.000.000 = \$160.000$$

Then we compute **total target cost**:

$$\text{Target cost} = \$560.000 - \$160.000 = \$400.000$$

Finally, we compute **target cost per unit**:

$$\text{Target cost per unit} = \frac{\$400.000}{80.000} = \boxed{\$5 \text{ per stapler}}$$

9.9 Exercise 17

Klopfer Company has developed a new industrial piece of equipment called the GWSD-49. The company is considering two methods of establishing a selling price for the GWSD-49—absorption cost-plus pricing and value-based pricing.

Klopfer's cost accounting system reports an absorption unit product cost for GWSD-49 of $15.000. Its markup percentage on absorption cost is 90%. The company's marketing managers have expressed concerns about the use of absorption cost-plus pricing because it seems to overlook the fact that the GWSD-49 offers superior performance relative to the comparable piece of equipment sold by Klopfer's primary competitor. More specifically, the GWSD-49 can be used for 25.000 hours before replacement. It only requires $4.500 of preventive maintenance during its useful life and it consumes $65 of electricity per 500 hours used.

These figures compare favorably to the competing piece of equipment that sells for $25.000, needs to be replaced after 12.500 hours of use, requires $5.000 of preventive maintenance during its useful life and consumes $95 of electricity per 500 hours used.

Required:

1. If Klopfer uses absorption cost-plus pricing, what price will it establish for the GWSD-49?
2. What is GWSD-49's economic value to the customer (EVC) over its 25.000-hour life?
3. If Klopfer uses value-based pricing, what range of possible prices should it consider when setting a price for the GWSD-49?

Solution (1)

$$\text{Selling price} = (1 + \text{Markup }\%) \times \text{Unit product cost} = 1{,}9 \times \$15.000 = \$28.500$$

Solution (2)

$$\text{EVC} = \text{Reference value} + \text{Differentiation value}$$

The reference value is the price of the competing product:

$$\text{Reference value} = \$25.000$$

The differentiation value captures the **economic advantage** of GWSD-49 compared to the competitor.

Preventive maintenance cost over 25.000 hours:

$$\text{Competing product} = \$5.000 \times \frac{25.000}{12.500} = \$10.000$$

$$\text{GWSD-49} = \$4.500$$

$$\text{Maintenance savings} = \$10.000 - \$4.500 = \$5.500$$

Capital cost savings (longer useful life):
The GWSD-49 lasts longer (25.000 hours vs 12.500 hours), so the customer avoids purchasing an additional fraction of the competing product:

$$\text{Additional usage} = 25.000 - 12.500 = 12.500 \text{ hours}$$

$$\text{Capital savings} = \$25.000 \times \frac{12.500}{12.500} = \$25.000$$

Electricity savings:

$$\text{Competing product} = \$95 \times \frac{25.000}{500} = \$4.750$$

$$\text{GWSD-49} = \$65 \times \frac{25.000}{500} = \$3.250$$

$$\text{Electricity savings} = \$4.750 - \$3.250 = \$1.500$$

Total differentiation value:

$$\text{Differentiation value} = \text{Maintenance savings} + \text{Capital savings} + \text{Electricity savings} =$$
$$= \$5.500 + \$25.000 + \$1.500 = \$32.000$$

Economic value to the customer:

$$
\begin{aligned}
\text{EVC} &= \text{Reference value} + \text{Differentiation value} = \\
&= \$25.000 + \$32.000 = \$57.000
\end{aligned}
$$

Solution (3)

The selling price should lie between the reference value and the EVC:

$$
\begin{aligned}
\text{Reference value} &\leq \text{Price} \leq \text{EVC} \\
\$25.000 &\leq \text{Price} \leq \$57.000
\end{aligned}
$$

9.10 Exercise 18

The postal service of Tobago, an island in the West Indies, obtains a significant portion of its revenues from sales of special souvenir sheets to stamp collectors. The postal service purchases the souvenir sheets from a supplier for \$1.20 each. Tobago has been selling the souvenir sheets for \$10.00 each and ordinarily sells about 95.000 units. To test the market, the postal service recently priced a new souvenir sheet at \$9.00 and sales increased to 111.150 units.

Required:

1. What total contribution margin did the postal service earn when it sold 95.000 sheets at a price of \$10.00 each?
2. By what percentage did the Tobago post office decrease its selling price? By what percentage did unit sales increase?
3. What total contribution margin did the postal service earn when it sold 111.150 sheets at a price of \$9.00 each?
4. What was the postal service's increase (decrease) in total contribution margin going from the higher price of \$10.00 to the lower price of \$9.00?
5. How many sheets would the postal service have to sell at the lower price of \$9.00 to equal the total contribution margin earned at the higher price of \$10.00?
6. What percentage increase in the number of sheets sold at \$9.00 must be achieved to equal the total contribution margin earned at the higher price of \$10.00?

Solution (1)

$$\text{CM} = (P - V) \times Q = (\$10 - \$1{,}20) \times 95.000 = \boxed{\$836.000}$$

Solution (2)

$$\text{Selling price decrease} = \frac{\$10 - \$9}{\$10} = 0{,}1 = \boxed{10\%}$$

$$\text{Unit sales increase} = \frac{111.150 - 95.000}{95.000} = 0{,}17 = \boxed{17\%}$$

Solution (3)

$$\text{CM} = (P - V) \times Q = (\$9 - \$1{,}20) \times 111.150 = \boxed{\$866.970}$$

Solution (4)

$$\Delta\text{CM} = \$866.970 - \$836.000 = \boxed{\$30.970}$$

Solution (5)

$$\begin{aligned} \text{CM} &= (P - V) \times Q \\ \$836.000 &= (\$9 - \$1{,}20) \times Q \end{aligned}$$

$$Q = \frac{\$836.000}{\$7{,}80} \approx \boxed{107.180}$$

Solution (6)

$$\text{Unit sales increase} = \frac{107.180 - 95.000}{95.000} \approx 0{,}128 = \boxed{12{,}8\%}$$

9.11 Exercise 19

Klayton Company wants to use absorption cost-plus pricing to set the selling price on a newly remodeled product. The company plans to invest $200.000 in operating assets to produce and sell 15.000 units. Its required return on investment (ROI) in its operating assets is 8%. The accounting department has provided cost estimates for the new product as follows:

	Per Unit	**Total**
Direct materials	$5.00	
Direct labor	$4.00	
Variable manufacturing overhead	$2.00	
Fixed manufacturing overhead		$75.000
Variable selling and administrative expenses	$1.50	
Fixed selling and administrative expenses		$45.500

Required:

1. What is the unit product cost for the remodeled product?
2. What is the markup percentage on absorption cost for the remodeled product?
3. What selling price would the company establish for its remodeled product using a markup percentage on absorption cost?
4. Suppose the company actually sold only 12.000 units (instead of its planned sales volume of 15.000 units) at the selling price that you derived in Requirement 3. What ROI did the company actually earn at this lower sales volume?
5. Assume that the company wants to raise the price of its newly remodeled product with the intention of achieving the product's desired ROI at the lower sales volume of 12.000 units. Using absorption cost-plus pricing, what would be the revised selling price at this lower sales volume?

Solution (1)

$$\begin{aligned}\text{Unit product cost} &= \text{DM} + \text{DL} + \text{Variable MOH} + \text{Fixed MOH} = \\ &= \$5 + \$4 + \$2 + \frac{\$75.000}{15.000} = \boxed{\$16}\end{aligned}$$

Solution (2)

$$\begin{aligned}\text{Markup \%} &= \frac{(\text{Required ROI} \times \text{Investment}) + \text{S\&A expenses}}{\text{Unit sales} \times \text{Unit product cost}} = \\ &= \frac{(8\% \times \$200.000) + (\$1{,}50 \times 15.000 + \$45.500)}{15.000 \times \$16} = \\ &= \frac{\$16.000 + \$68.000}{\$240.000} = \boxed{35\%}\end{aligned}$$

Solution (3)

$$\text{Selling price} = (1 + \text{Markup \%}) \times \text{Unit product cost} = 1{,}35 \times \$16 = \boxed{\$21{,}60}$$

Solution (4)

$$\begin{aligned}\text{Unit product cost} &= \text{DM} + \text{DL} + \text{Variable MOH} + \text{Fixed MOH} = \\ &= \$5 + \$4 + \$2 + \frac{\$75.000}{12.000} = \$17{,}25\end{aligned}$$

$$\begin{aligned}\text{Sales} &= \$21{,}60 \times 12.000 = \$259.200 \\ \text{Cost of goods sold} &= \$17{,}25 \times 12.000 = \$207.000 \\ \text{S\&A expenses} &= \$1{,}50 \times 12.000 + \$45.500 = \$63.500\end{aligned}$$

$$\begin{aligned}\text{ROI} &= \frac{\text{Net operating loss}}{\text{Initial investment}} = \\ &= \frac{\$259.200 - \$207.000 - \$63.500}{\$200.000} = -0{,}0565 \approx \boxed{-5{,}6\ \%}\end{aligned}$$

Solution (5)

$$\text{Markup \%} = \frac{(\text{Required ROI} \times \text{Investment}) + \text{S\&A expenses}}{\text{Unit sales} \times \text{Unit product cost}} =$$

$$= \frac{(8\% \times \$200.000) + \$63.500}{12.000 \times \$17{,}25} =$$

$$= \frac{\$16.000 + \$63.500}{\$207.000} \approx \boxed{38{,}4\%}$$

$$\text{Selling price} = (1 + \text{Markup \%}) \times \text{Unit product cost} = 1{,}384 \times \$17{,}25 \approx \boxed{\$23{,}87}$$